Containers & OpenShift Workshop

Alexander Mamani - IBM
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containers are created from container images

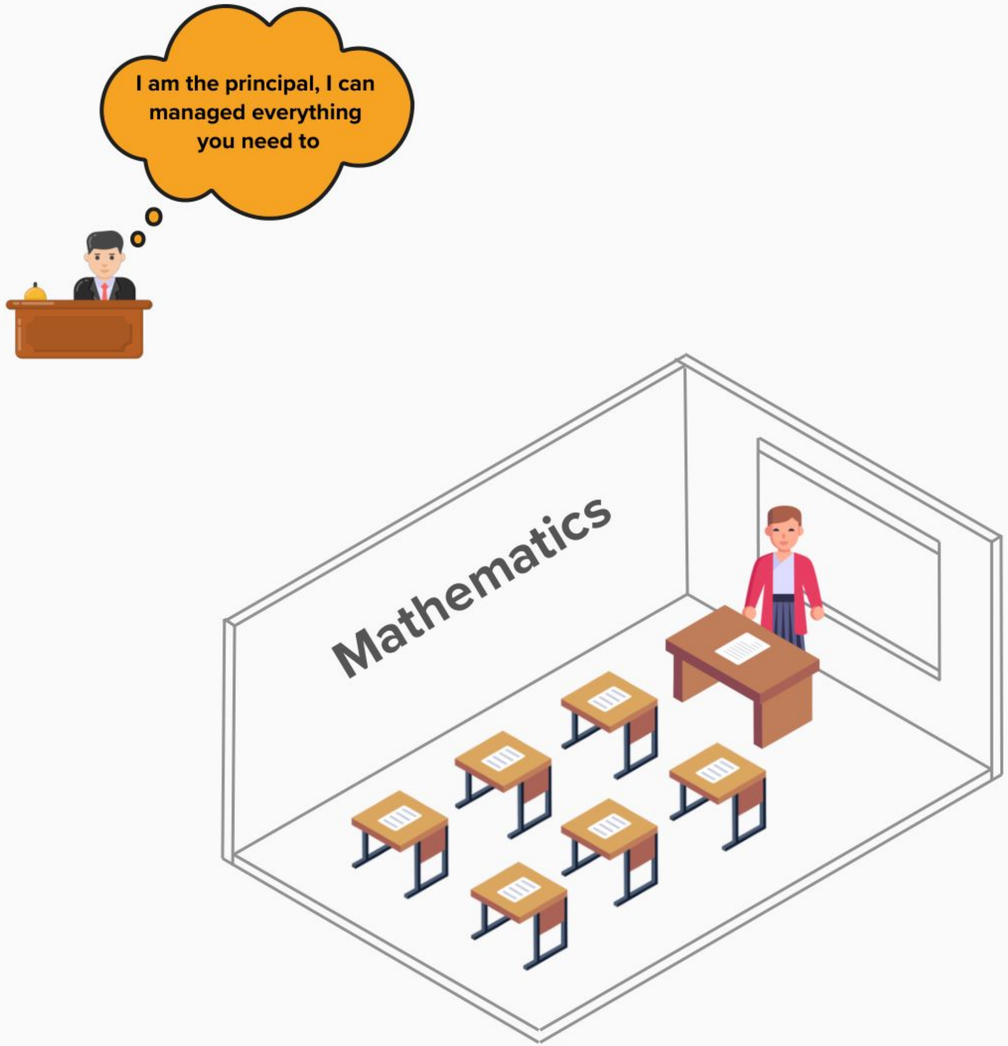
What is kubernetes?

containers are created from container images

Kubernetes is a portable, extensible, open source platform for managing containerized workloads and services that facilitate both declarative configuration and automation. It has a large, rapidly growing ecosystem. Kubernetes services, support, and tools are widely available.

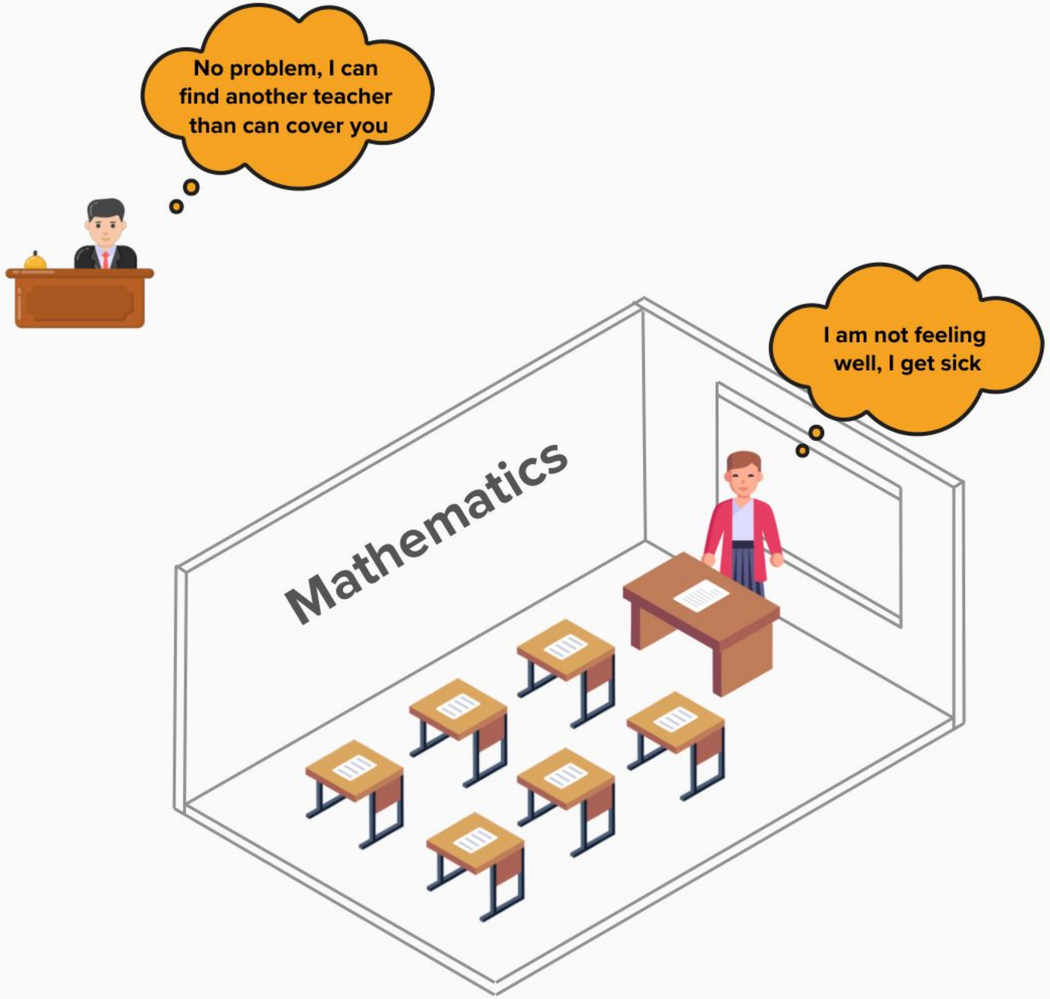
Let’s compare it with a real-life example

1. KUBERNETES



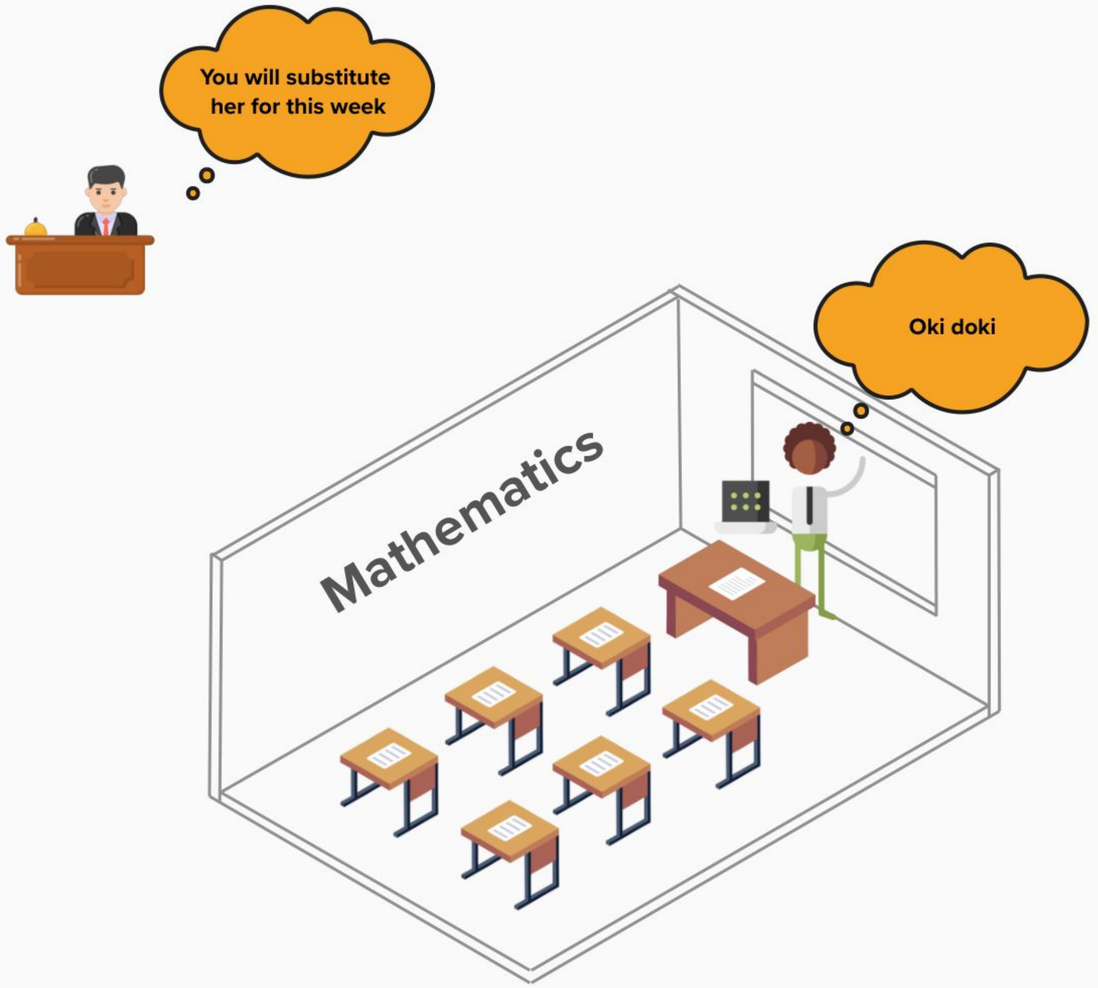
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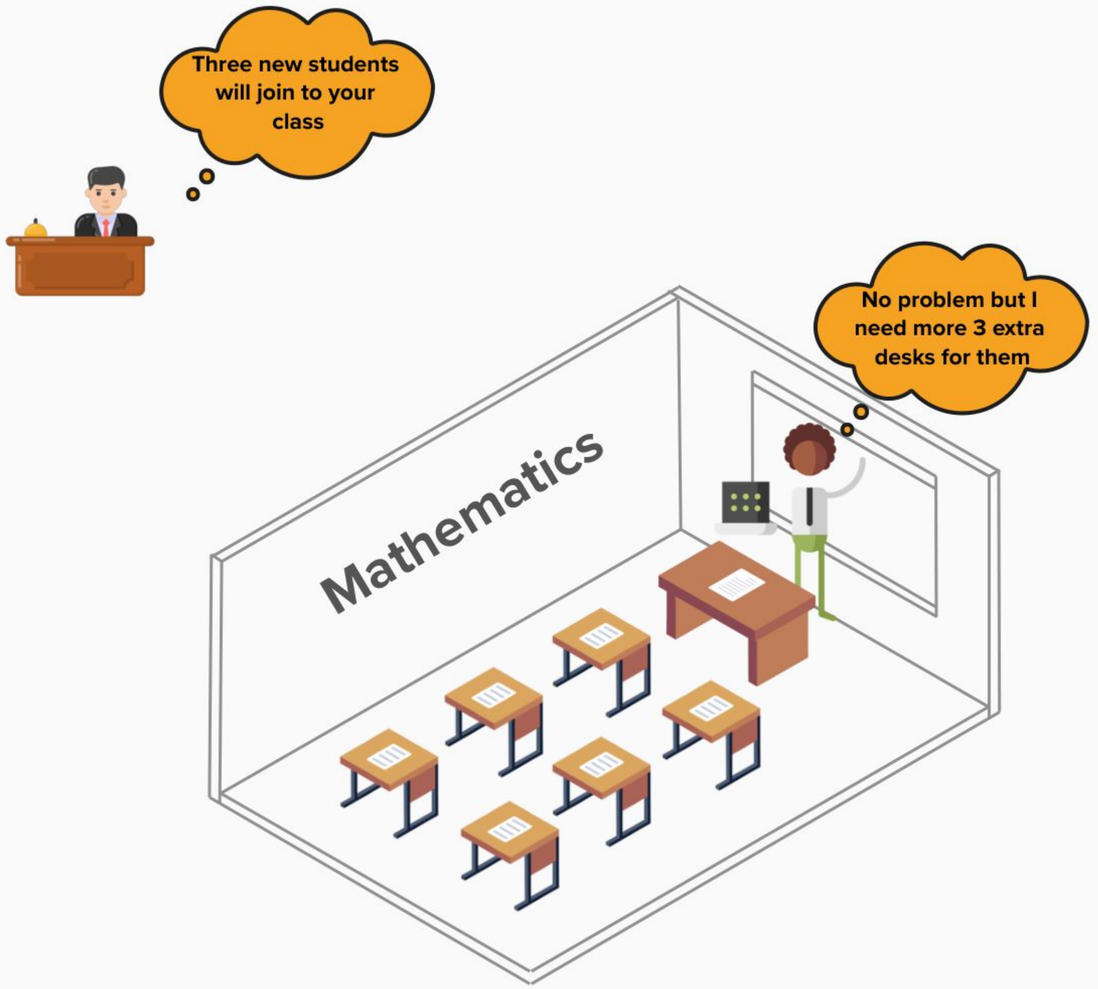
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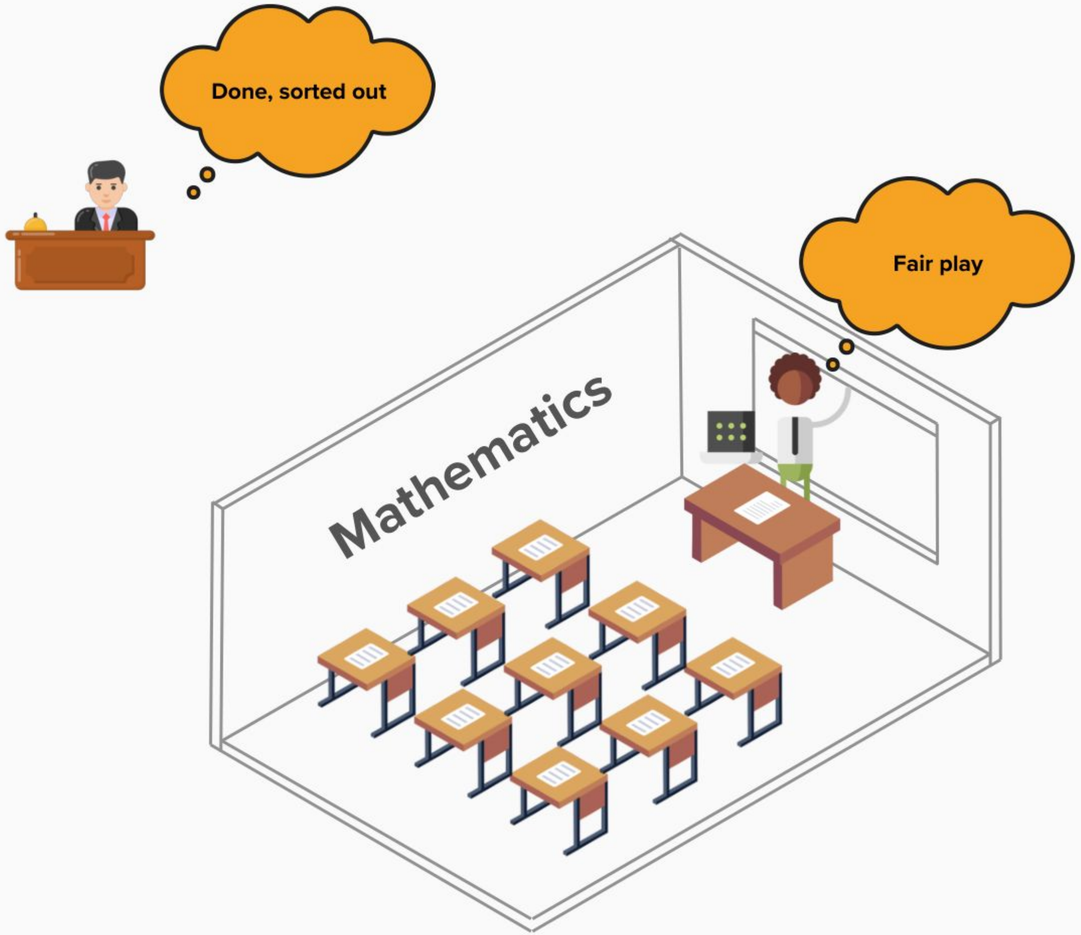
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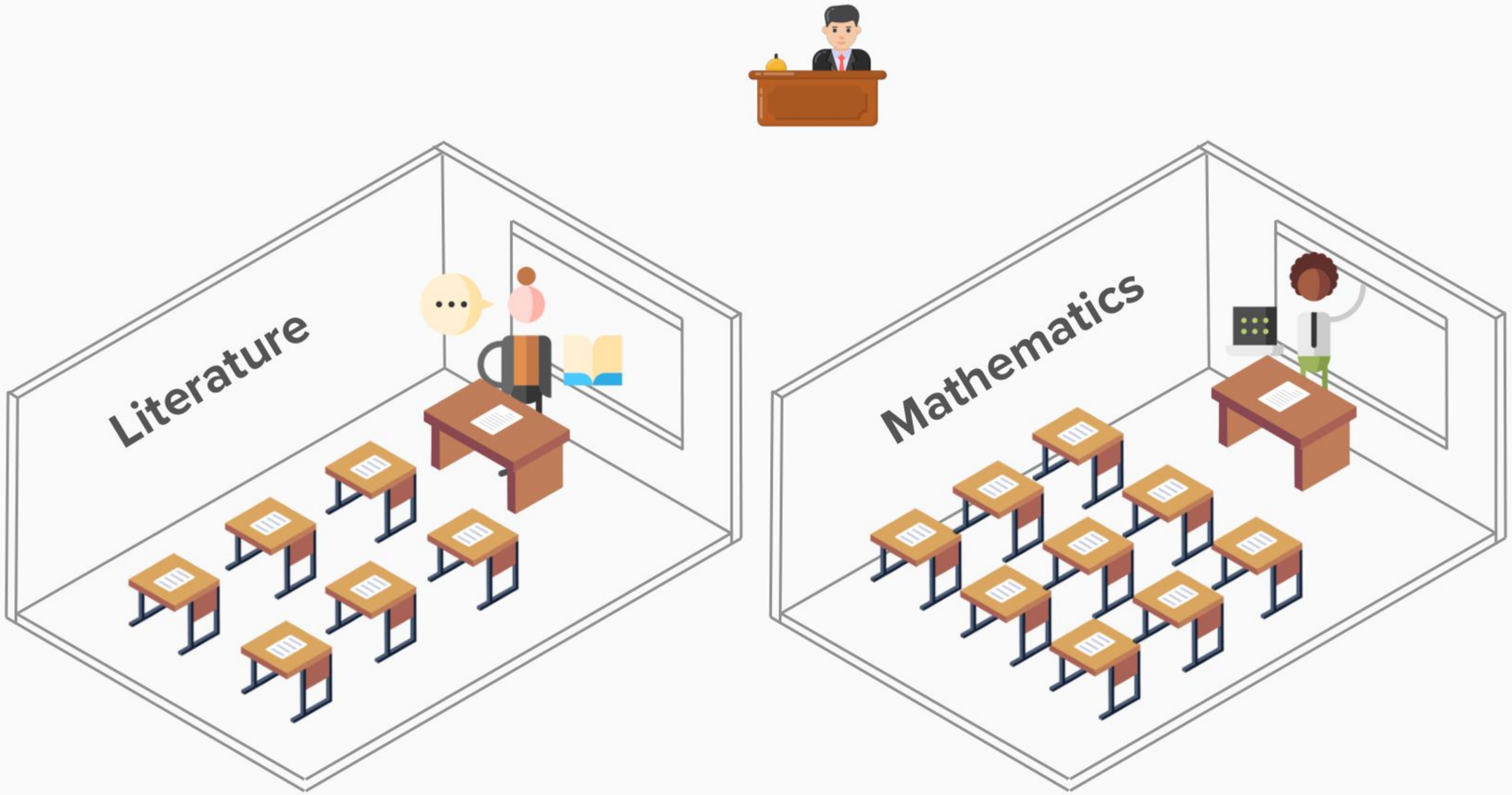
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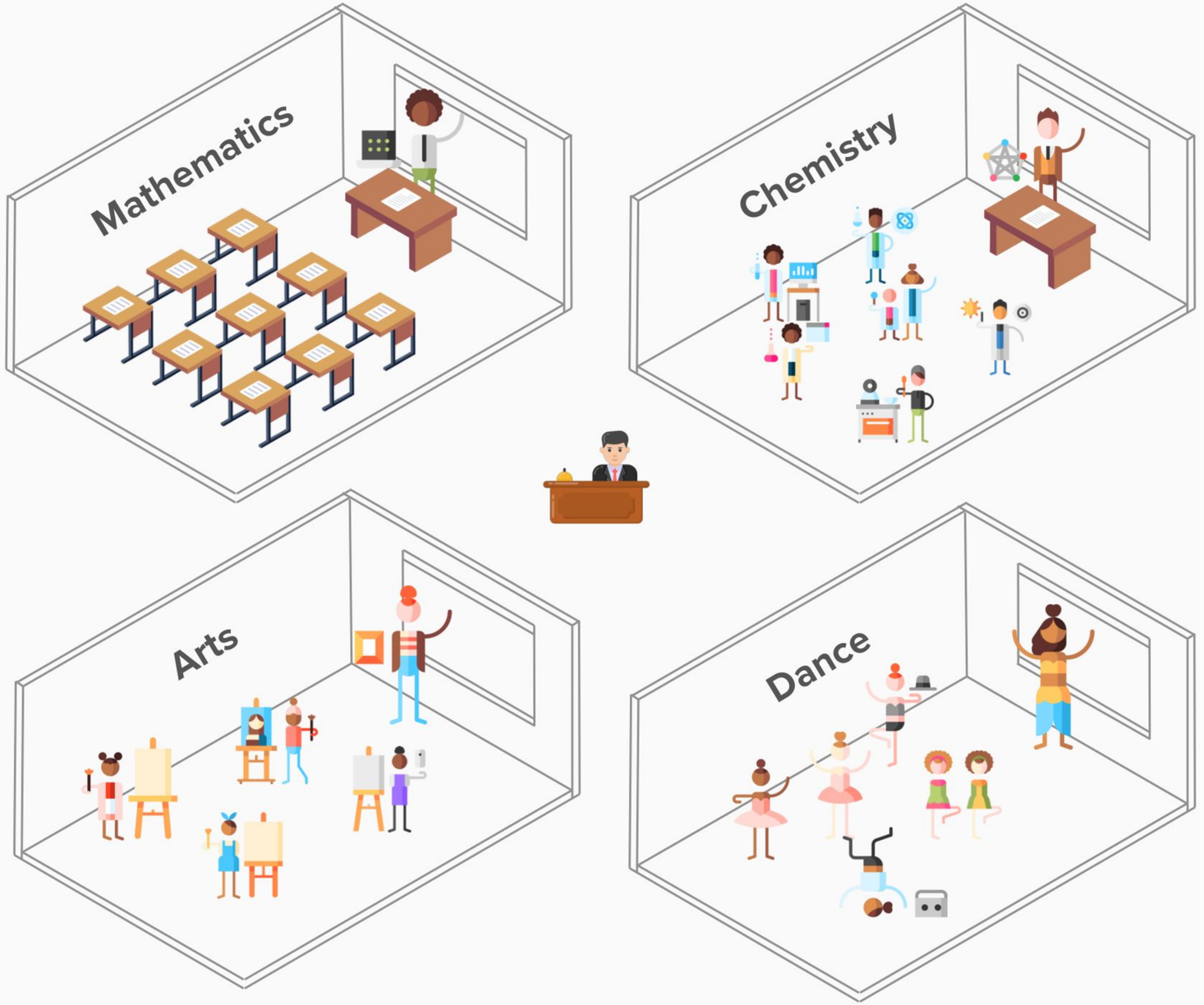
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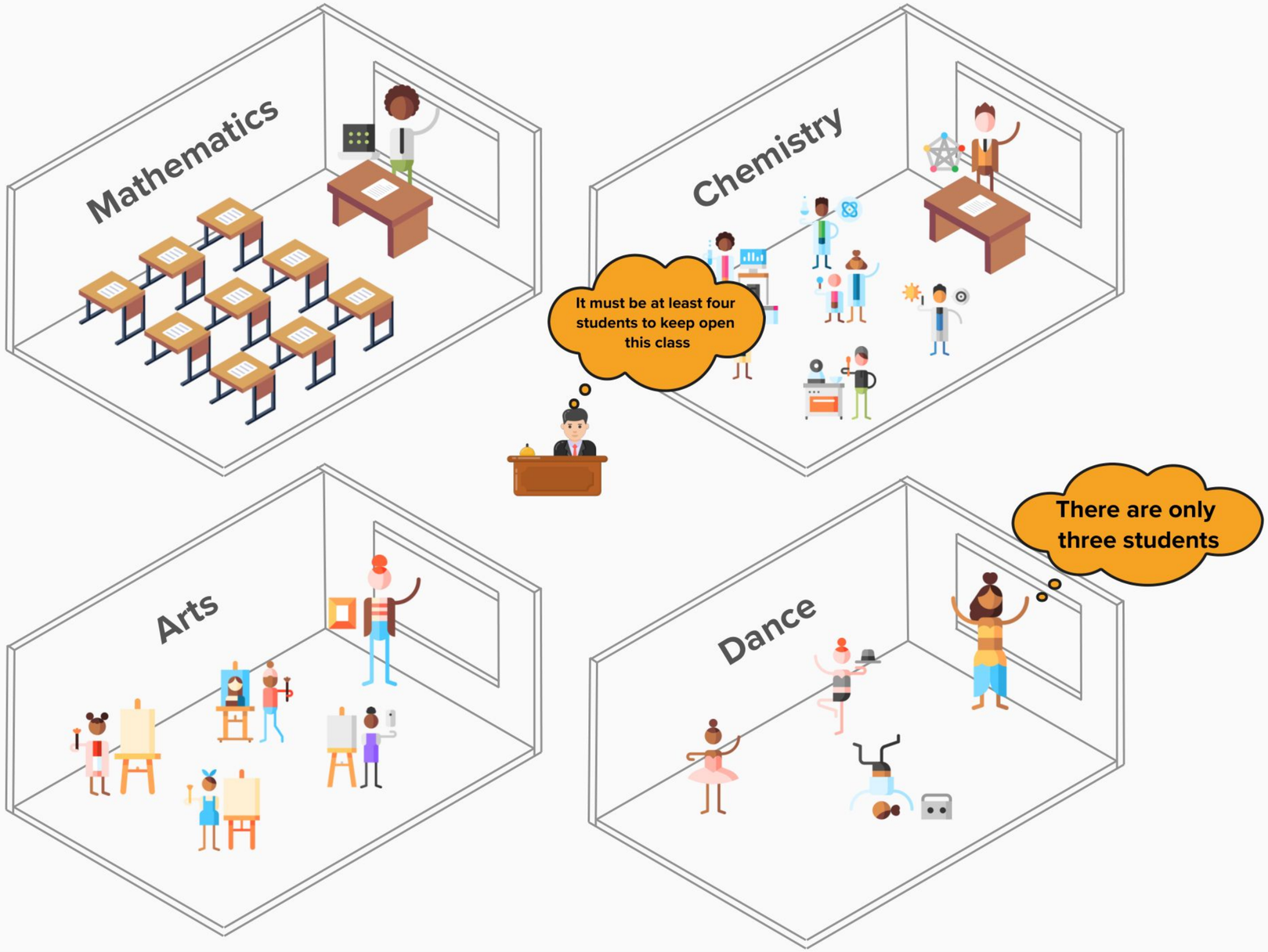
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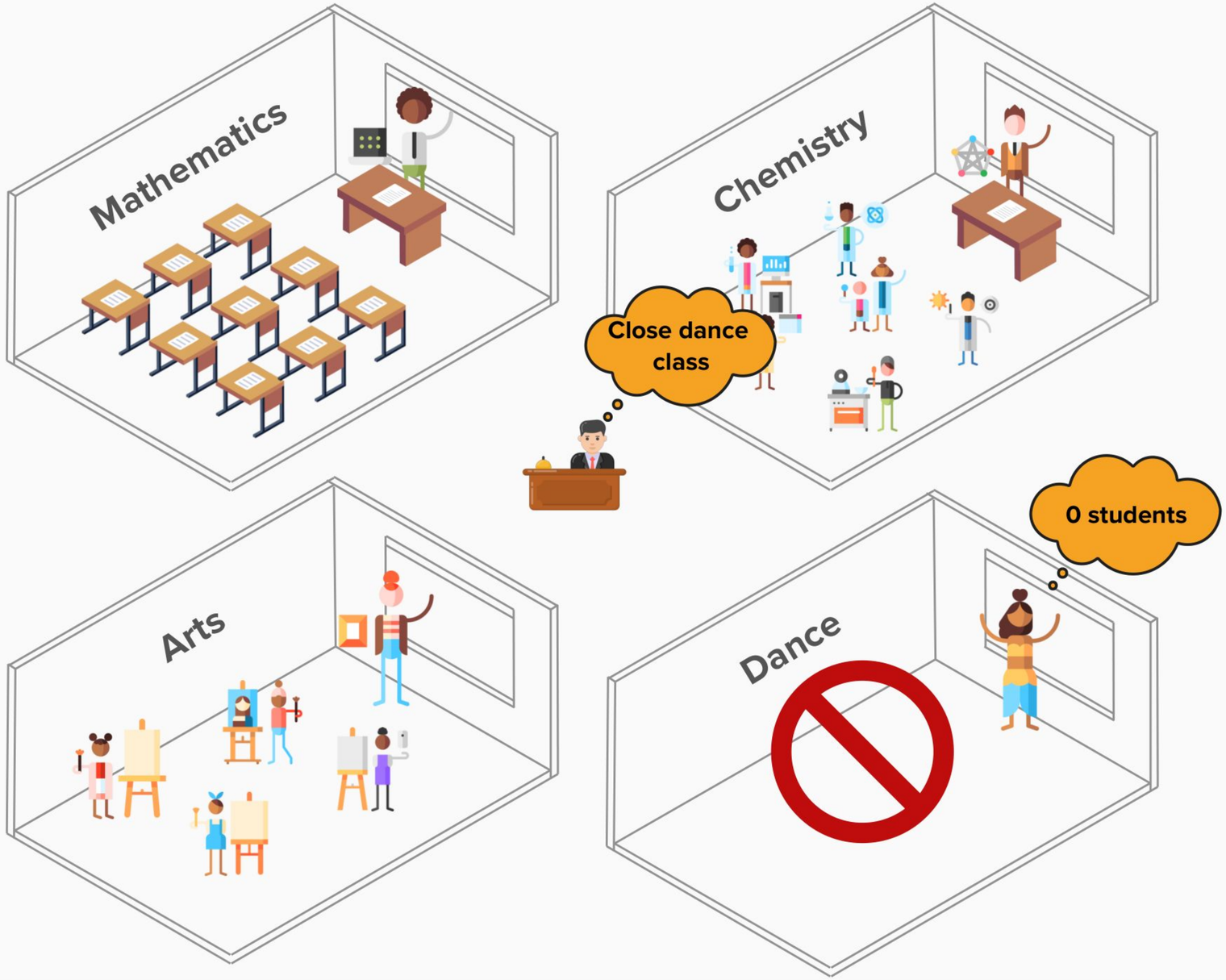
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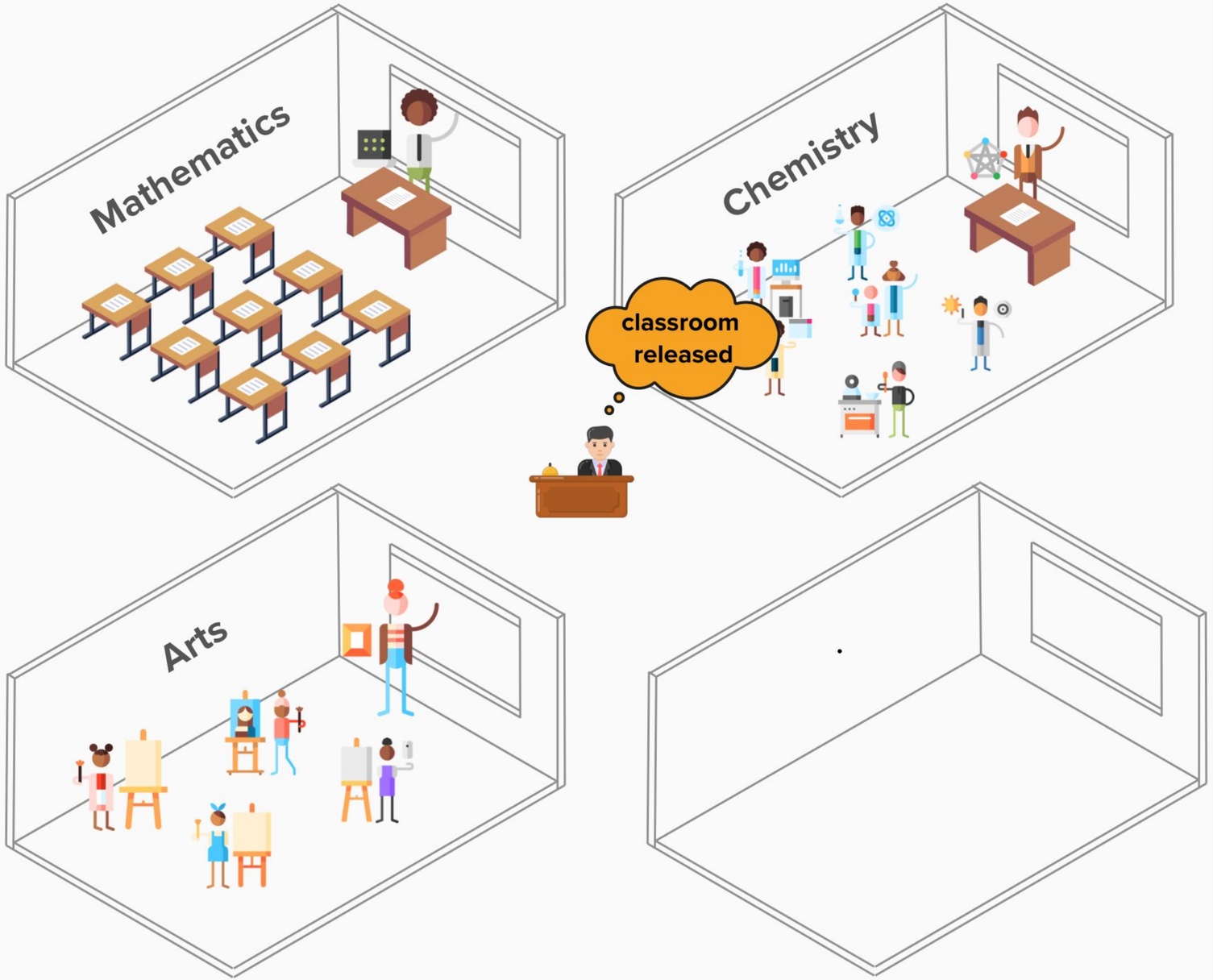
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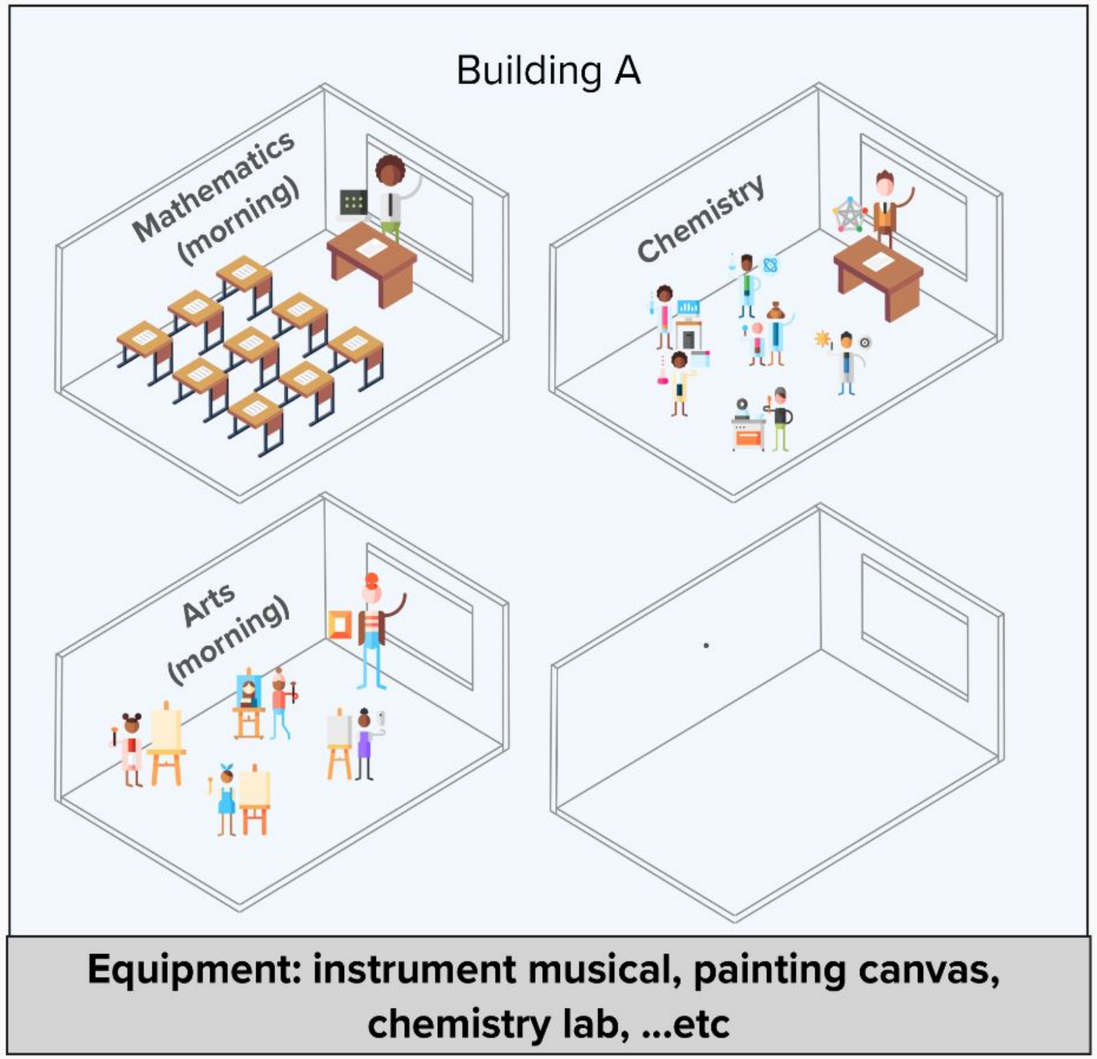
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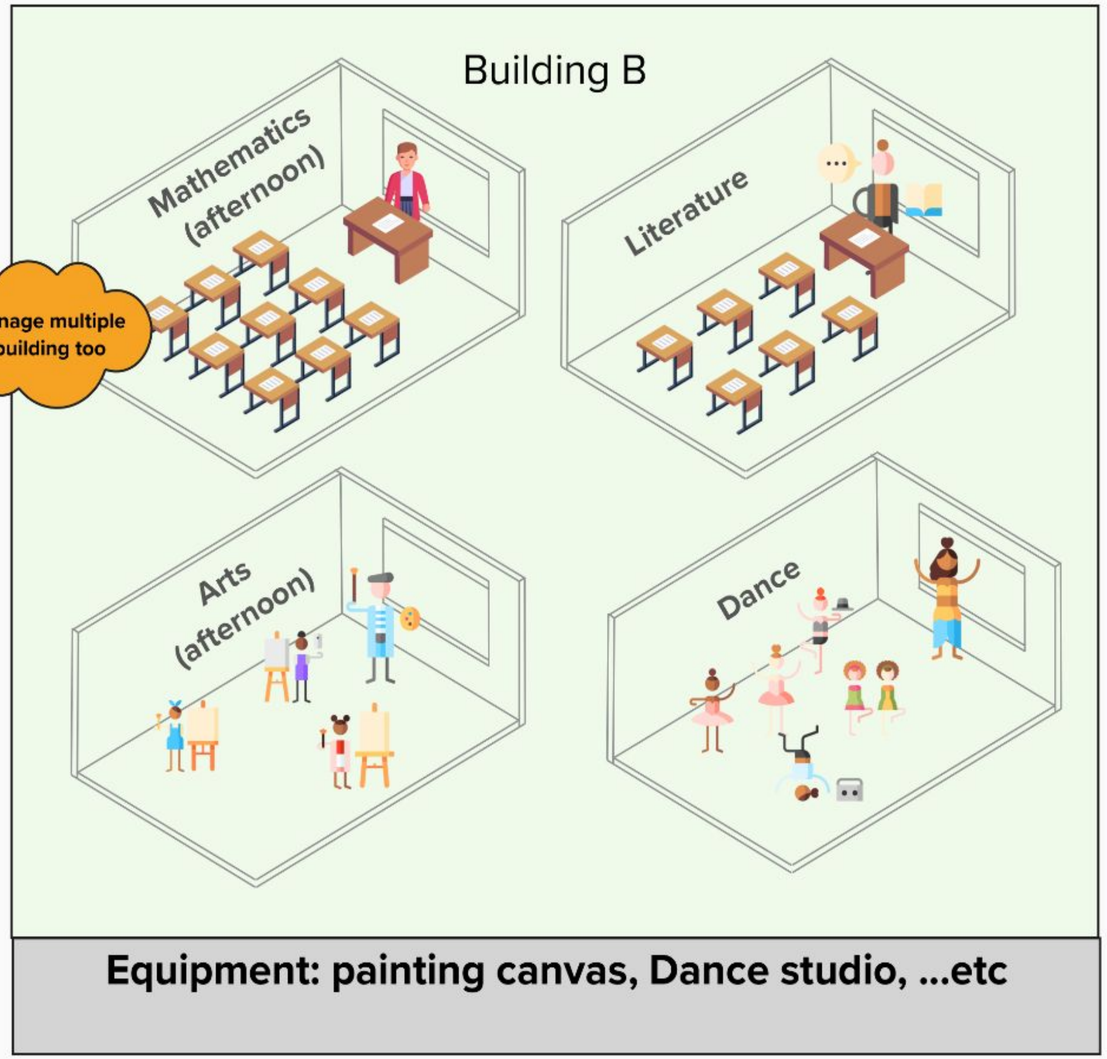
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Equipment: instrument musical, painting canvas, chemistry lab, ...etc

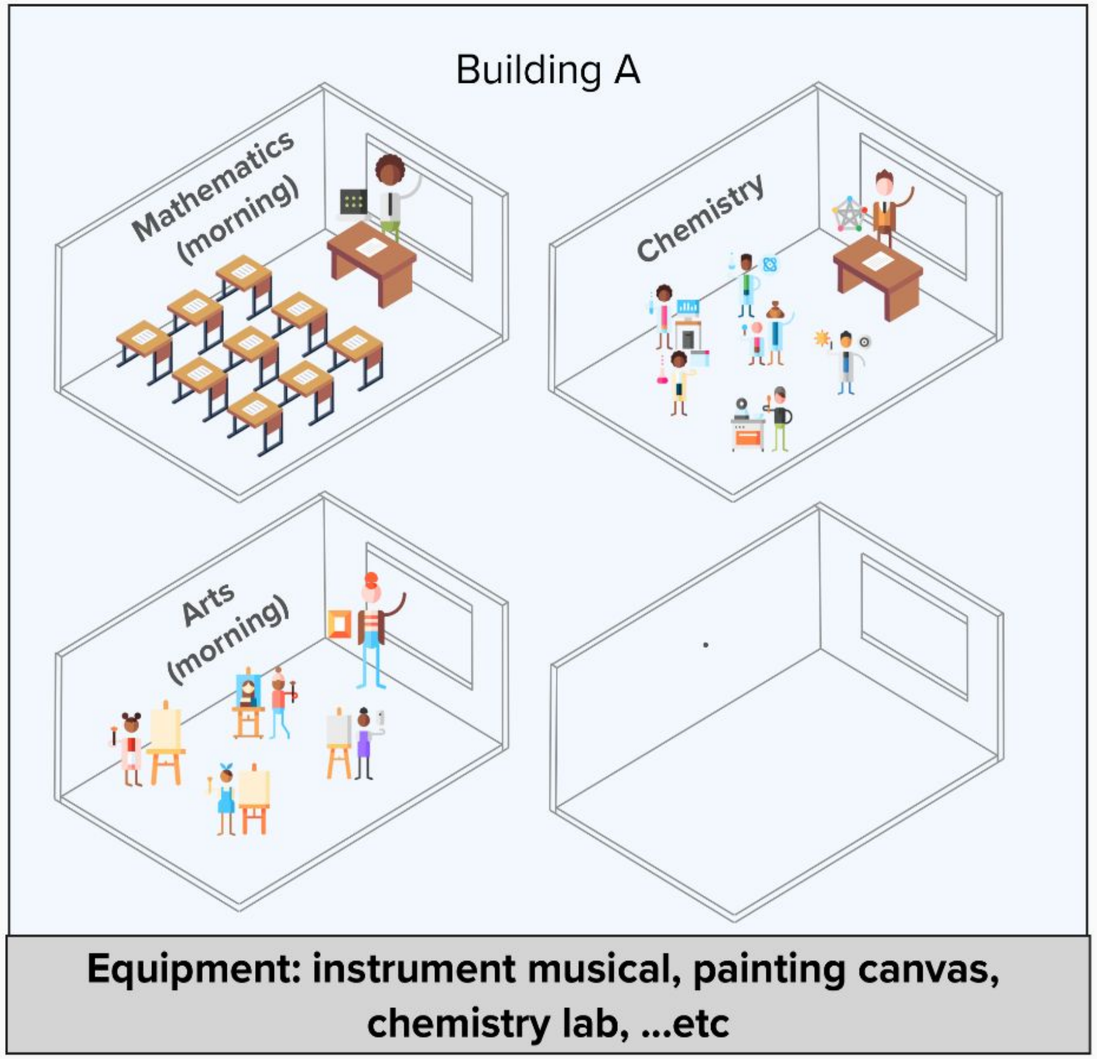




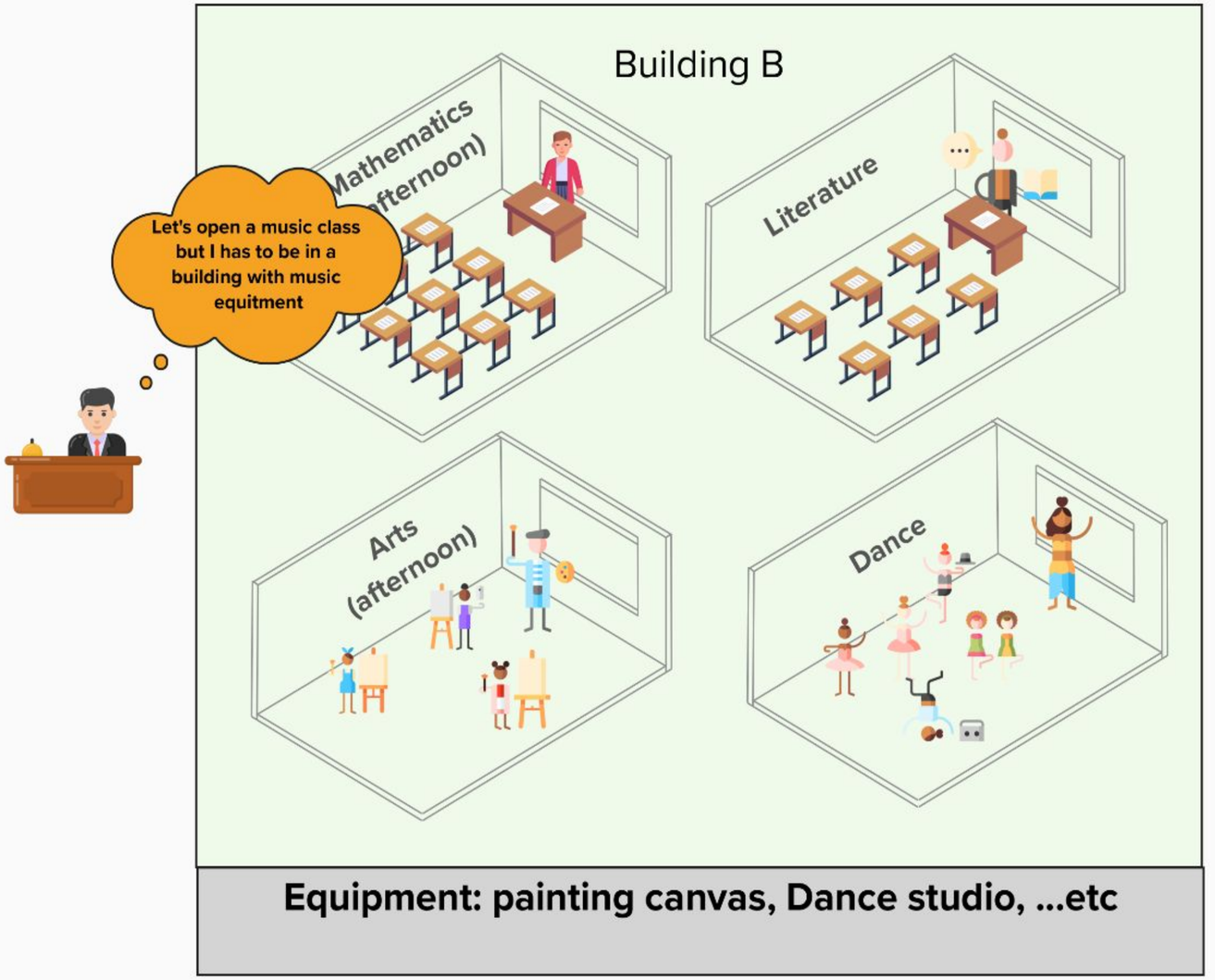
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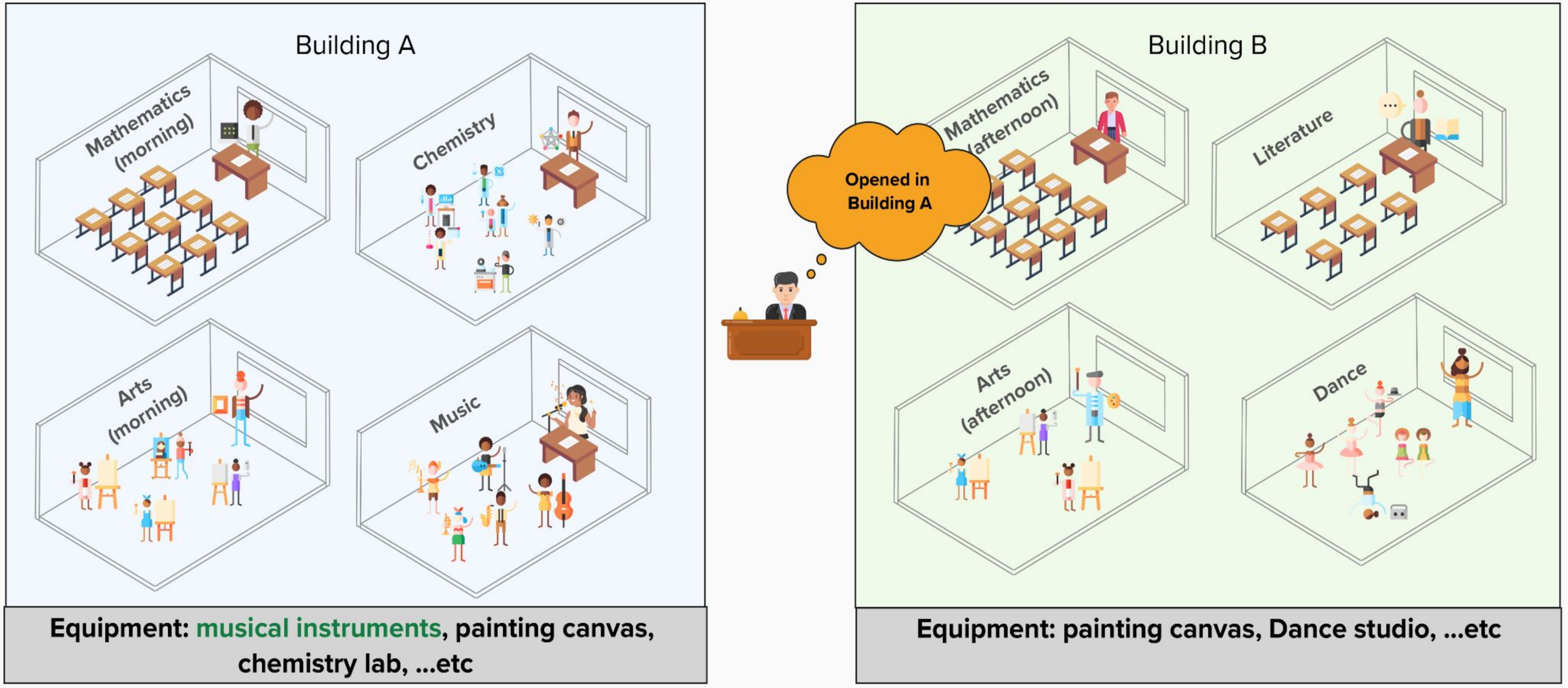
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Equipment: painting canvas, Dance studio, ...etc

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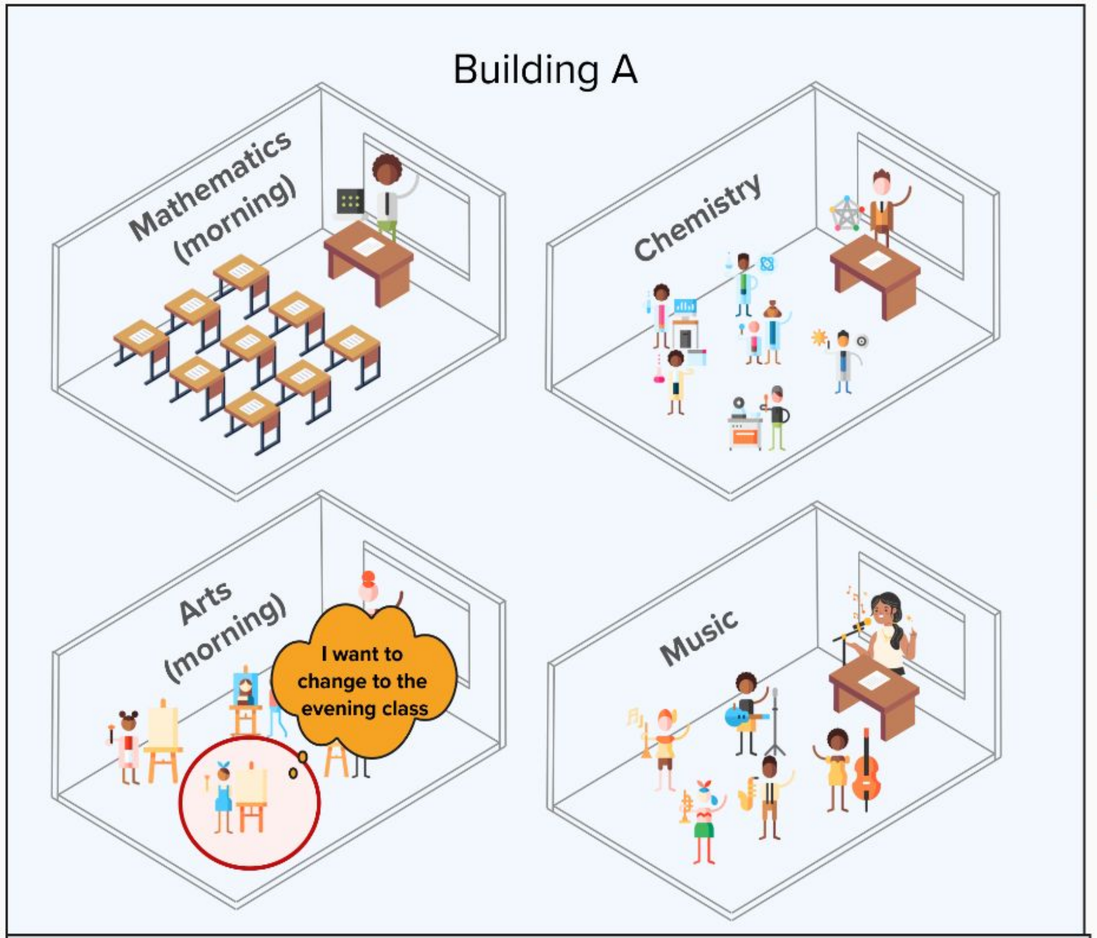


Equipment: musical instruments, painting canvas, chemistry lab, ...etc

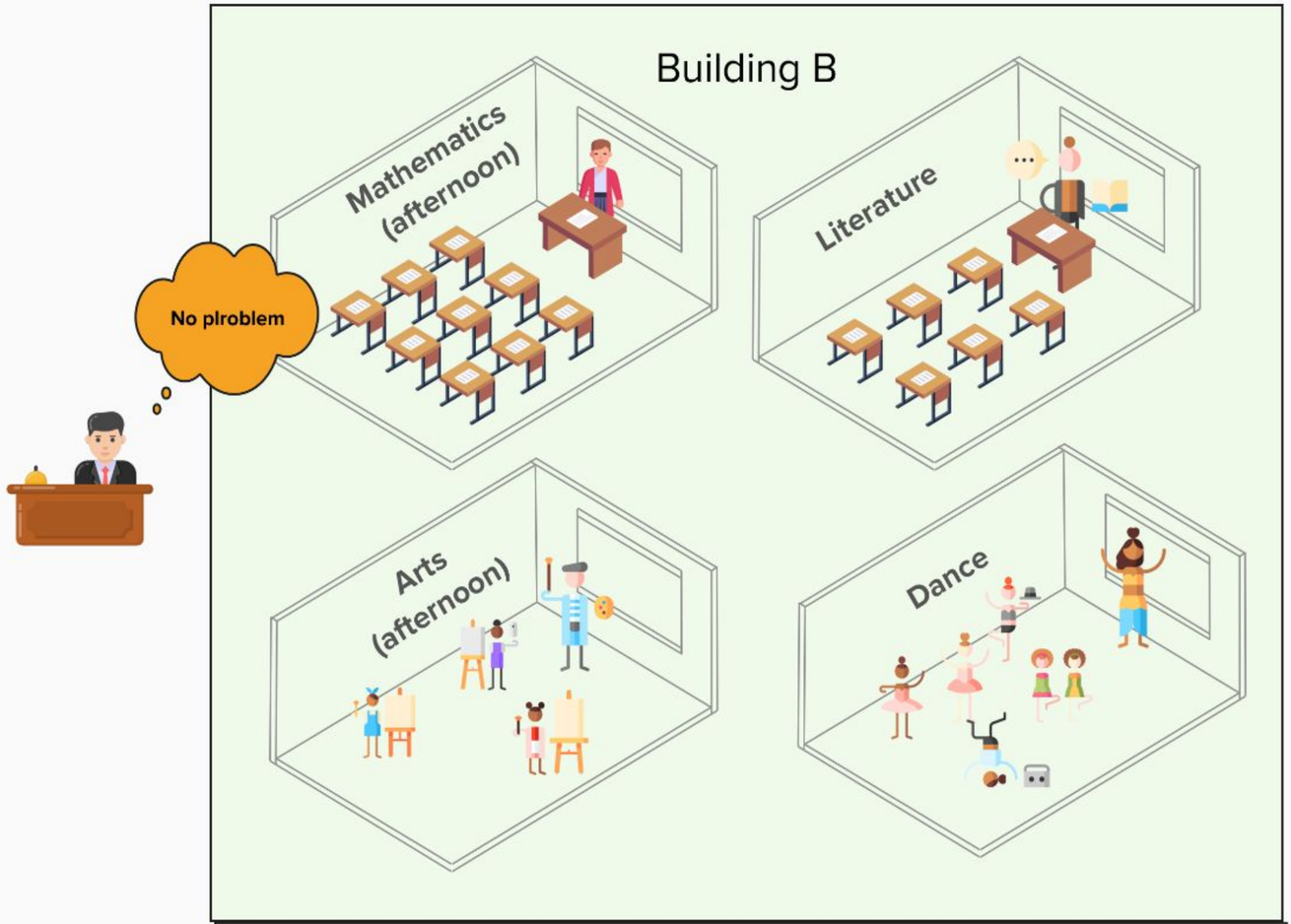
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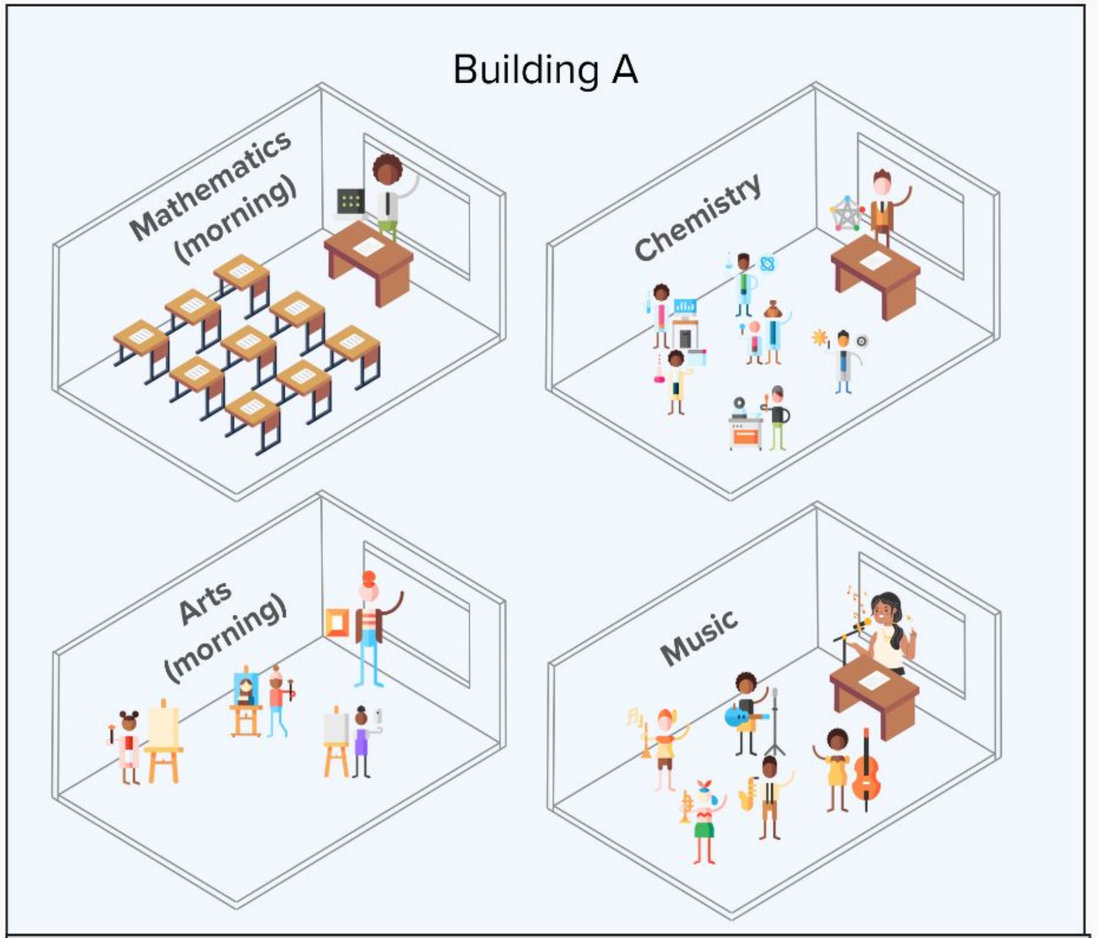
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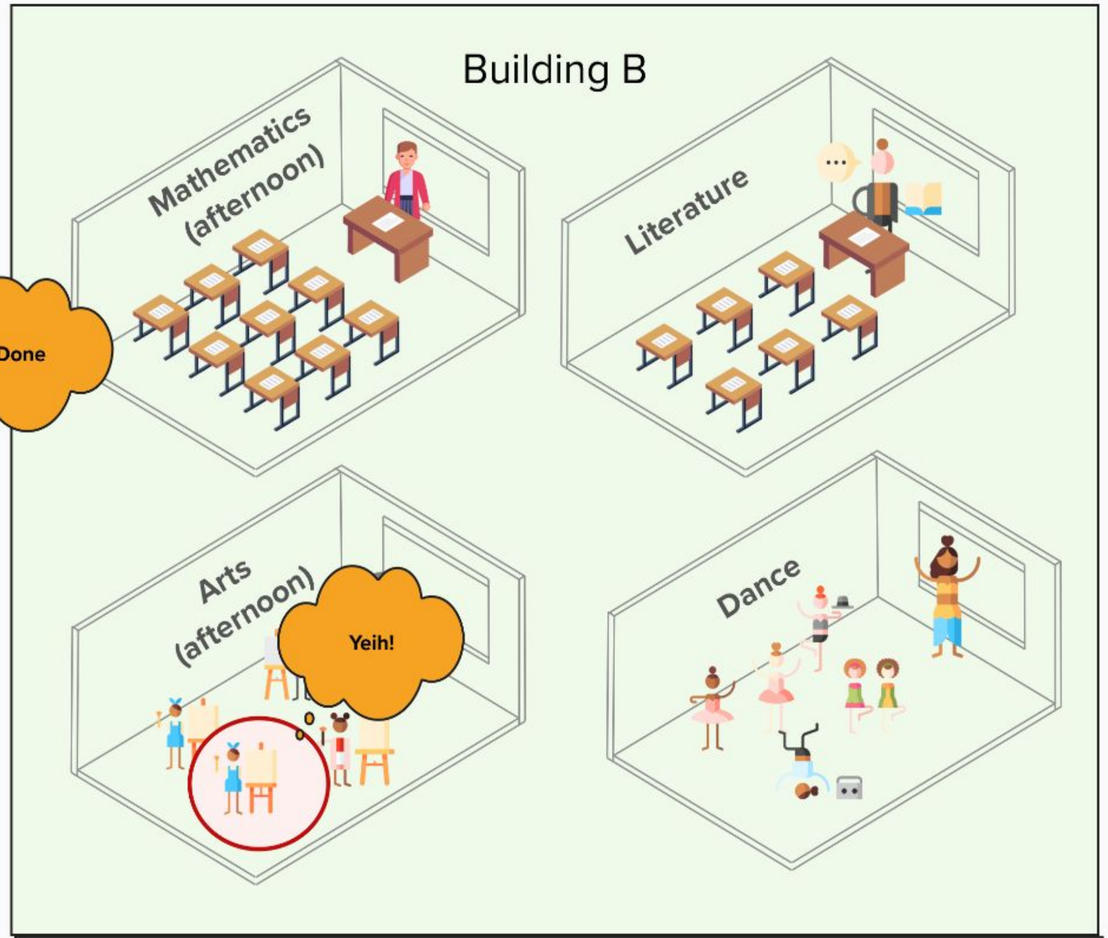
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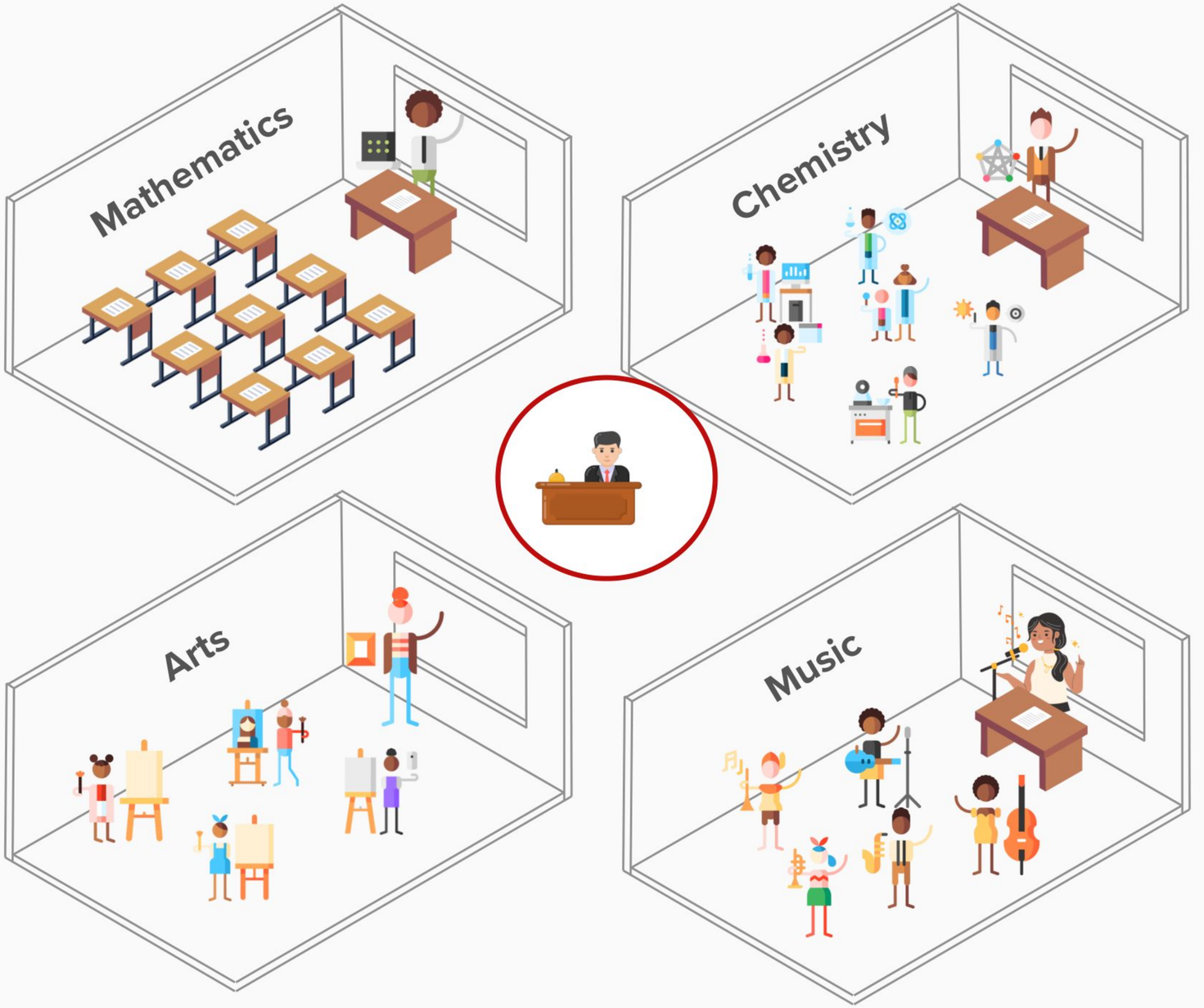
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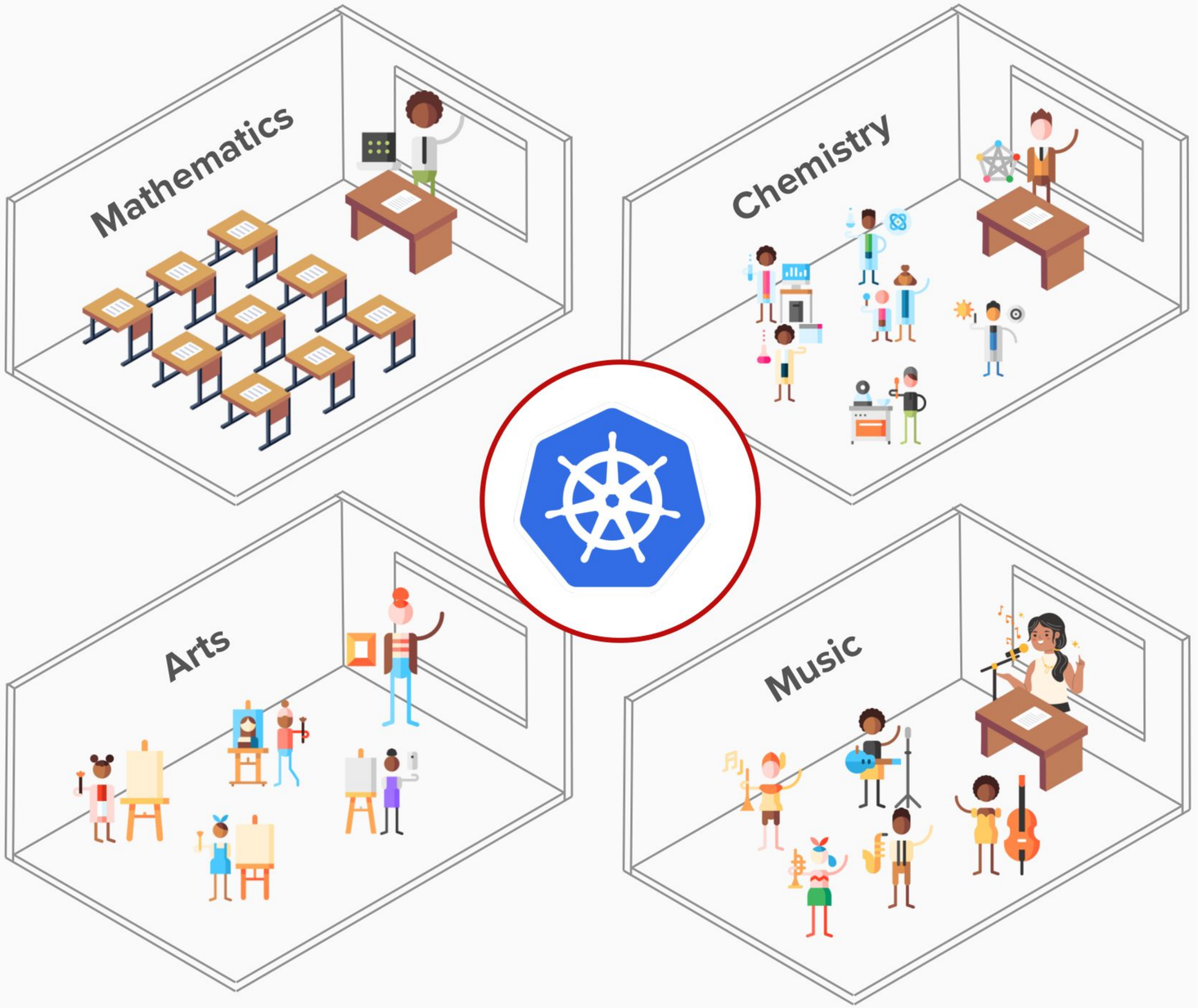
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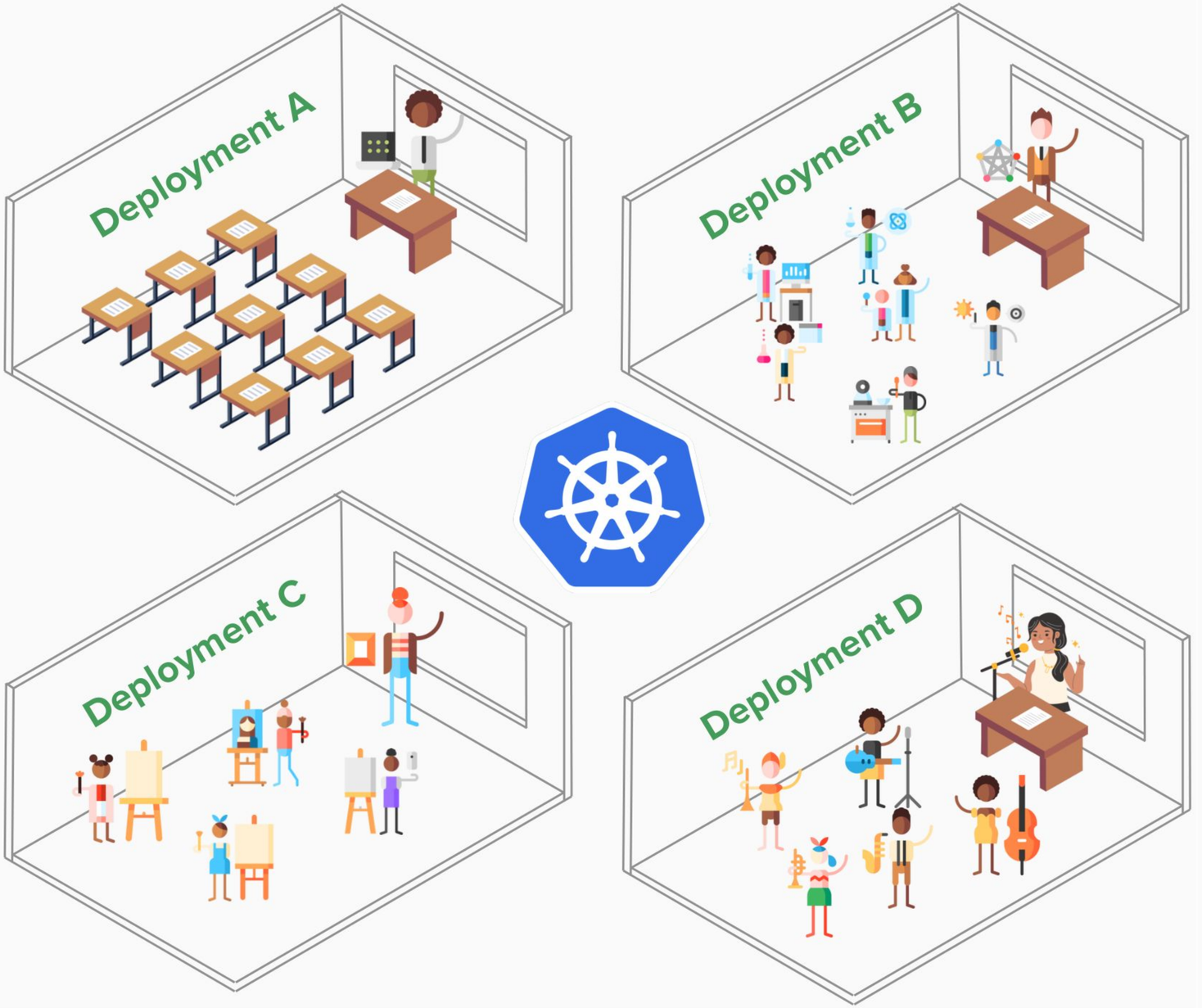
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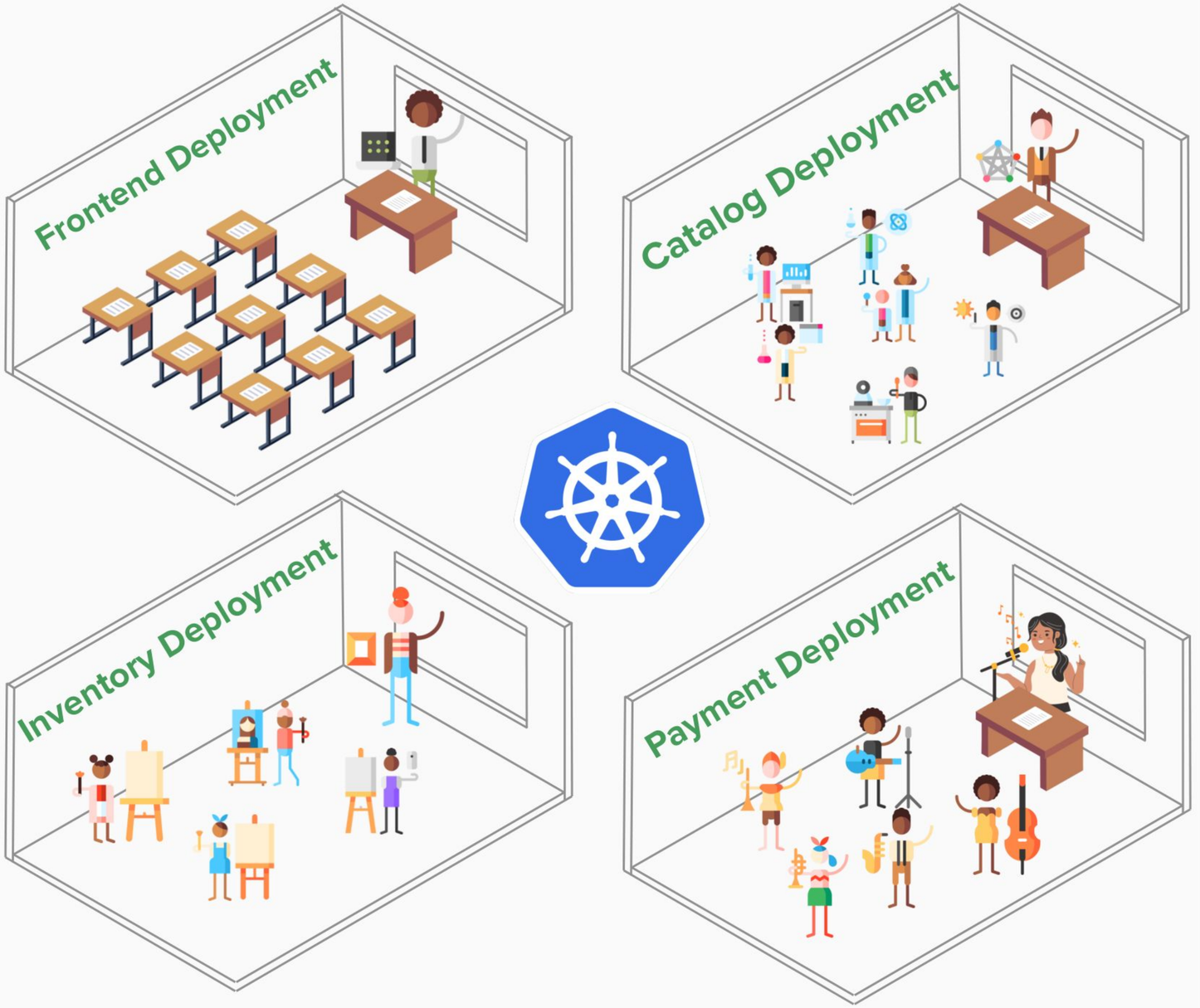
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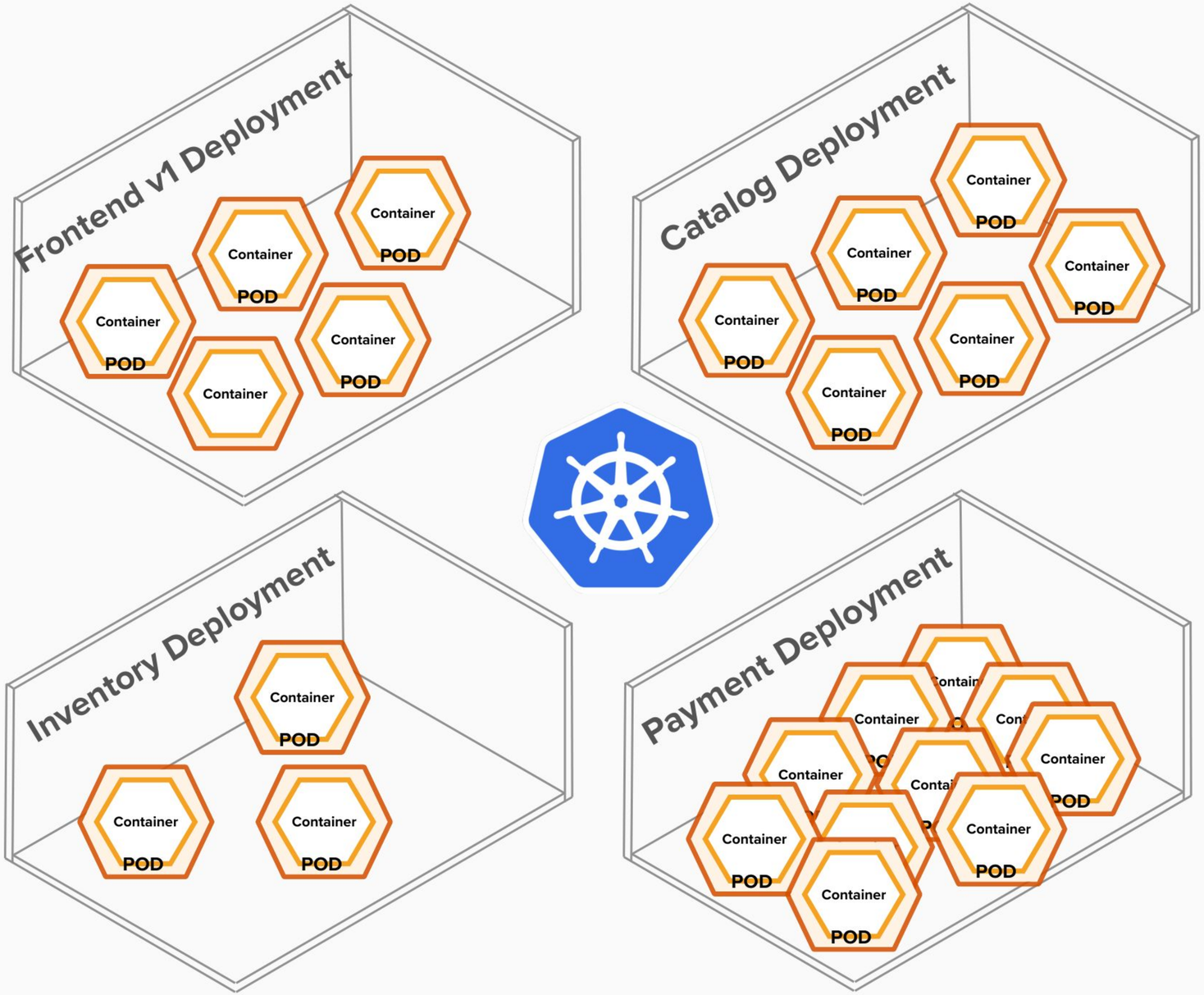


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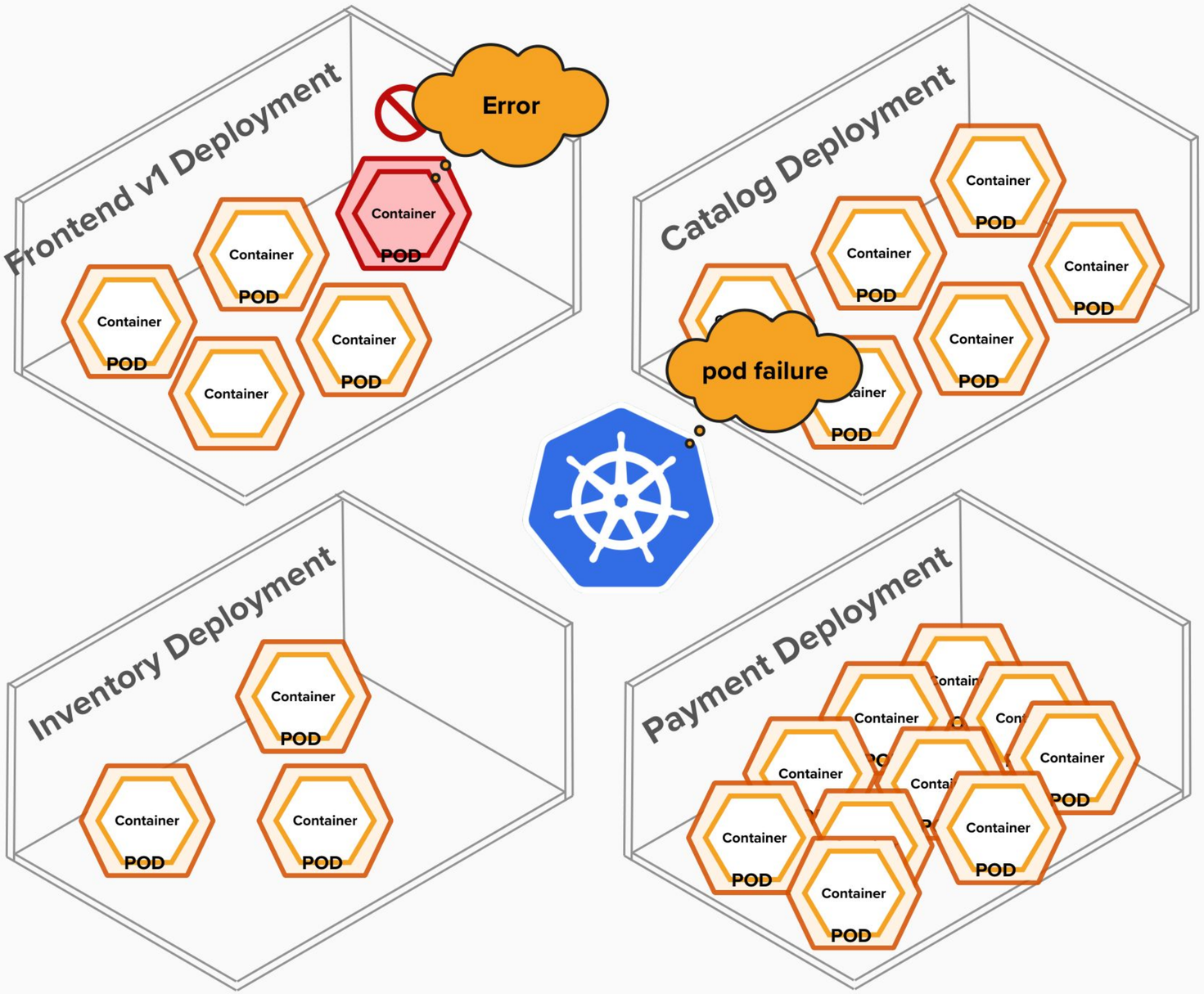
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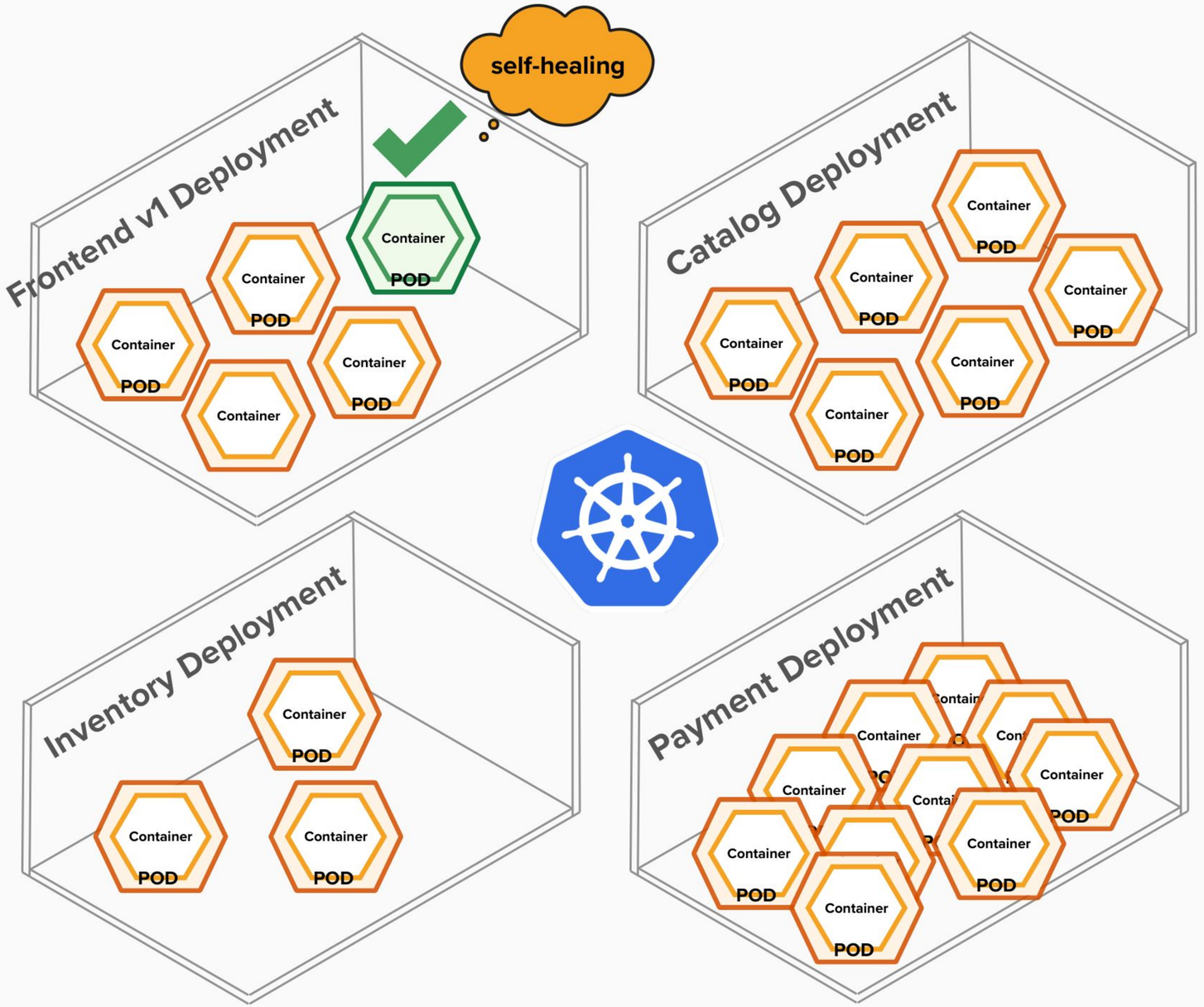
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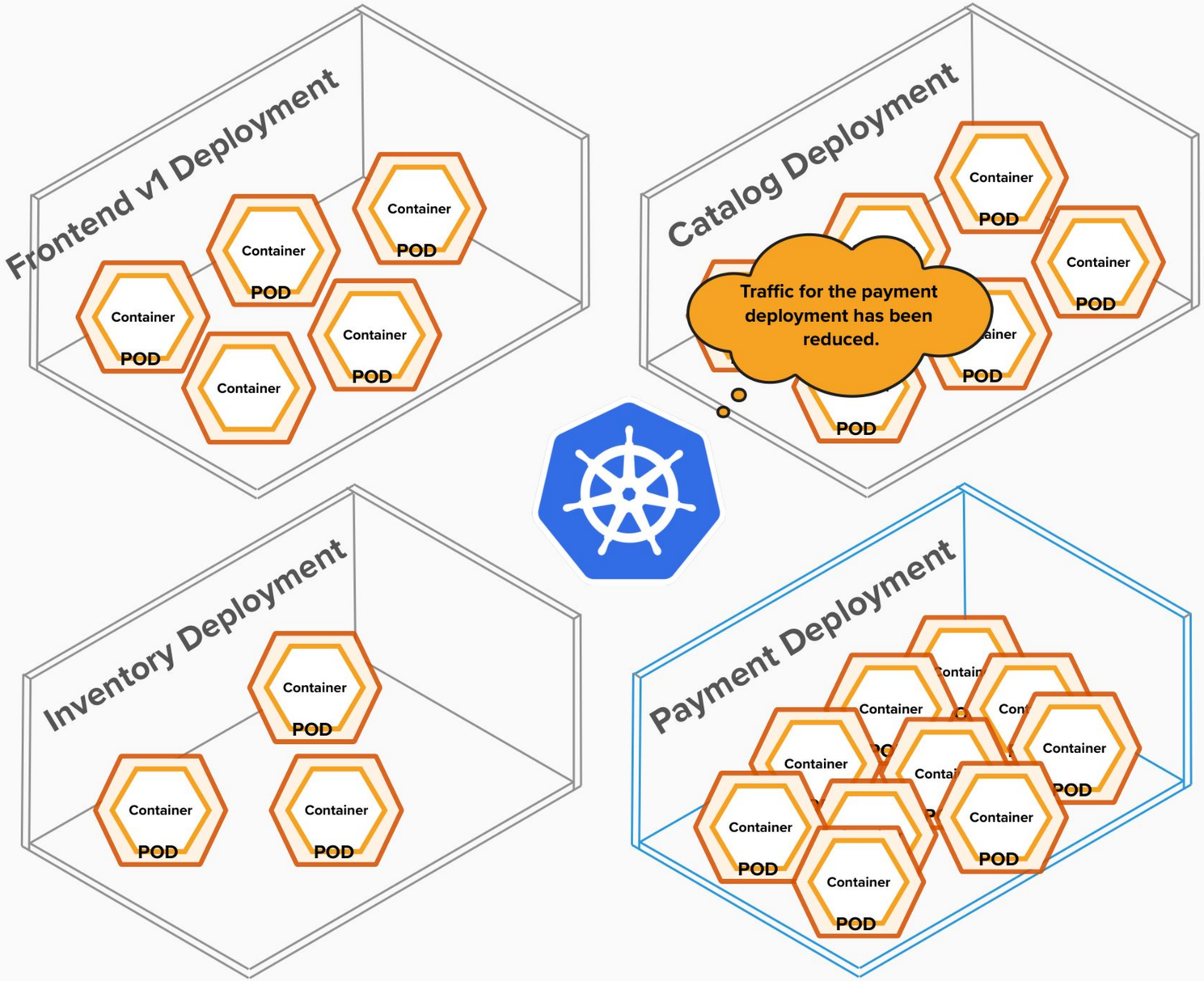


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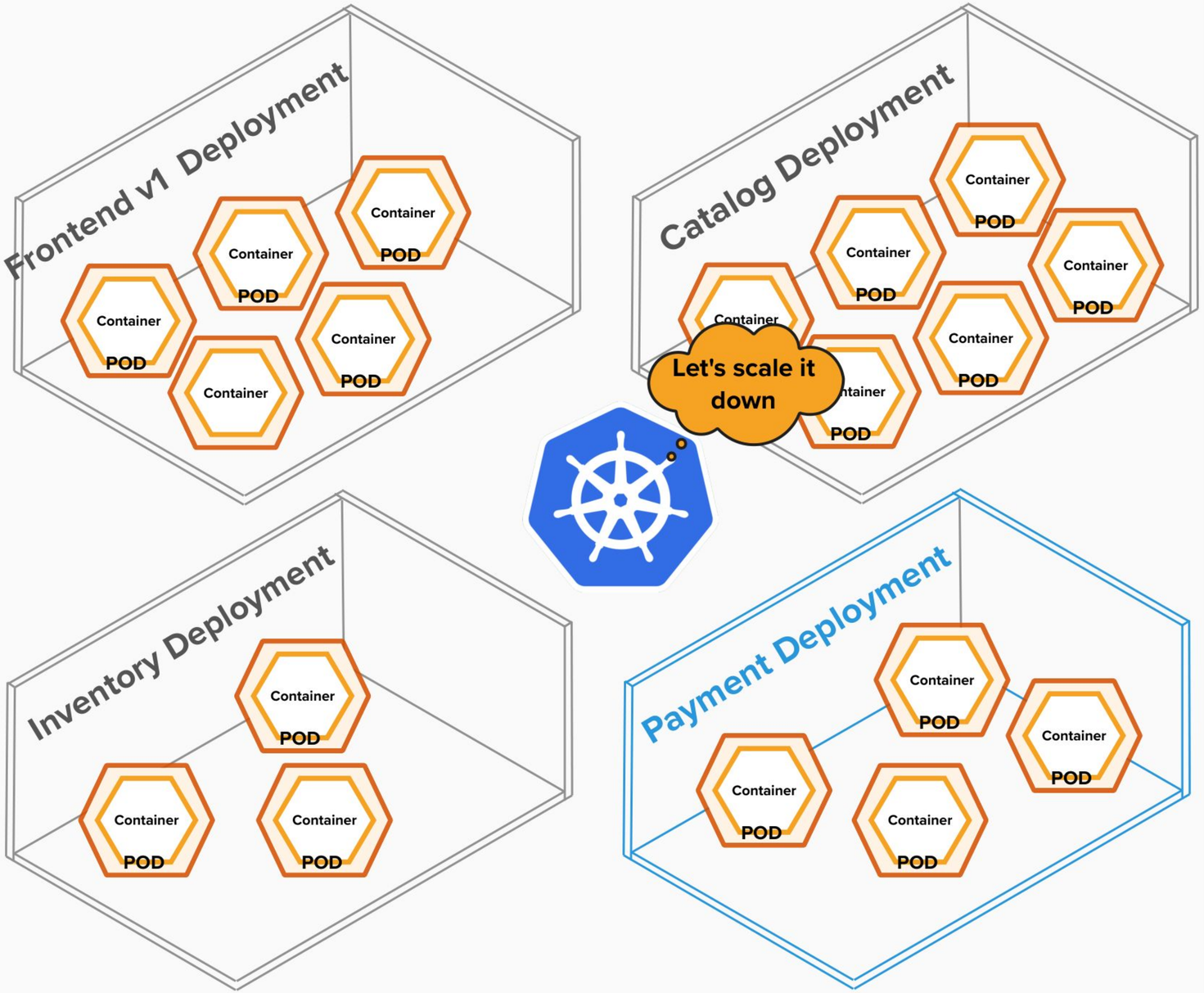


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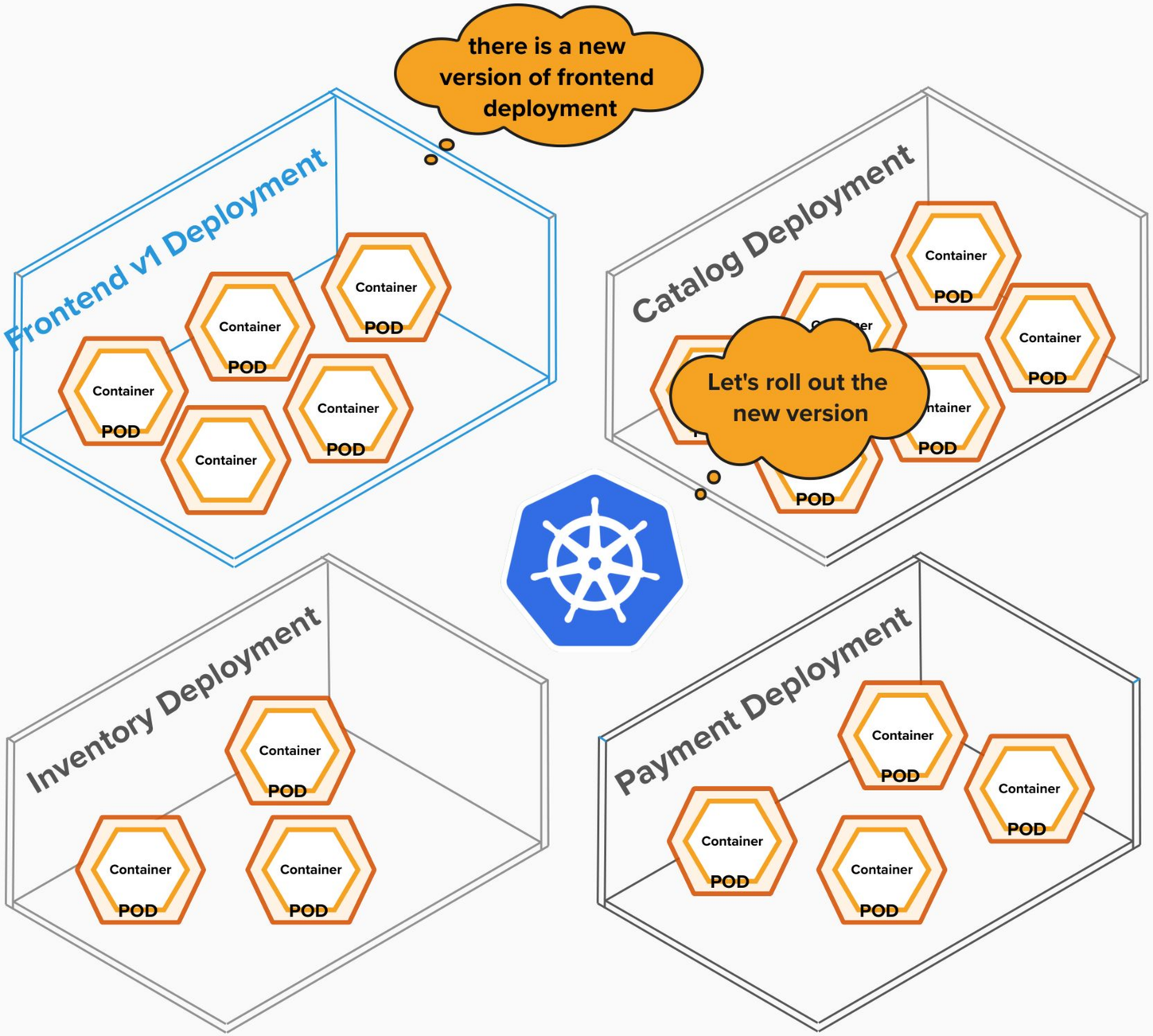
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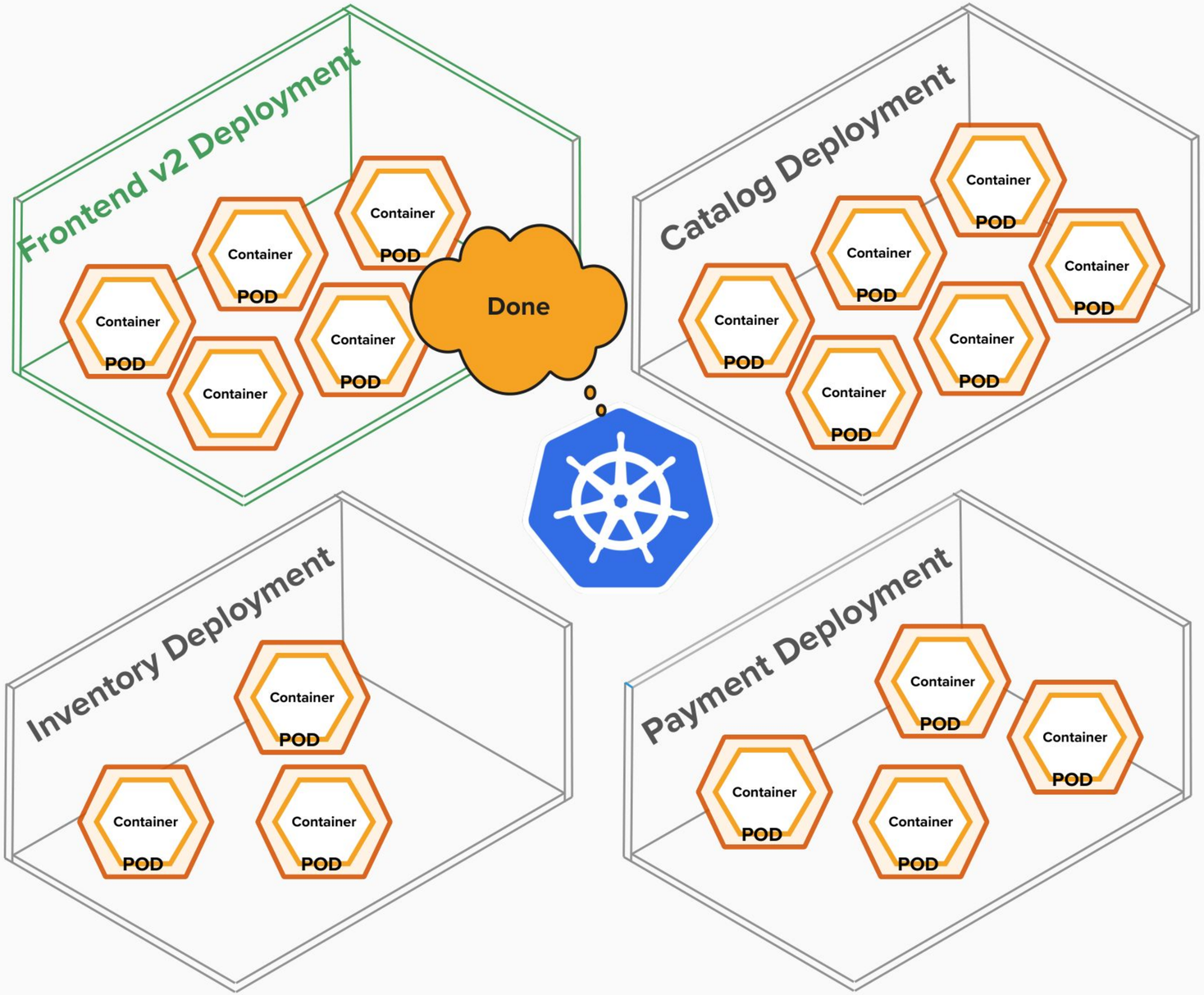


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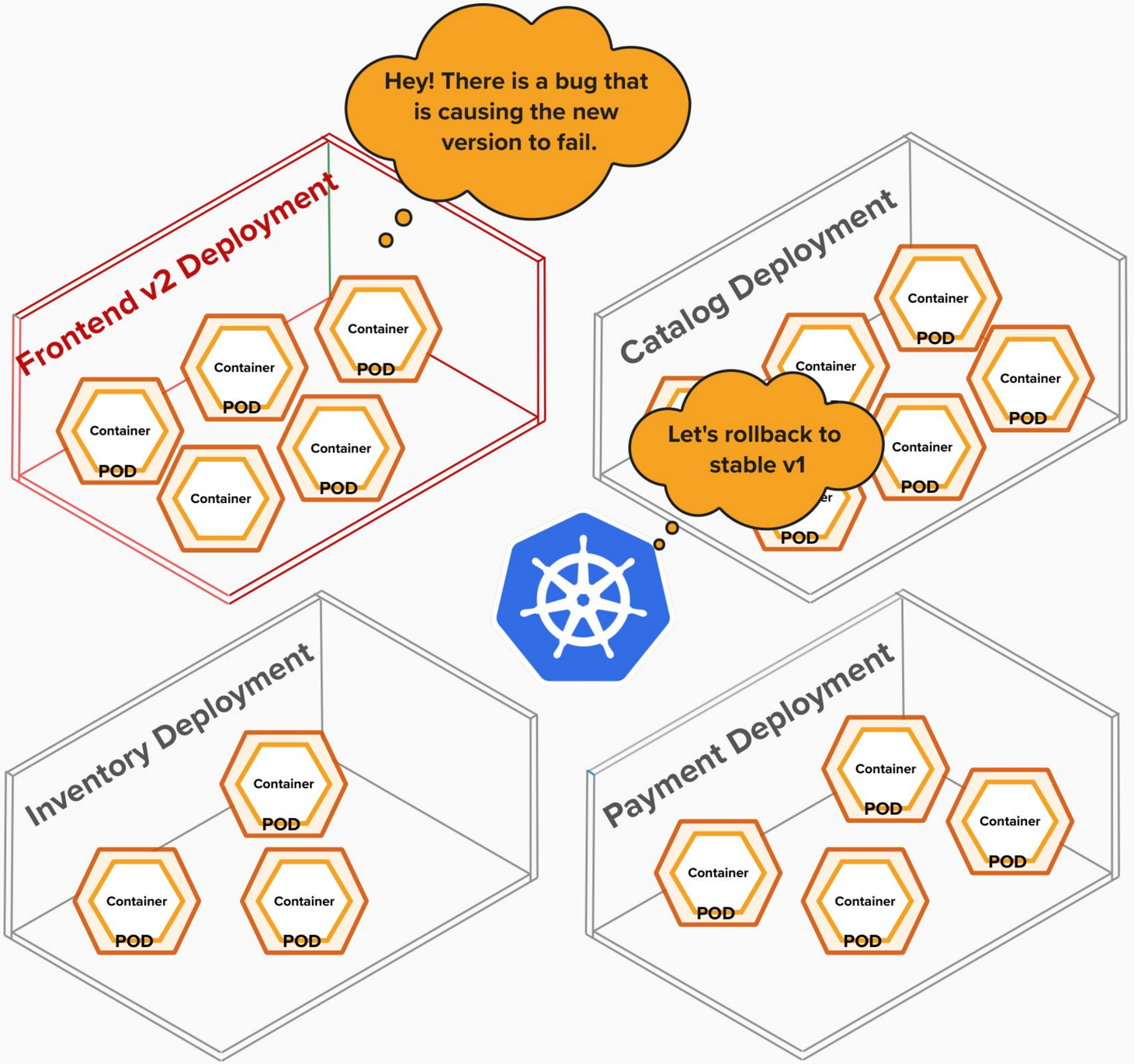


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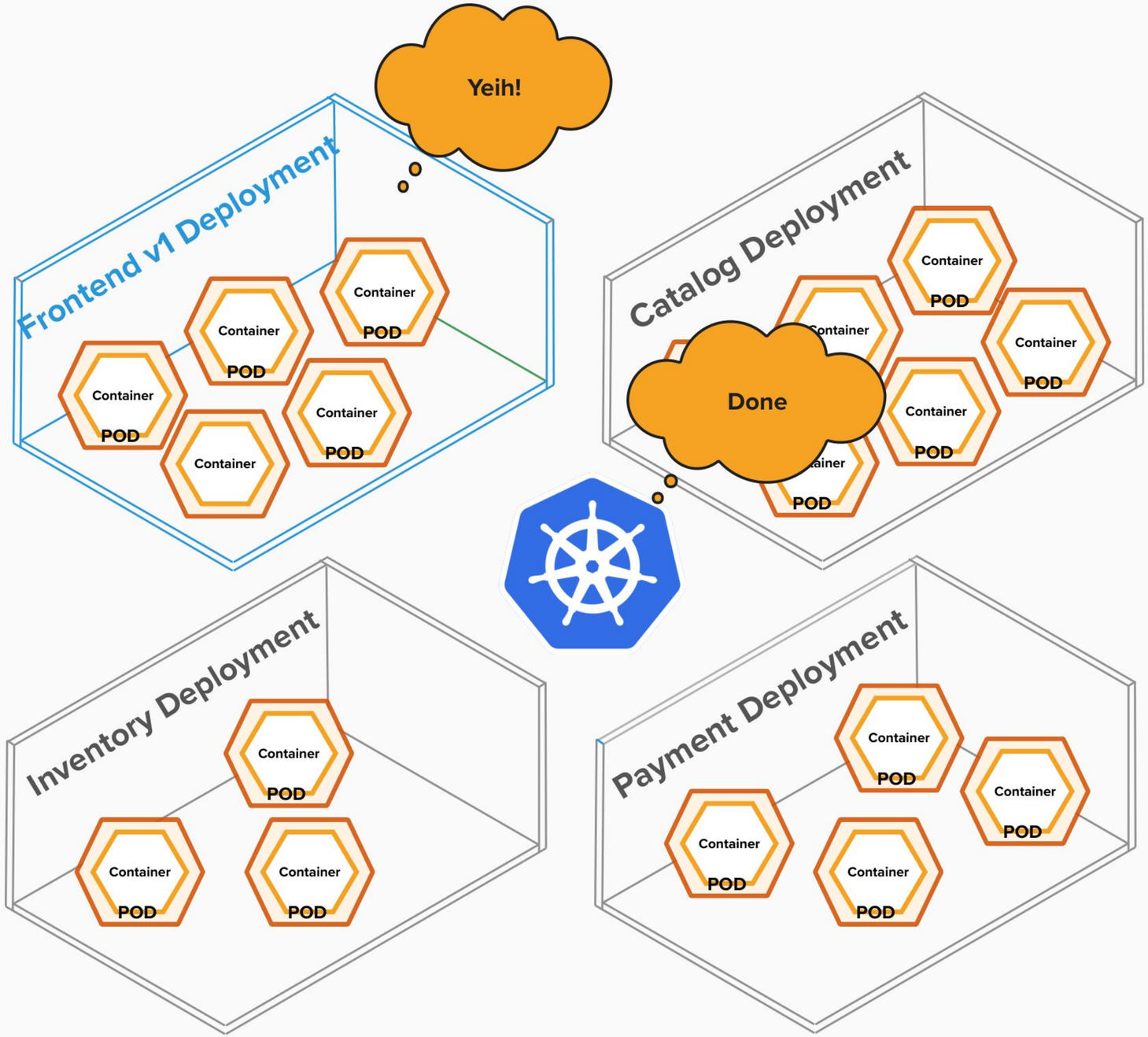


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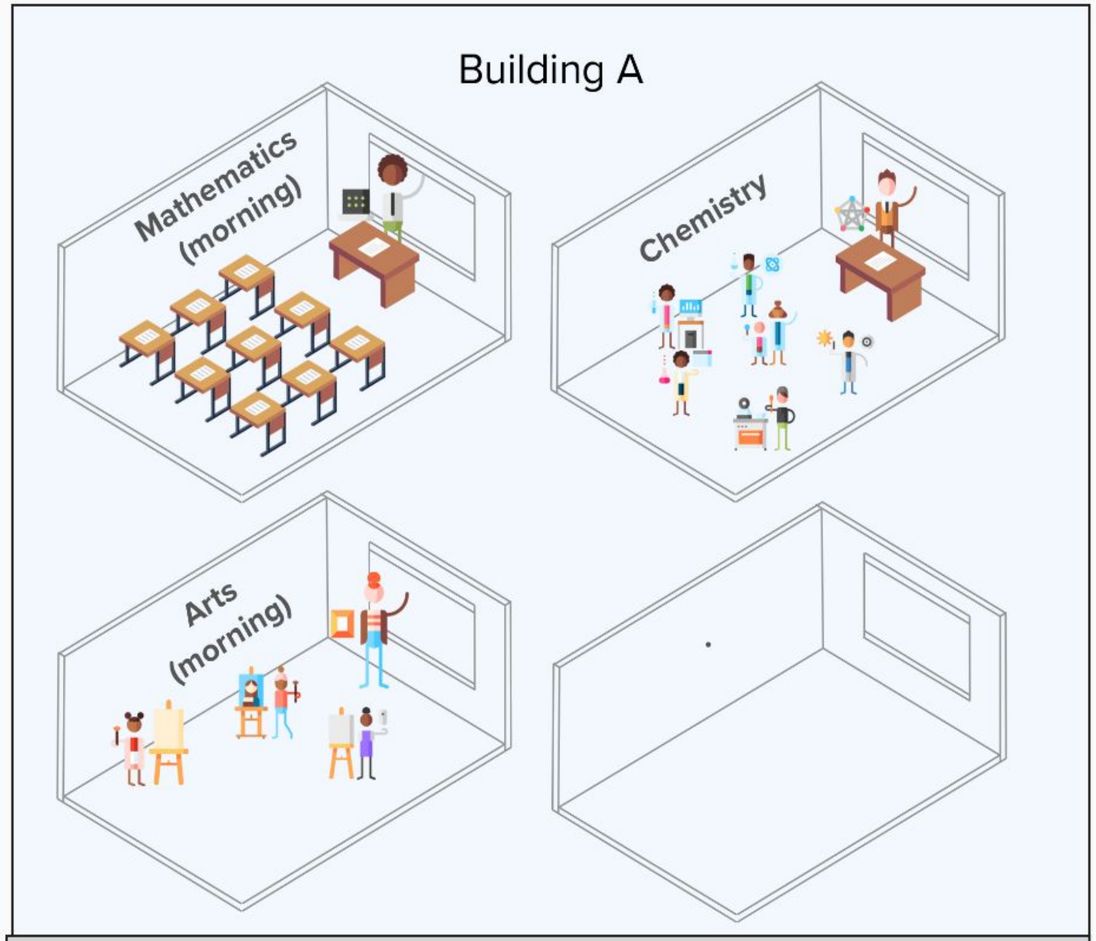


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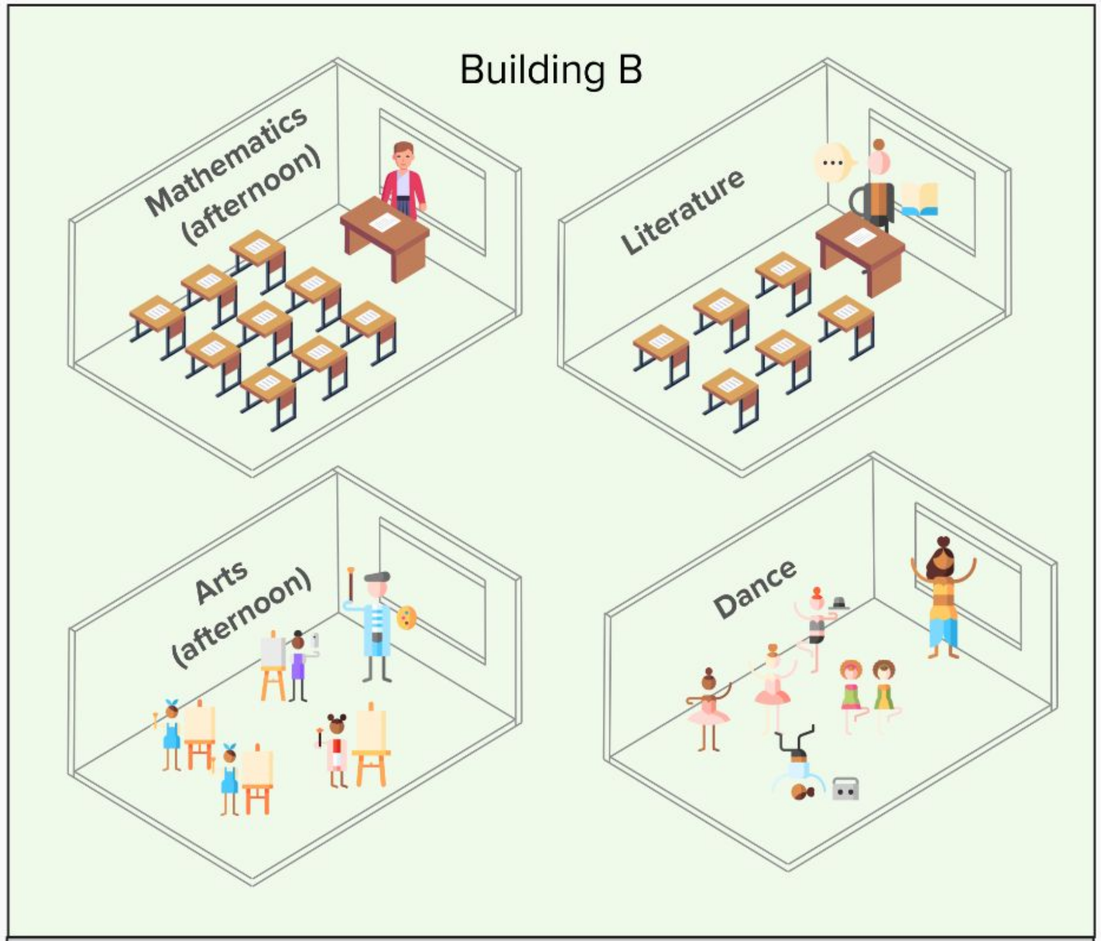


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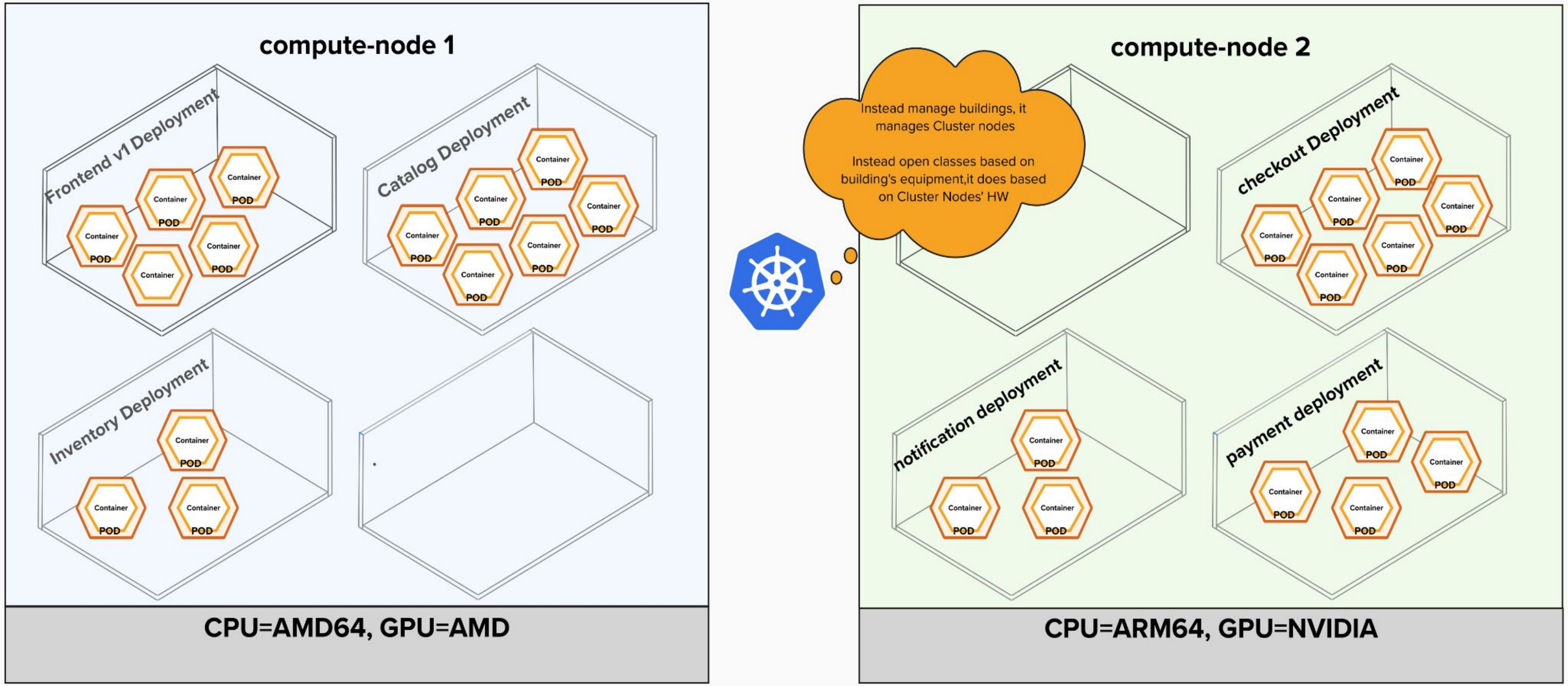




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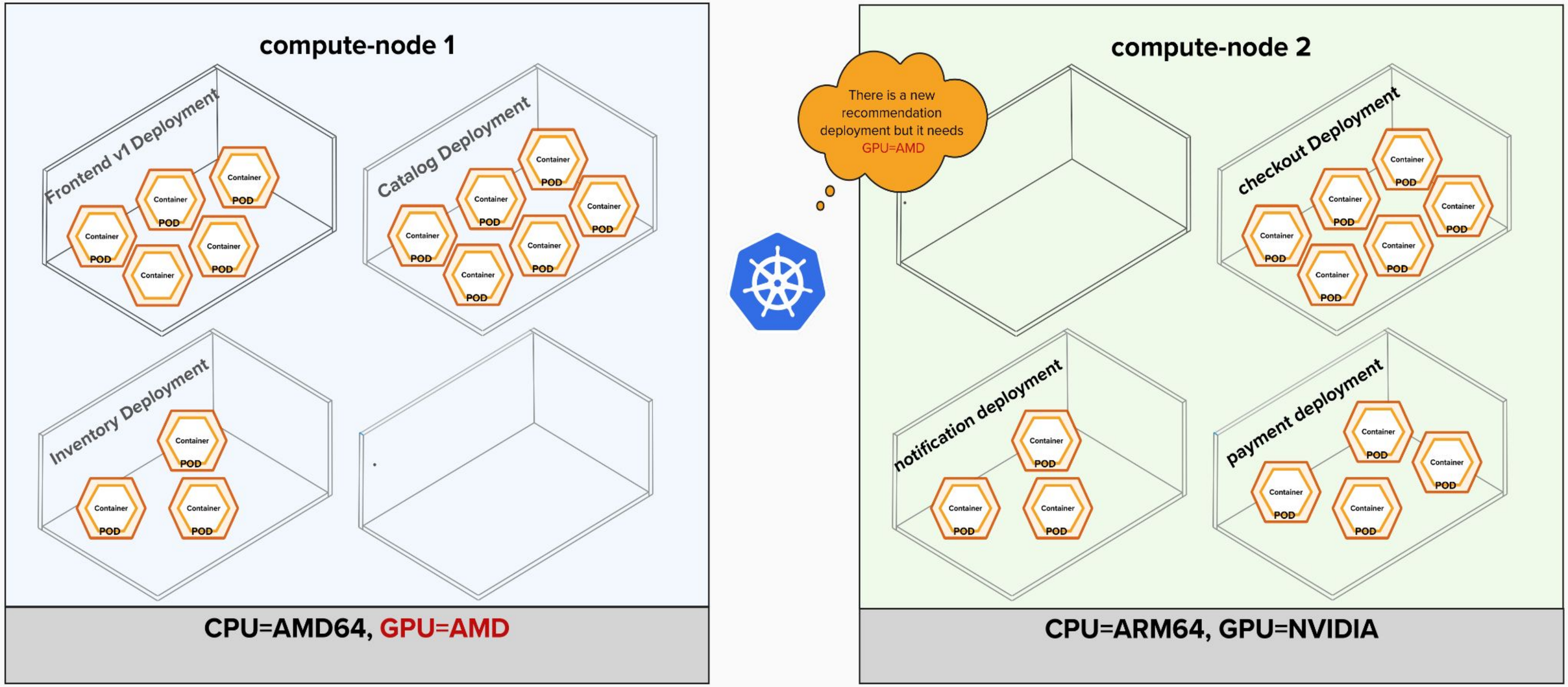
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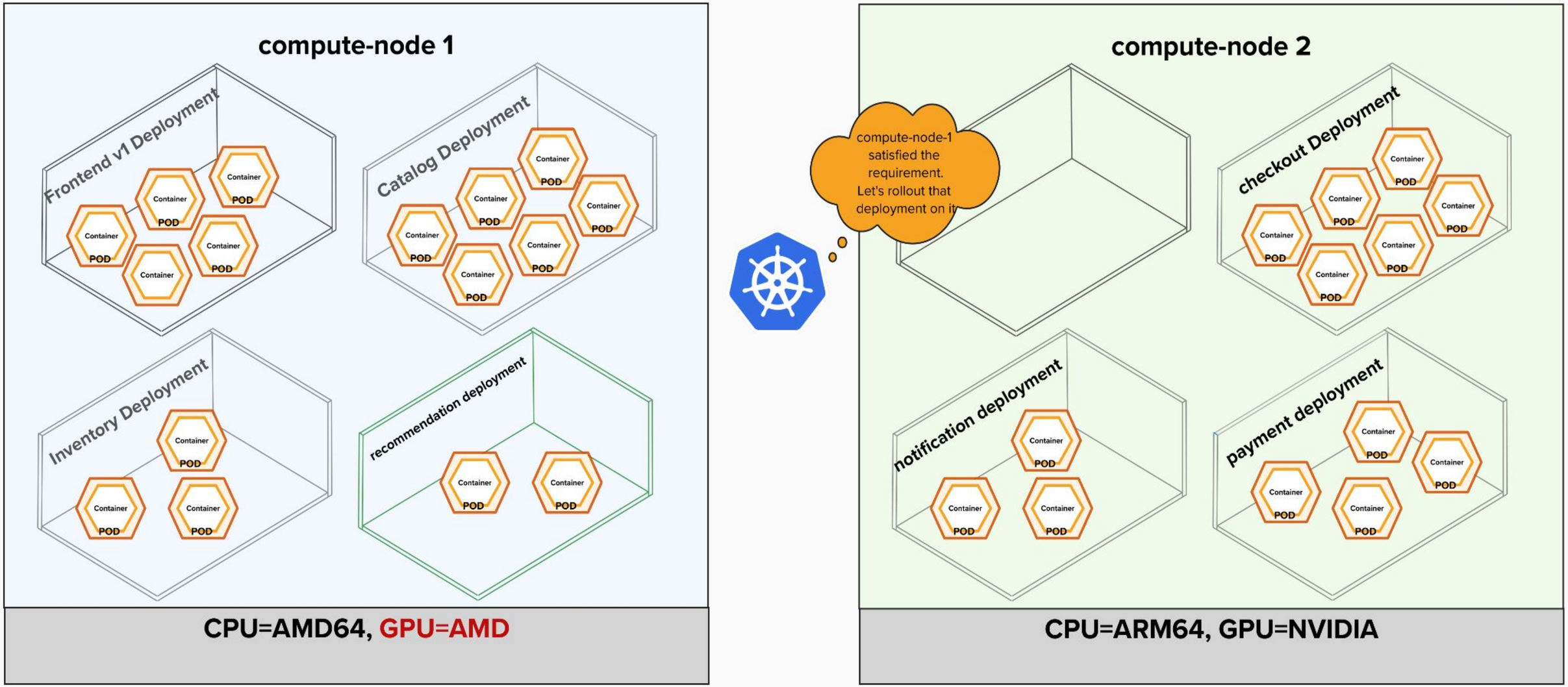
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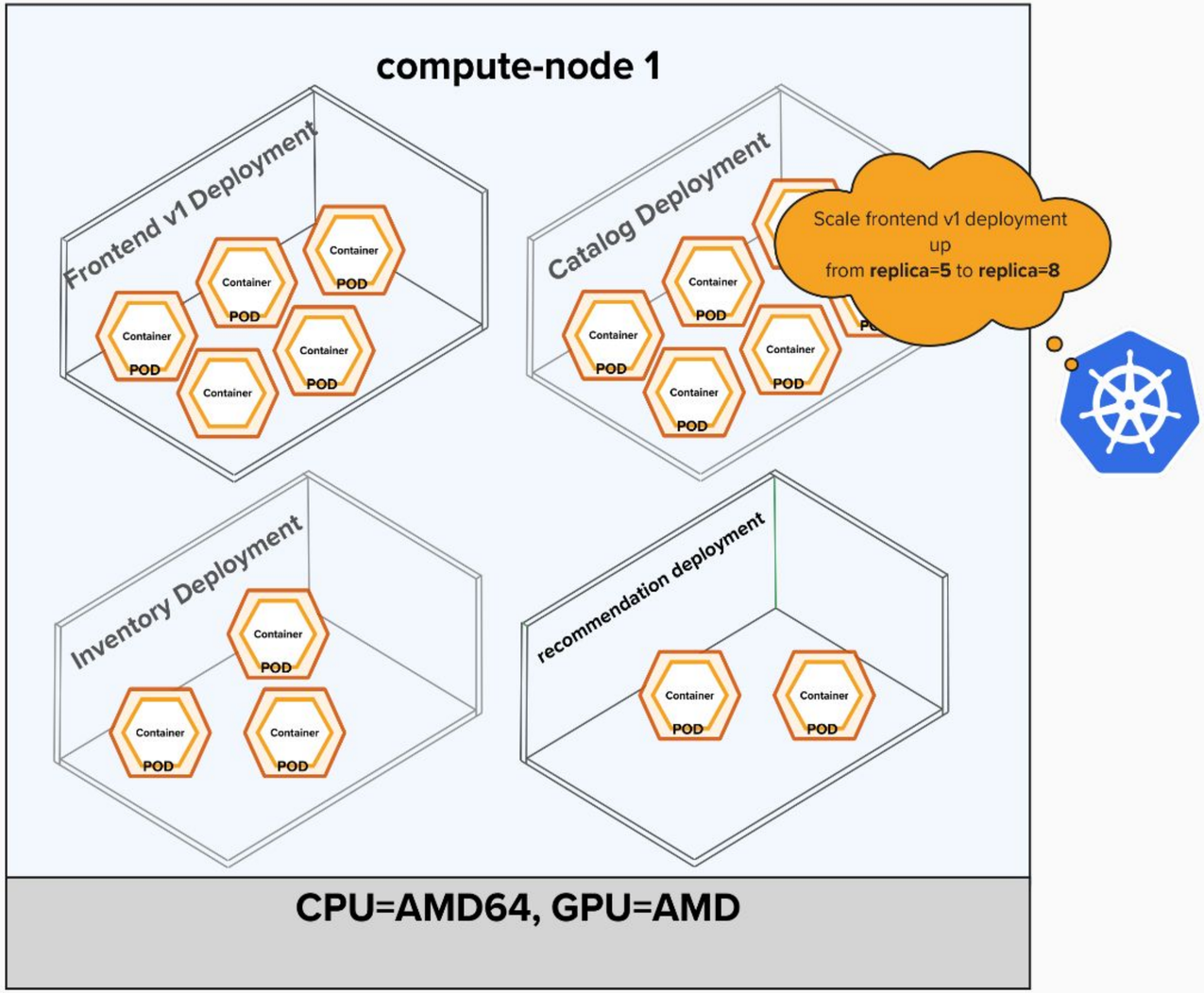
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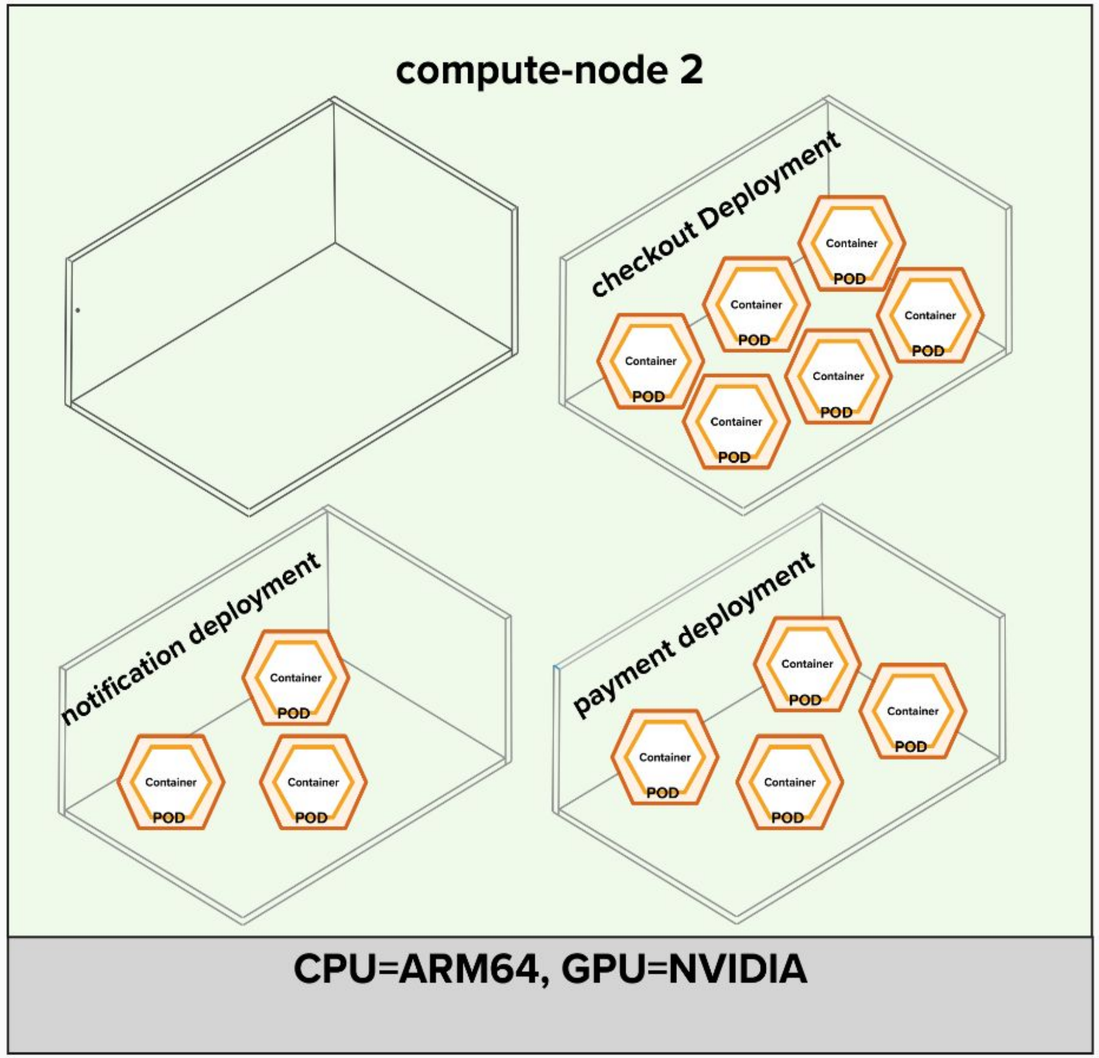
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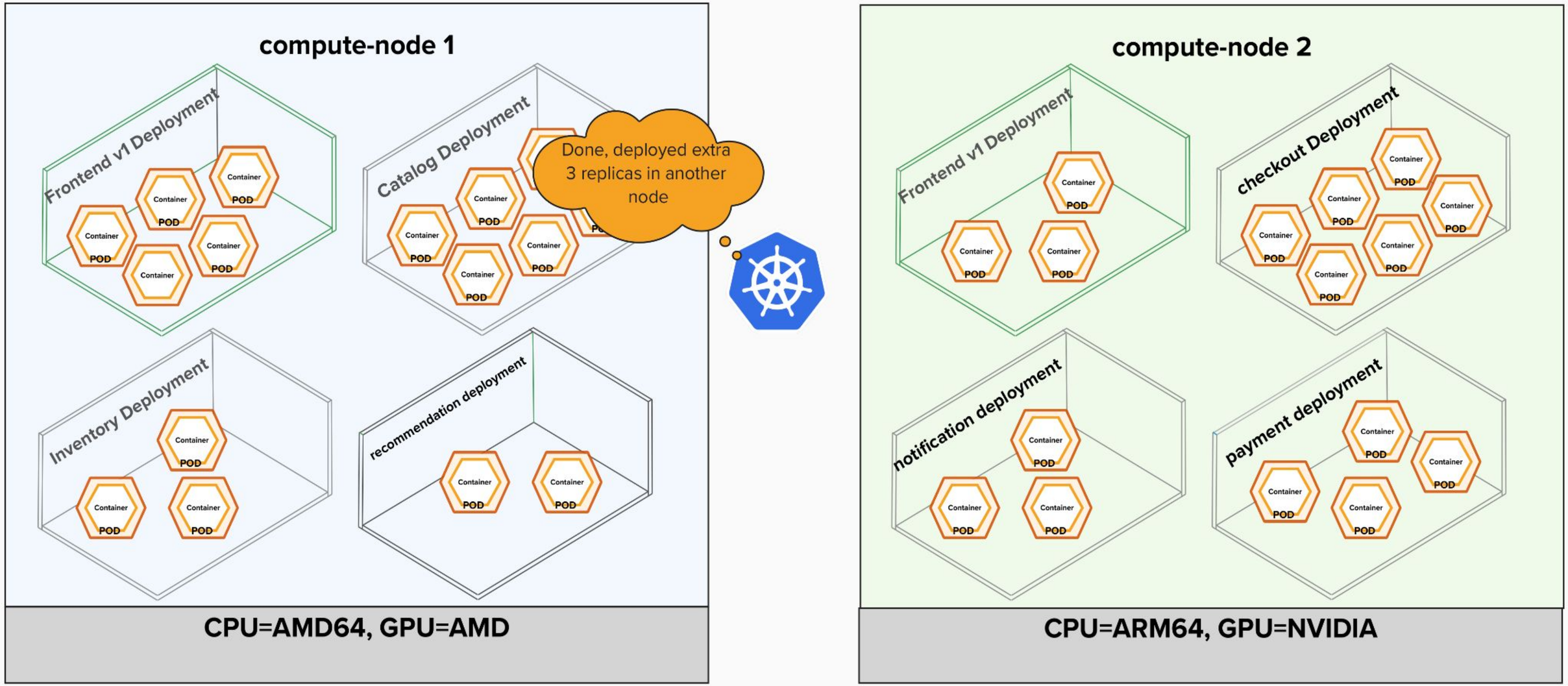
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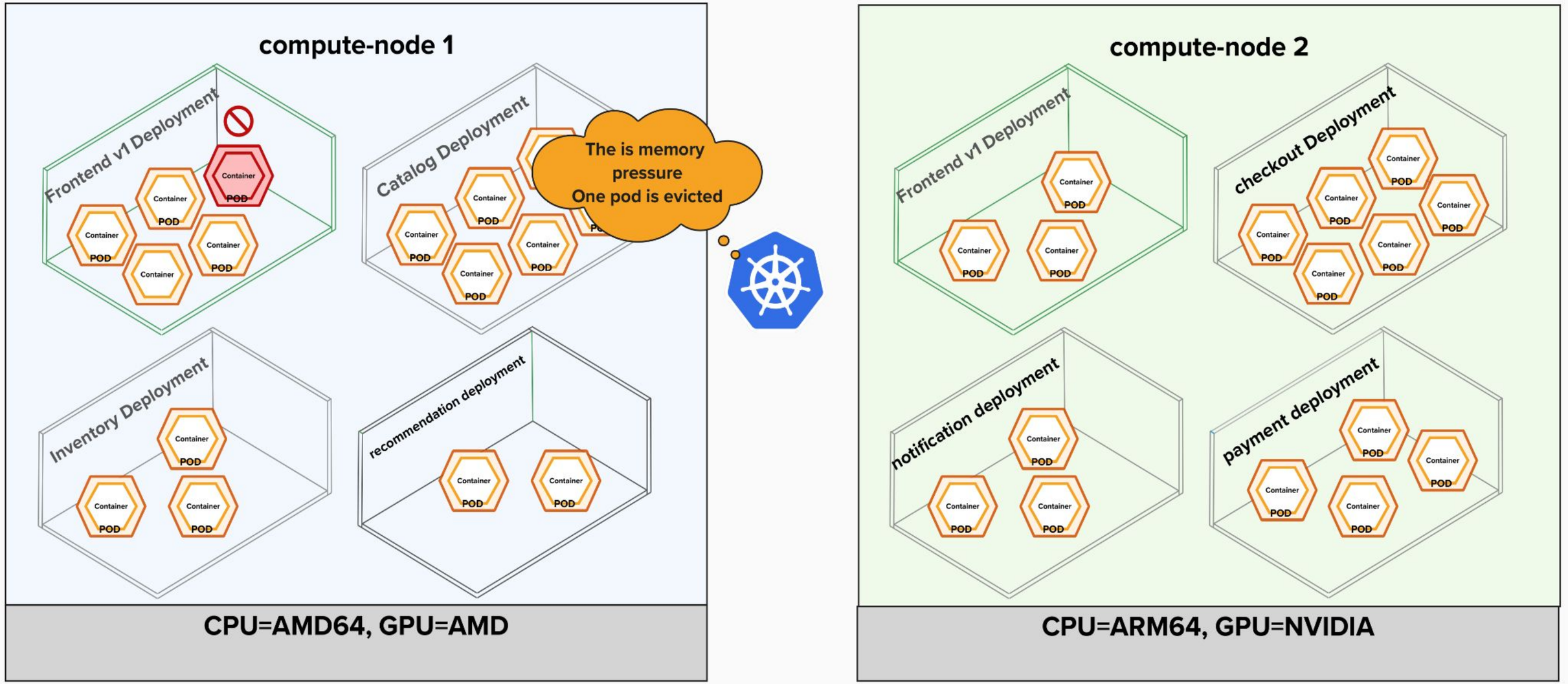
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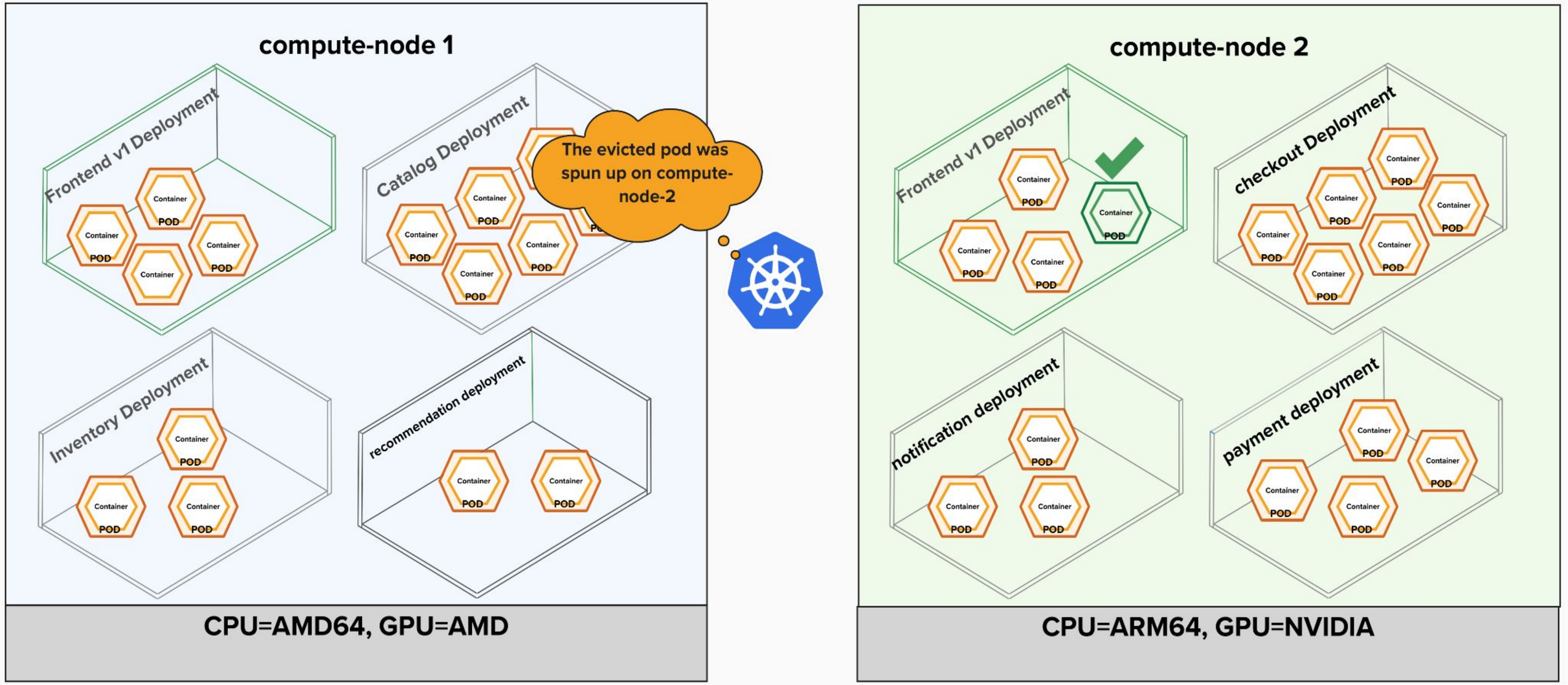
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WHAT IS KUBERNETES

- Kubernetes is a portable, extensible, open source platform for managing containerized workloads and services that facilitate both declarative configuration and automation. It has a large, rapidly growing ecosystem. Kubernetes services, support, and tools are widely available.
- Created by three Google employees initially during the summer of 2014; grew exponentially and became the first project to get donated to the CNCF.
- Hit the first production-grade version v1.0.1 in July 2015. Has continually released a new minor version every three months since v1.2.0 in March 2016. Lately v1.13.0 was released in December 2018.

Decouples Infrastructure and Scaling

- **All services** within Kubernetes are natively Load Balanced.
- Can scale up and down dynamically.
- Used both to enable self-healing and seamless upgrading or rollback of applications.

Self Healing

- Kubernetes will **ALWAYS** try and steer the cluster to its desired state.
- **Me:** “I want 3 healthy instances of redis to always be running.”
- **Kubernetes:** “Okay, I’ll ensure there are always 3 instances up and running.”
- **Kubernetes:** “Oh look, one has died. I’m going to attempt to spin up a new one.”

containers are created from container images

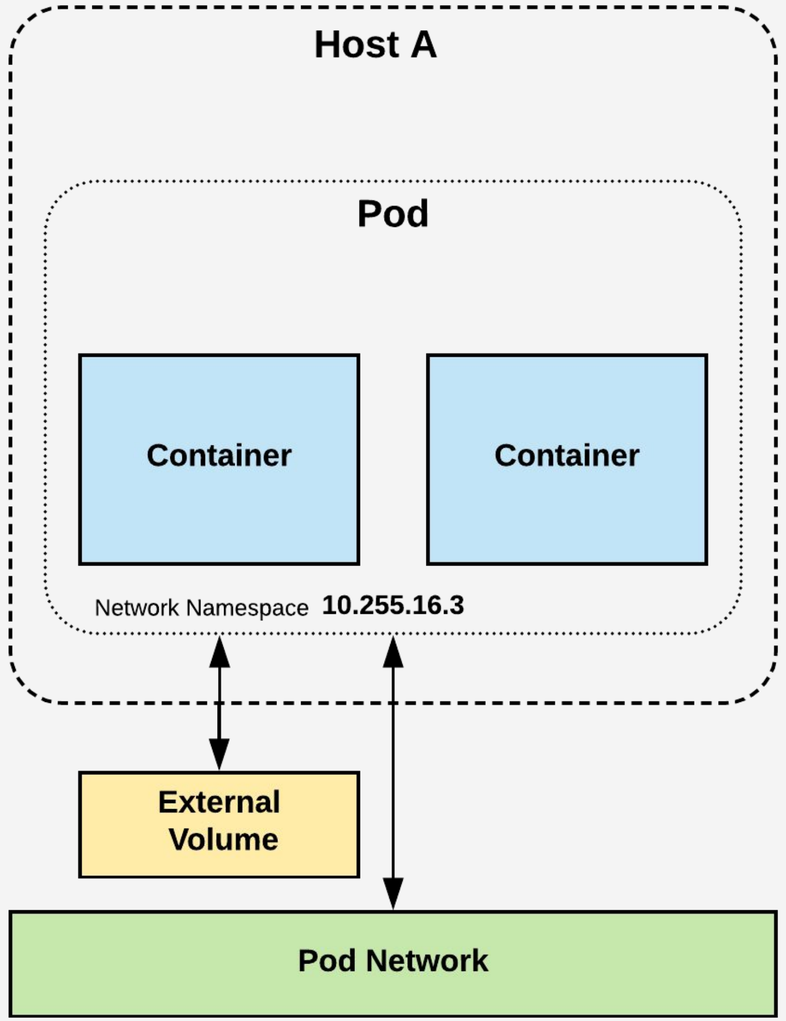
A Couple
Key Concepts...





Pods

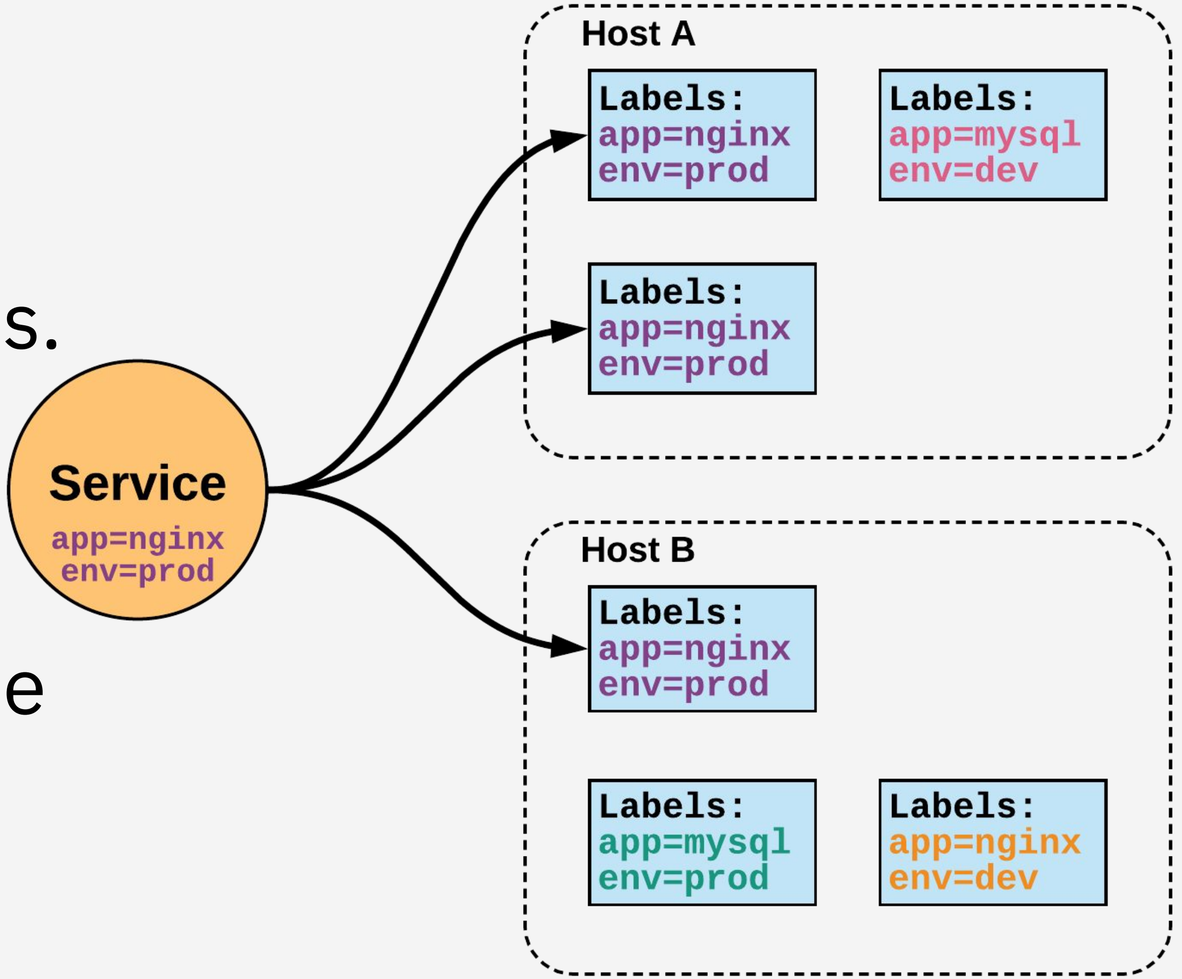
- **Atomic unit** or smallest "*unit of work*"of Kubernetes.
- Pods are **one or MORE containers** that share volumes and namespace.
- **They are also ephemeral!**



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Services

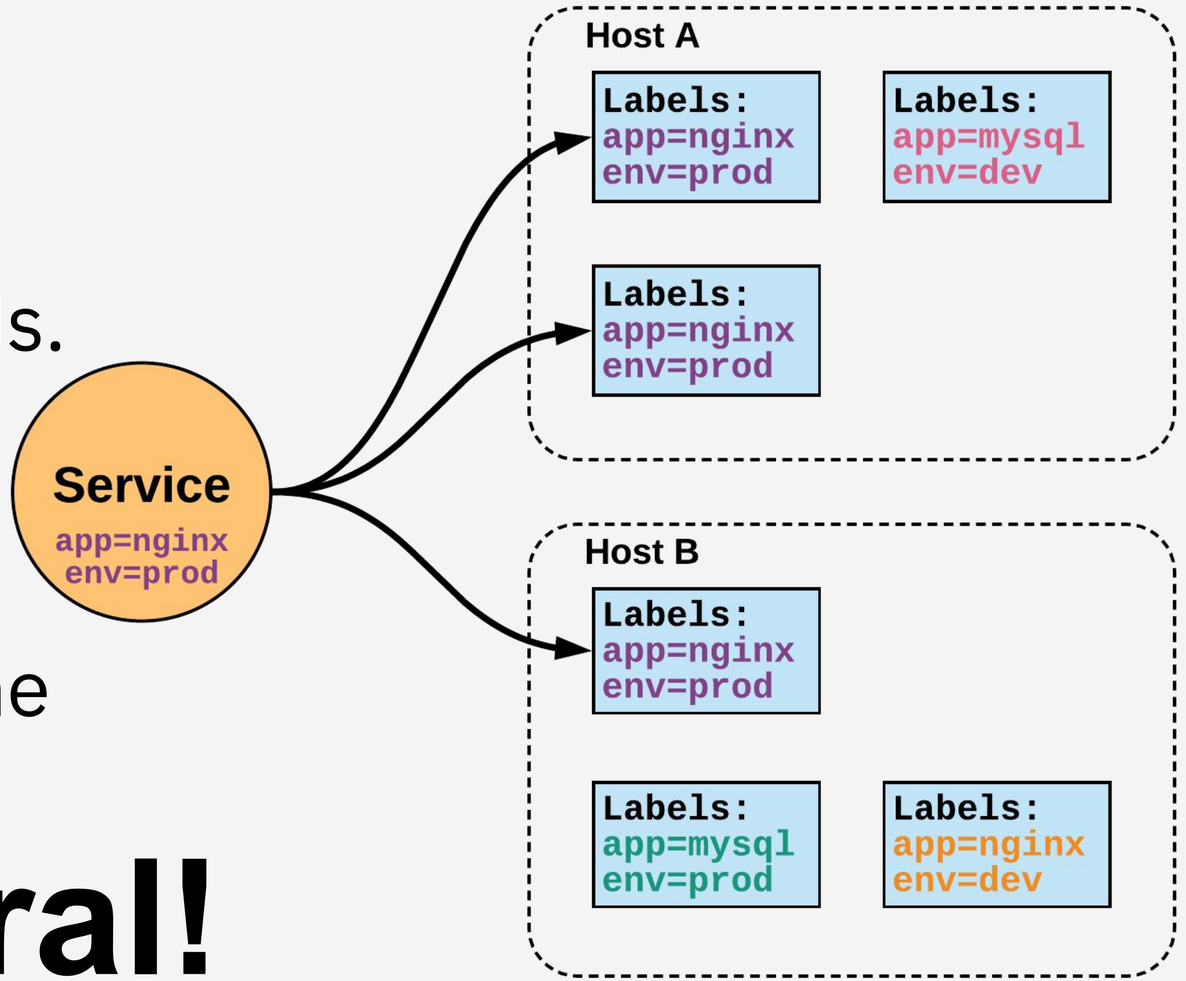
- **Unified method of accessing** the exposed workloads of Pods.
- **Durable resource**
  - static cluster IP
  - static namespaced DNS name



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Services

- **Unified method of accessing** the exposed workloads of Pods.
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NOT Ephemeral!

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containers are created from container images

Architecture Overview

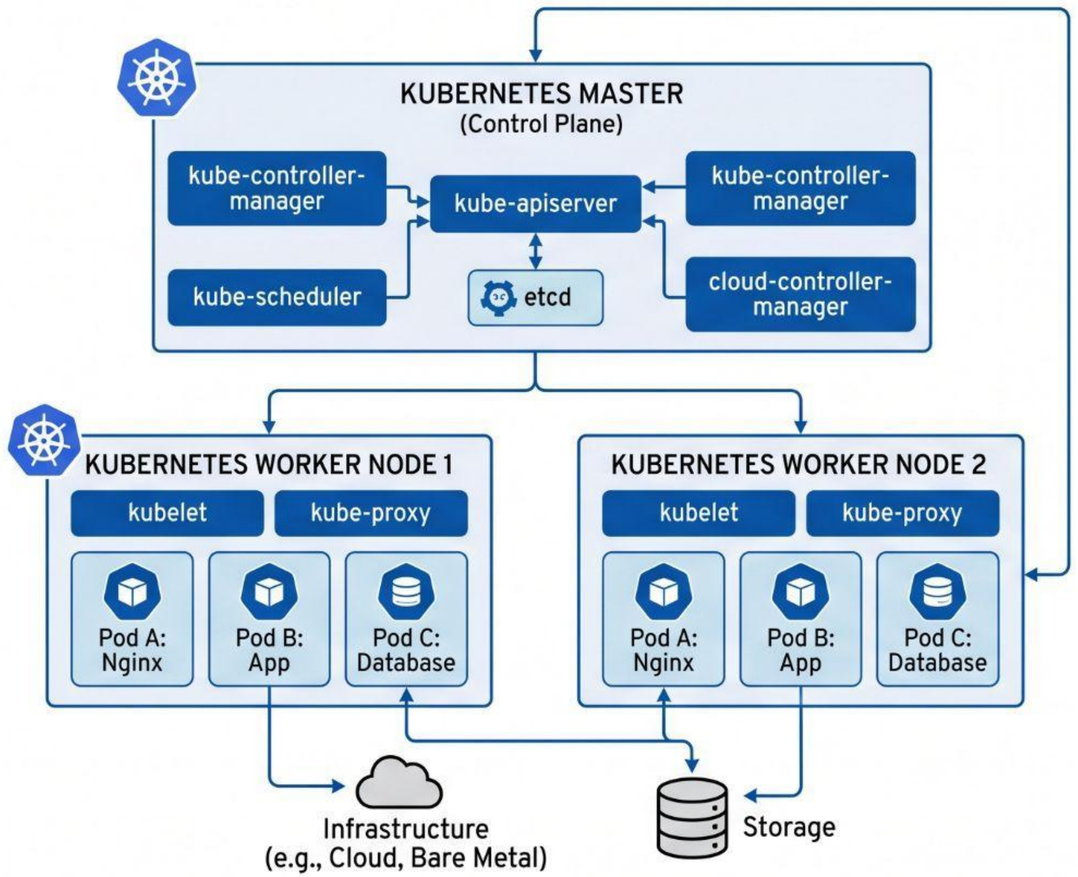


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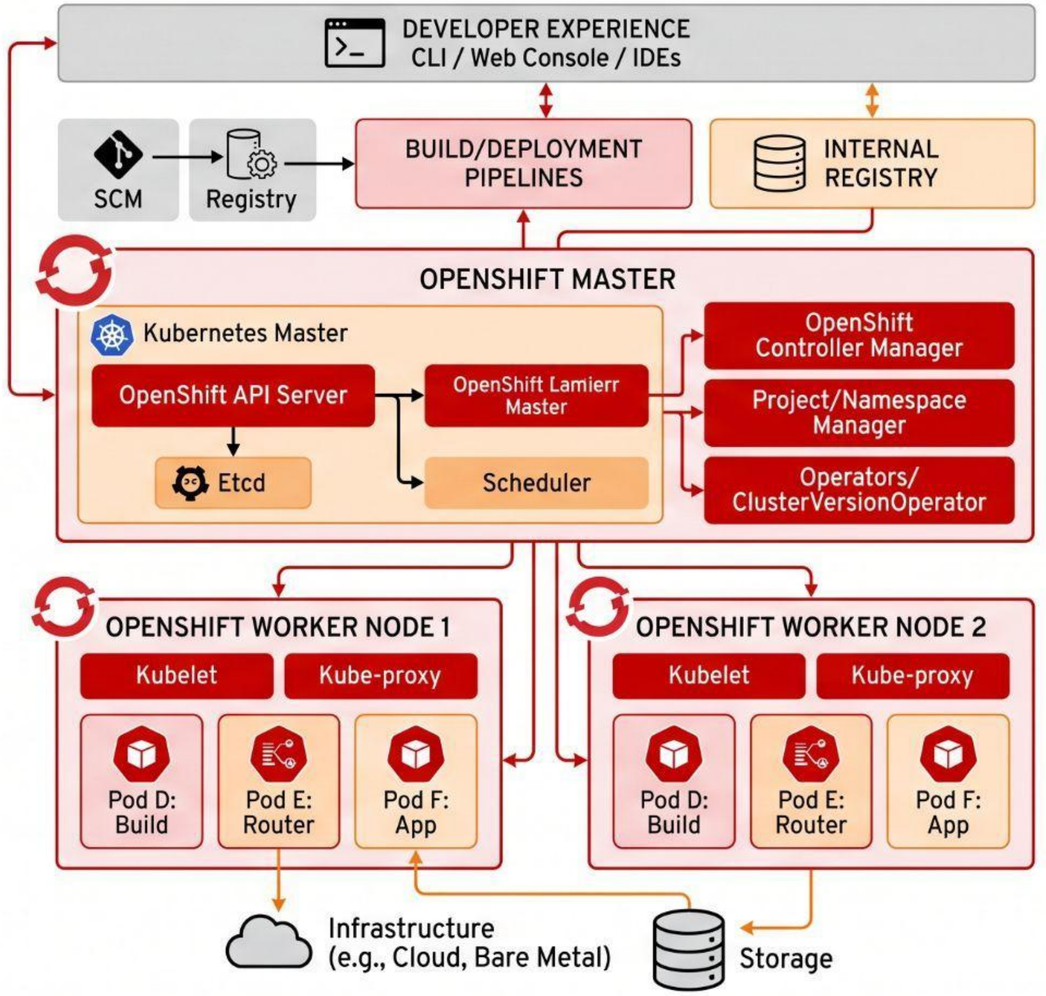


COMPARING KUBERNETES AND OPENSHIFT ARCHITECTURES

KUBERNETES ARCHITECTURE



OPENSHIFT ARCHITECTURE



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containers are created from container images

Core Objects and API

- Namespaces
- Pods
- Labels
- Selectors
- Services

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Namespaces

Namespaces are a logical cluster or environment, and are the primary method of partitioning a cluster or scoping access.

```
apiVersion: v1
kind: Namespace
metadata:
  name: prod
  labels:
    app: MyBigWebApp
```

```
$ kubectl get ns --show-labels
NAME          STATUS    AGE       LABELS
default       Active    11h       <none>
kube-public   Active    11h       <none>
kube-system   Active    11h       <none>
prod          Active    6s        app=MyBigWebApp
```

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Pod Examples

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-example
spec:
  containers:
  - name: nginx
    image: nginx:stable-alpine
    ports:
    - containerPort: 80
```

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Key Pod Container Attributes

- `name` - The name of the container
- `image` - The container image
- `ports` - array of ports to expose. Can be granted a friendly name and protocol may be specified
- `env` - array of environment variables
- `command` – Entrypoint array (equiv to Docker `ENTRYPOINT`)
- `args` - Arguments to pass to the command (equiv to Docker `CMD`)

Container

```
name: nginx
image: nginx:stable-alpine
ports:
  - containerPort: 80
    name: http
    protocol: TCP
env:
  - name: MYVAR
    value: isAwesome
command: [“/bin/sh”, “-c”]
args: [“echo ${MYVAR}”]
```

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Key Pod Container Attributes

- `name` - The name of the container
- `image` - The container image
- `ports` - array of ports to expose. Can be granted a friendly name and protocol may be specified
- `env` - array of environment variables
- `command` – Entrypoint array (equiv to Docker `ENTRYPOINT`)
- `args` - Arguments to pass to the command (equiv to Docker `CMD`)

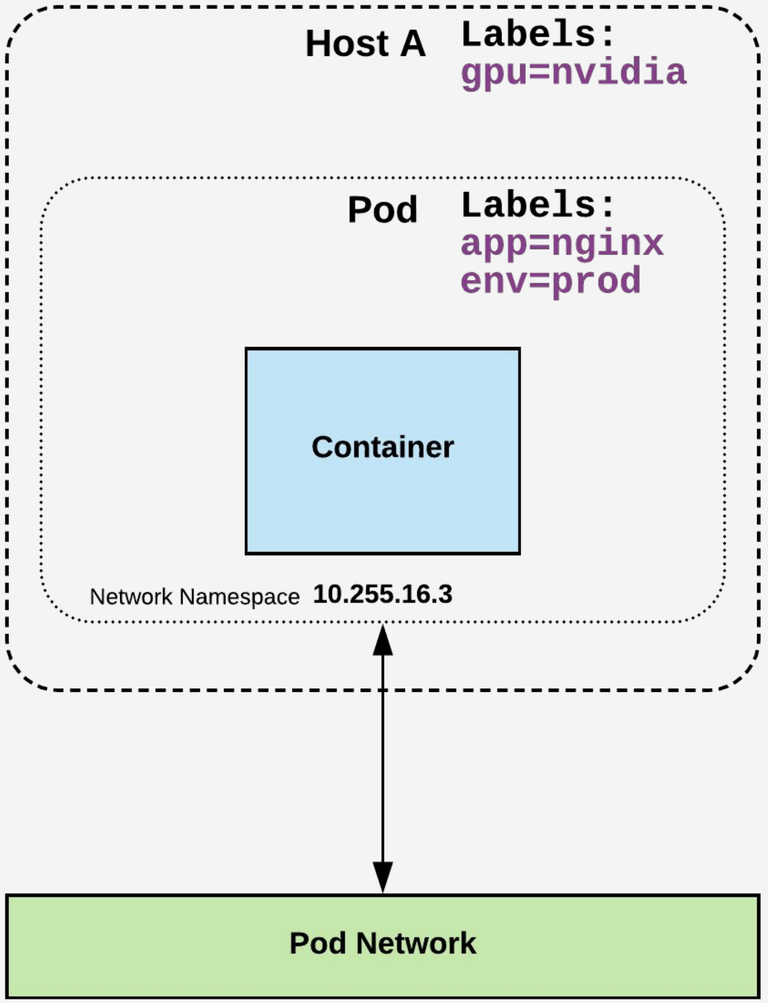
Container

```
name: nginx
image: nginx:stable-alpine
ports:
  - containerPort: 80
    name: http
    protocol: TCP
env:
  - name: MYVAR
    value: isAwesome
command: [“/bin/sh”, “-c”]
args: [“echo ${MYVAR}”]
```

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Labels

- key-value pairs that are used to identify, describe and group together related sets of objects or resources.
- **NOT** characteristic of uniqueness.
- Have a strict syntax with a slightly limited character set*.



Selectors

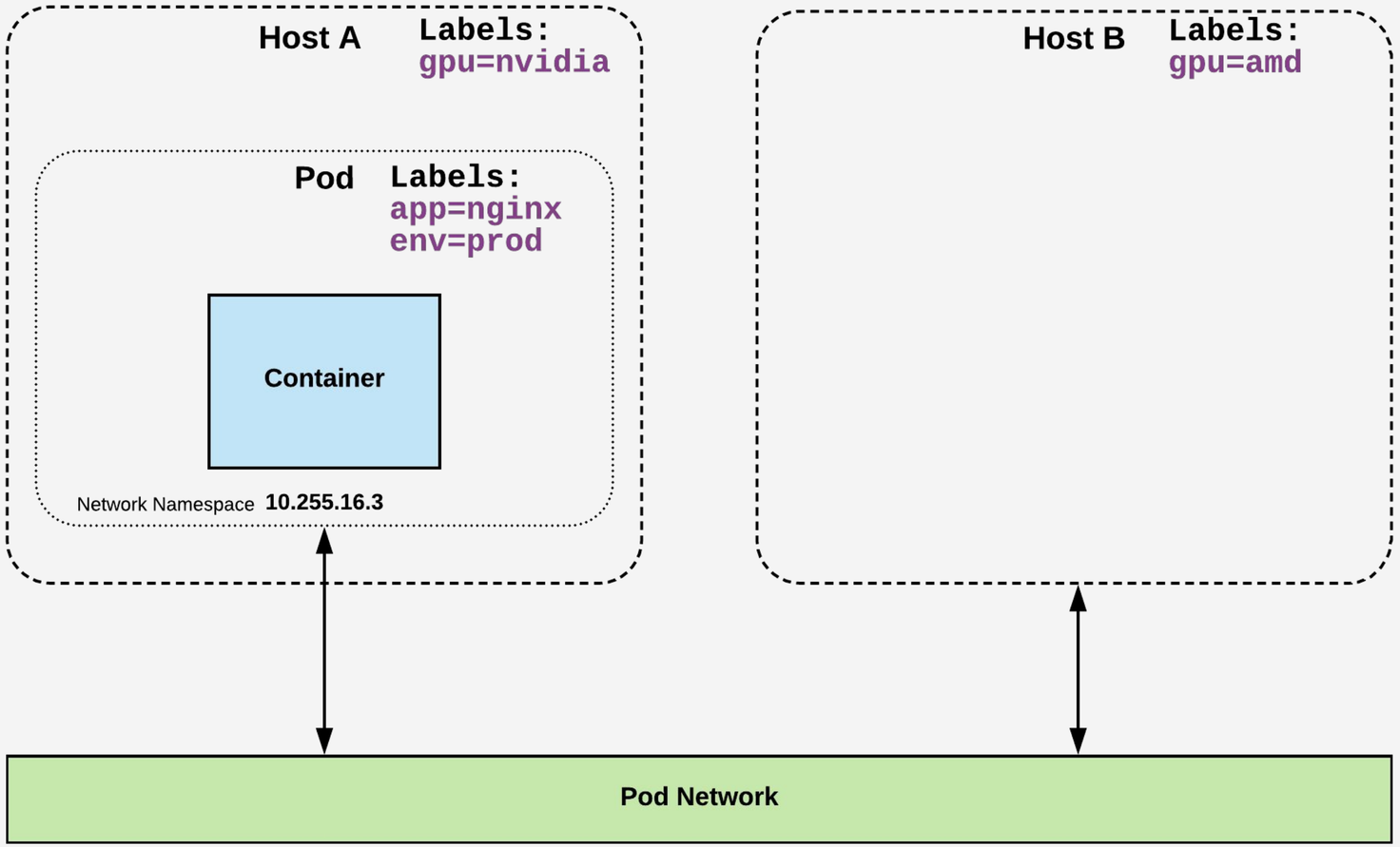
Selectors use labels to filter or select objects, and are used throughout Kubernetes.

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-label-example
  labels:
    app: nginx
    env: prod
spec:
  containers:
  - name: nginx
    image: nginx:stable-alpine
    ports:
    - containerPort: 80
  nodeSelector:
    gpu: nvidia
```

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Selector Example

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod-label-example
  labels:
    app: nginx
    env: prod
spec:
  containers:
  - name: nginx
    image: nginx:stable-alpine
    ports:
    - containerPort: 80
  nodeSelector:
    gpu: nvidia
```



Selector Types

Equality based selectors allow for simple filtering (=,==, or !=).

```
selector:
  matchLabels:
    gpu: nvidia
```

Set-based selectors are supported on a limited subset of objects. However, they provide a method of filtering on a set of values, and supports multiple operators including: in, notin, and exist.

```
selector:
  matchExpressions:
    - key: gpu
      operator: in
      values: [“nvidia”]
```

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Services

- **Unified method of accessing** the exposed workloads of Pods.
- **Durable resource** (unlike Pods)
  - static cluster-unique IP
  - static namespaced DNS name

```
<service name>.<namespace>.svc.cluster.local
```

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Services

- Target Pods using **equality based selectors.**
- Uses **kube-proxy** to provide simple load-balancing.
- **kube-proxy** acts as a daemon that creates **local entries** in the host’s iptables for every service.

Service Types

There are 4 major service types:

- ClusterIP (default)
- NodePort
- LoadBalancer
- ExternalName

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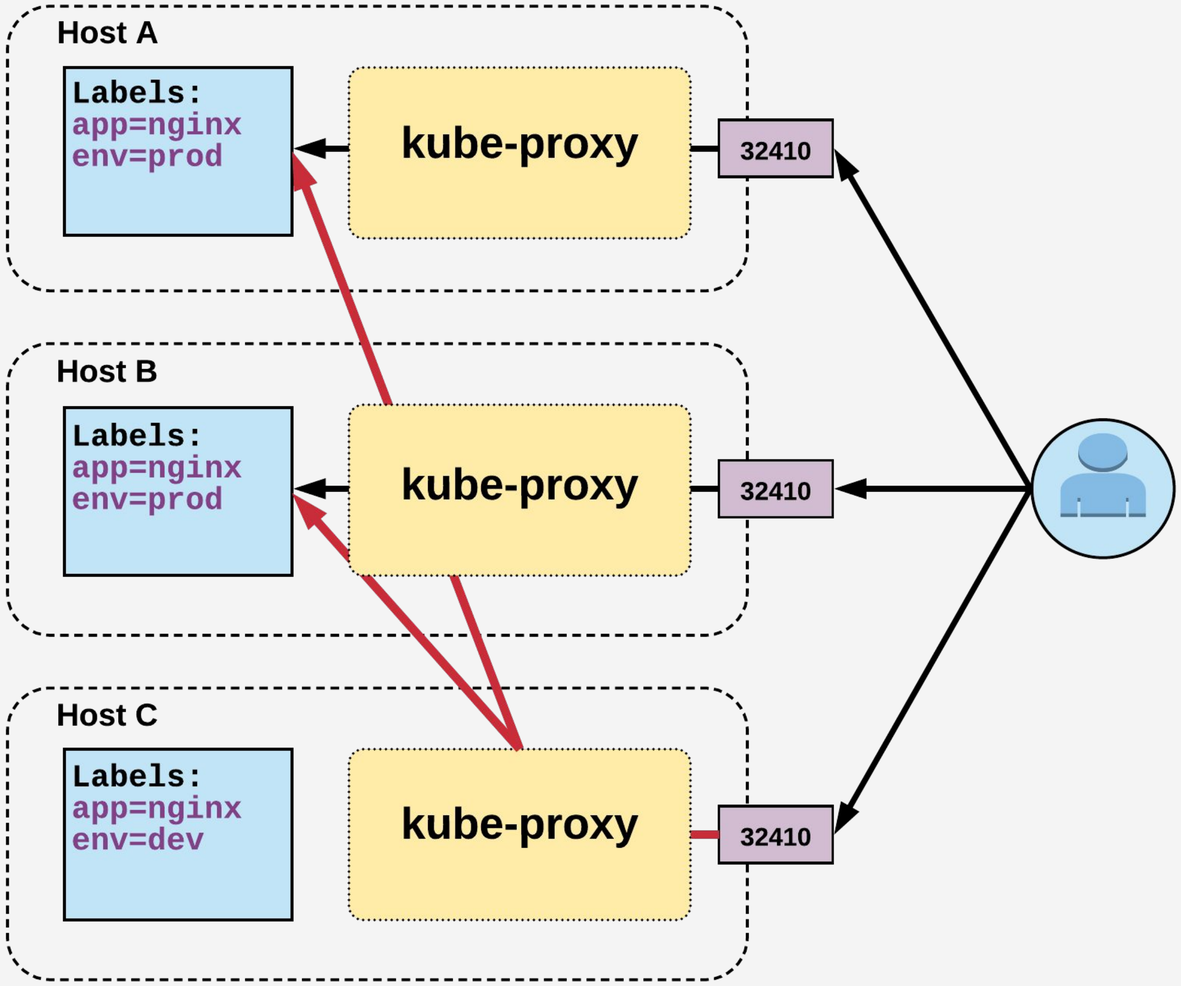
NodePort Service

- **NodePort** services extend the **ClusterIP** service.
- Exposes a port on every node's IP.
- Port can either be statically defined, or dynamically taken from a range between 30000-32767.

```yaml
apiVersion: v1
kind: Service
metadata:
  name: example-prod
spec:
  type: NodePort
  selector:
    app: nginx
    env: prod
  ports:
  - nodePort: 32410
    protocol: TCP
    port: 80
    targetPort: 80
```

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NodePort Service




```
Name:           example-prod
Selector:       app=nginx,env=prod
Type:           NodePort
IP:             10.96.28.176
Port:           <unset>  80/TCP
TargetPort:     80/TCP
NodePort:       <unset>  32410/TCP
Endpoints:      10.255.16.3:80,
                10.255.16.4:80
```

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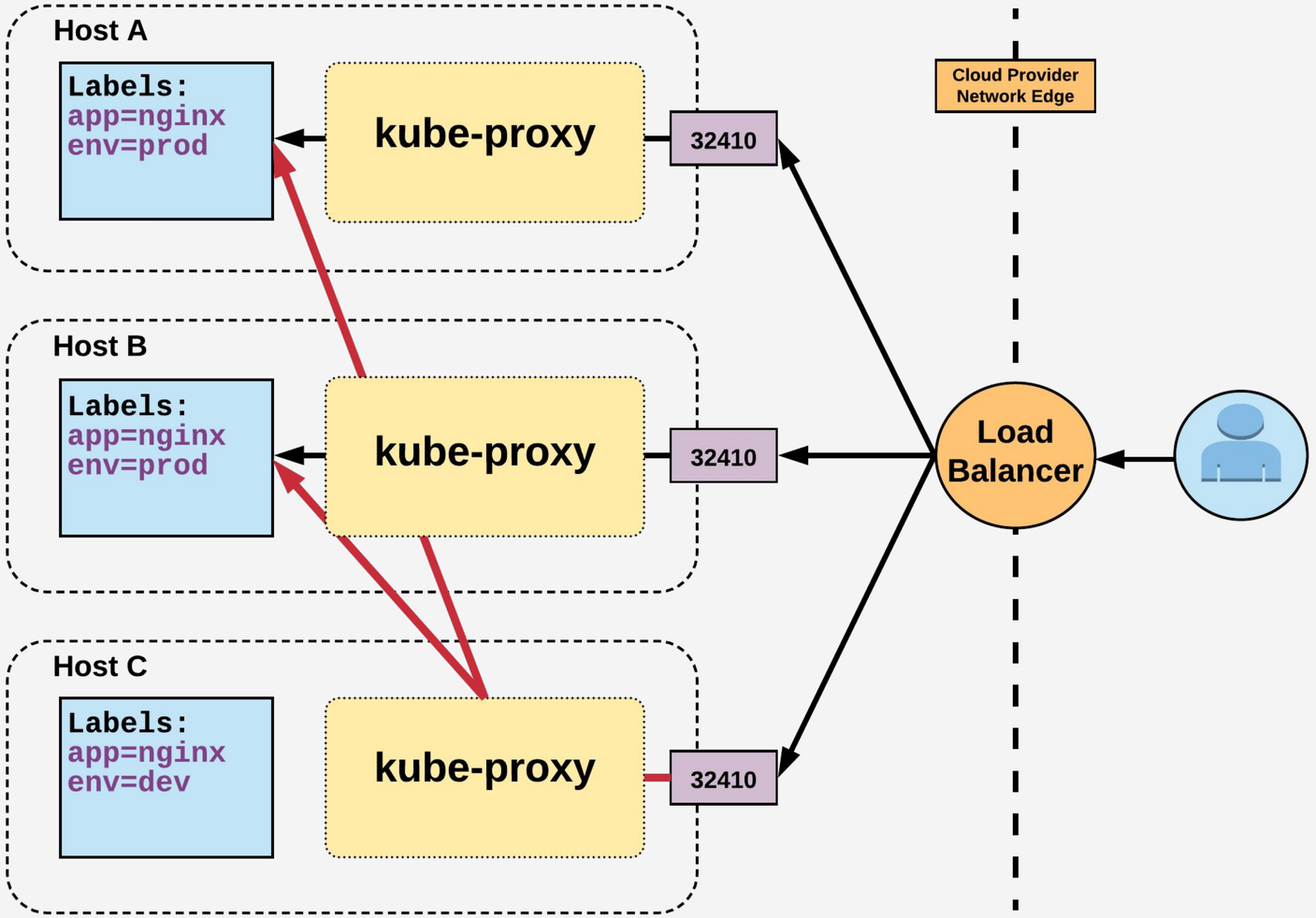
LoadBalancer Service

- **LoadBalancer** services extend **NodePort.**
- Works in conjunction with an external system to map a cluster external IP to the exposed service.

```
apiVersion: v1
kind: Service
metadata:
  name: example-prod
spec:
  type: LoadBalancer
  selector:
    app: nginx
    env: prod
  ports:
    protocol: TCP
    port: 80
    targetPort: 80
```

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LoadBalancer Service




```
Name:           example-prod
Selector:       app=nginx,env=prod
Type:           LoadBalancer
IP:             10.96.28.176
LoadBalancer
Ingress:        172.17.18.43
Port:           <unset>  80/TCP
TargetPort:     80/TCP
NodePort:       <unset>  32410/TCP
Endpoints:      10.255.16.3:80,
                10.255.16.4:80
```

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Ingress – Name Based Routing

- An API object that manages external access to the services in a cluster
- Provides load balancing, SSL termination and name/path-based virtual hosting
- Gives services externally-reachable URLs

```
apiVersion: extensions/v1beta1
kind: Ingress
metadata:
  name: name-virtual-host-ingress
spec:
  rules:
  - host: first.bar.com
    http:
      paths:
      - backend:
          serviceName: service1
          servicePort: 80
  - host: second.foo.com
    http:
      paths:
      - backend:
          serviceName: service2
          servicePort: 80
  - http:
      paths:
      - backend:
          serviceName: service3
          servicePort: 80
```

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Ingress – Path Based Routing

```
apiVersion: extensions/v1beta1
kind: Ingress
metadata:
  name: simple-fanout-example
spec:
  rules:
  - host: foo.bar.com
    http:
      paths:
      - path: /foo
        backend:
          serviceName: service1
          servicePort: 4200
      - path: /bar
        backend:
          serviceName: service2
          servicePort: 8080
```

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containers are created from container images

Core workloads

- ReplicaSet
- Deployment
- DaemonSet
- StatefulSet
- Job
- CronJob

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ReplicaSet

- Primary method of managing pod replicas and their lifecycle.
- Includes their scheduling, scaling, and deletion.
- Their job is simple: **Always ensure the desired number of pods are running.**



ReplicaSet

- replicas: The desired number of instances of the Pod.
- selector:The label selector for the **ReplicaSet** will manage **ALL** Pod instances that it targets; whether it’s desired or not.

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: rs-example
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
      env: prod
  template:
    <pod template>
```

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ReplicaSet

```yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: rs-example
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
      env: prod
  template:
    metadata:
      labels:
        app: nginx
        env: prod
    spec:
      containers:
      - name: nginx
        image: nginx:stable-alpine
        ports:
        - containerPort: 80
```

```
$ kubectl get pods
NAME                READY     STATUS    RESTARTS   AGE
rs-example-9l4dt    1/1       Running   0          1h
rs-example-b7bcg    1/1       Running   0          1h
rs-example-mkll2    1/1       Running   0          1h
```

```
$ kubectl describe rs rs-example
Name:         rs-example
Namespace:    default
Selector:     app=nginx,env=prod
Labels:       app=nginx
              env=prod
Annotations:  <none>
Replicas:     3 current / 3 desired
Pods Status:  3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=nginx
           env=prod
  Containers:
   nginx:
    Image:        nginx:stable-alpine
    Port:         80/TCP
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
Events:
  Type    Reason            Age   From                   Message
  ----    ------            ----  ----                   -------
  Normal  SuccessfulCreate  16s   replicaset-controller  Created pod: rs-example-mkll2
  Normal  SuccessfulCreate  16s   replicaset-controller  Created pod: rs-example-b7bcg
  Normal  SuccessfulCreate  16s   replicaset-controller  Created pod: rs-example-9l4dt
```

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Deployment

- Way of managing Pods via **ReplicaSets.**
- Provide rollback functionality and update control.
- Updates are managed through the **pod-template-hash** label.
- Each iteration creates a unique label that is assigned to both the **ReplicaSet** and subsequent Pods.

Deployment

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.7.9
        ports:
        - containerPort: 80
```

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Deployment

- revisionHistoryLimit: The number of previous iterations of the Deployment to retain.
- strategy: Describes the method of updating the Pods based on the type. Valid options are Recreate or RollingUpdate.
  - Recreate: All existing Pods are killed before the new ones are created.
  - RollingUpdate: Cycles through updating the Pods according to the parameters: maxSurge and maxUnavailable.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: deploy-example
spec:
  replicas: 3
  revisionHistoryLimit: 3
  selector:
    matchLabels:
      app: nginx
      env: prod
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  template:
    <pod template>
```

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Deployment

```
$ kubectl create deployment test --image=nginx
$ kubectl set image deployment test nginx=nginx:1.9.1 --record

$ kubectl rollout history deployment test
deployments "test"
REVISION  CHANGE-CAUSE
1         <none>
2         kubectl set image deployment test nginx=nginx:1.9.1 --record=true

$ kubectl annotate deployment test kubernetes.io/change-cause="image updated to 1.9.1"

$ kubectl rollout undo deployment test
$ kubectl rollout undo deployment test --to-revision=2

$ kubectl rollout history deployment test
deployments "test"
REVISION  CHANGE-CAUSE
2         kubectl set image deployment test nginx=nginx:1.9.1 --record=true
3         <none>

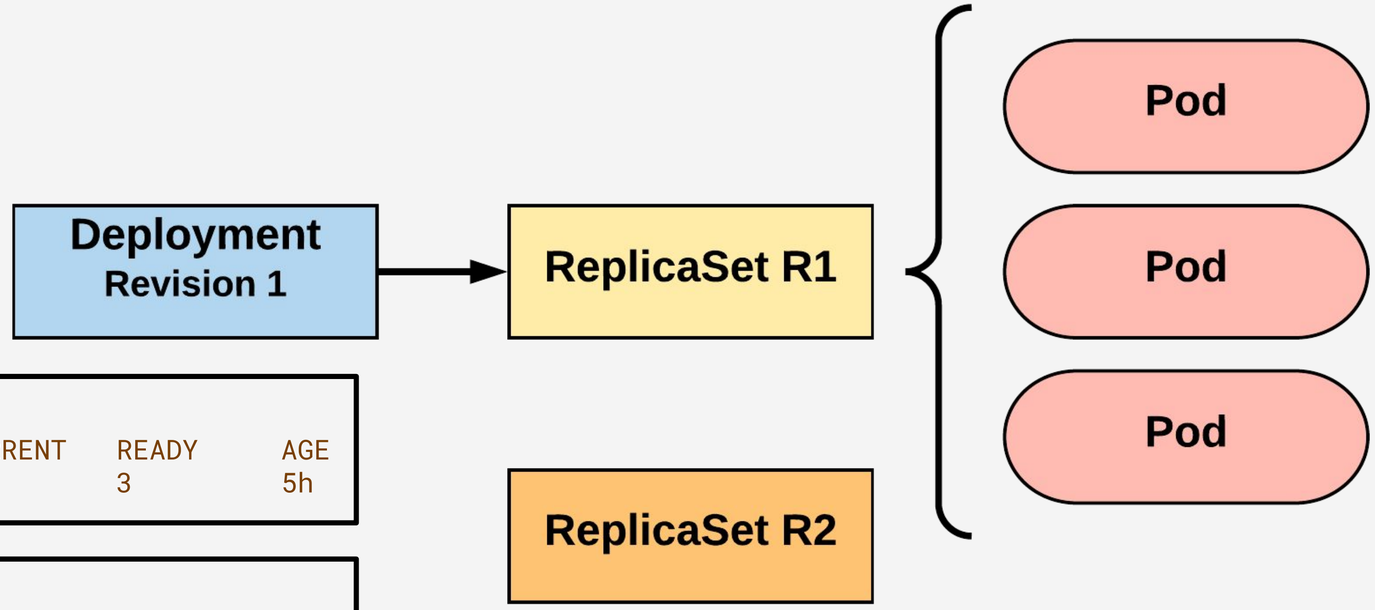
kubectl scale deployment test --replicas=10
kubectl rollout pause deployment test
kubectl rollout resume deployment test
```

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RollingUpdate Deployment

Updating pod template generates a new **ReplicaSet** revision.

R1 pod-template-hash:
676677fff
R2 pod-template-hash:
54f7ff7d6d




```
$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY     AGE
mydep-6766777fff              3         3         3         5h
```

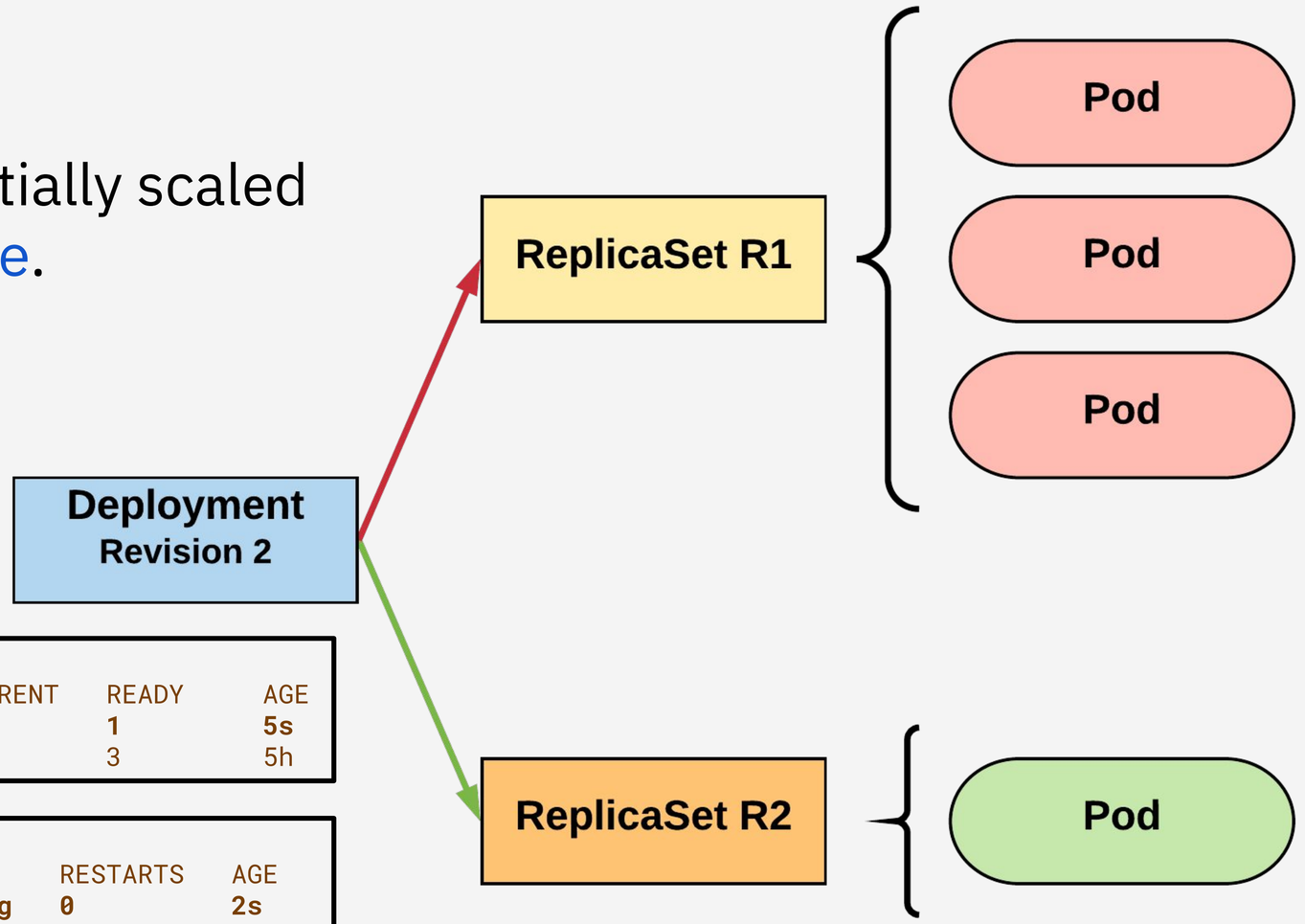
```
$ kubectl get pods
NAME                          READY     STATUS    RESTARTS   AGE
mydep-6766777fff-9r2zn        1/1       Running   0          5h
mydep-6766777fff-hsfz9        1/1       Running   0          5h
mydep-6766777fff-sjxhf        1/1       Running   0          5h
```

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RollingUpdate Deployment

New **ReplicaSet** is initially scaled up based on maxSurge.

R1 pod-template-hash:
676677fff
R2 pod-template-hash:
54f7ff7d6d




```
$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY     AGE
mydep-54f7ff7d6d              1         1         1         5s
mydep-6766777fff              2         3         3         5h
```

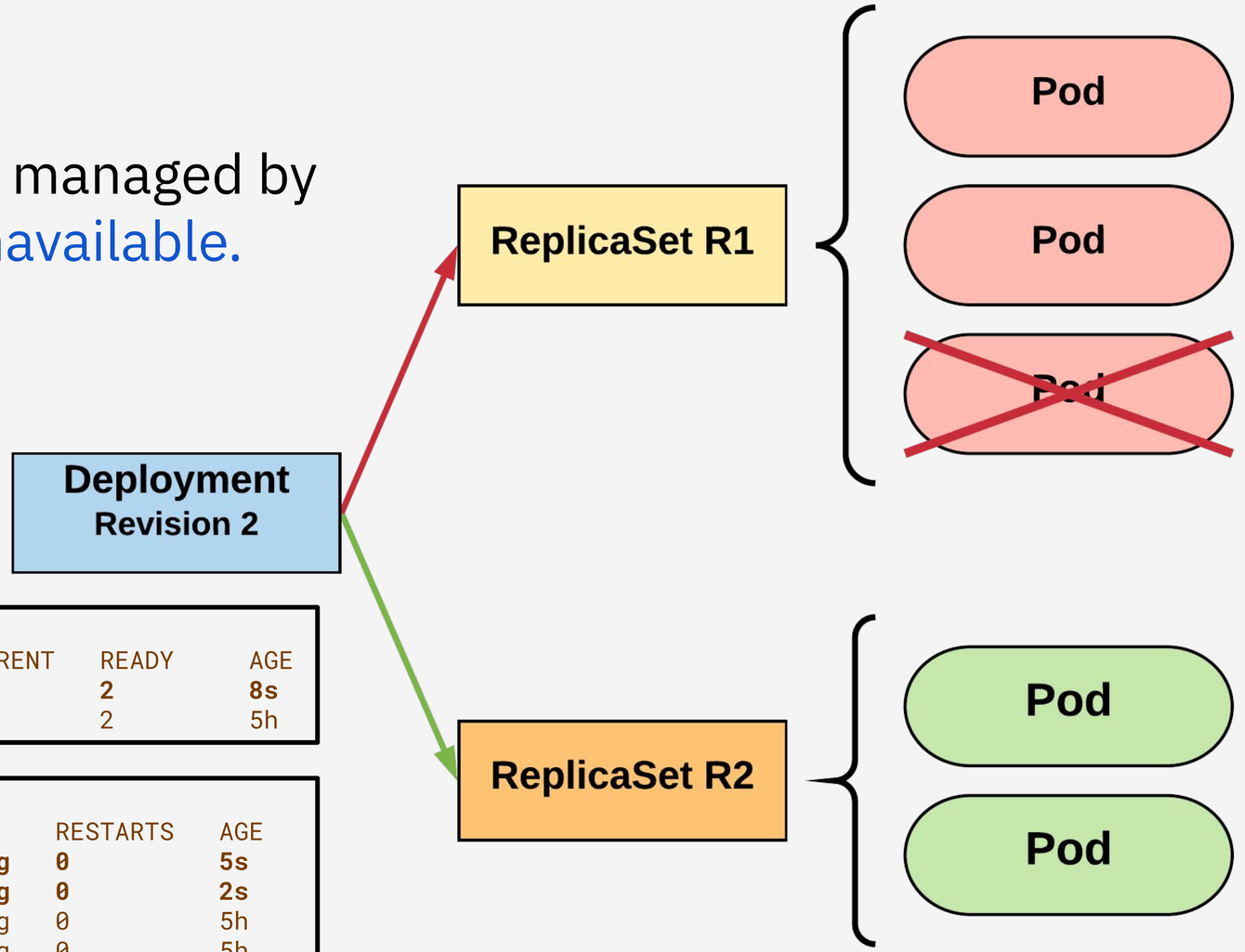
```
$ kubectl get pods
NAME                          READY     STATUS    RESTARTS   AGE
mydep-54f7ff7d6d-9gvll        1/1       Running   0          2s
mydep-6766777fff-9r2zn        1/1       Running   0          5h
mydep-6766777fff-hsfz9        1/1       Running   0          5h
mydep-6766777fff-sjxhf        1/1       Running   0          5h
```

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RollingUpdate Deployment

Phase out of old Pods managed by maxSurge and maxUnavailable.

R1 pod-template-hash:
676677fff
R2 pod-template-hash:
54f7ff7d6d




```
$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY     AGE
mydep-54f7ff7d6d              2         2         2         8s
mydep-6766777fff              2         2         2         5h
```

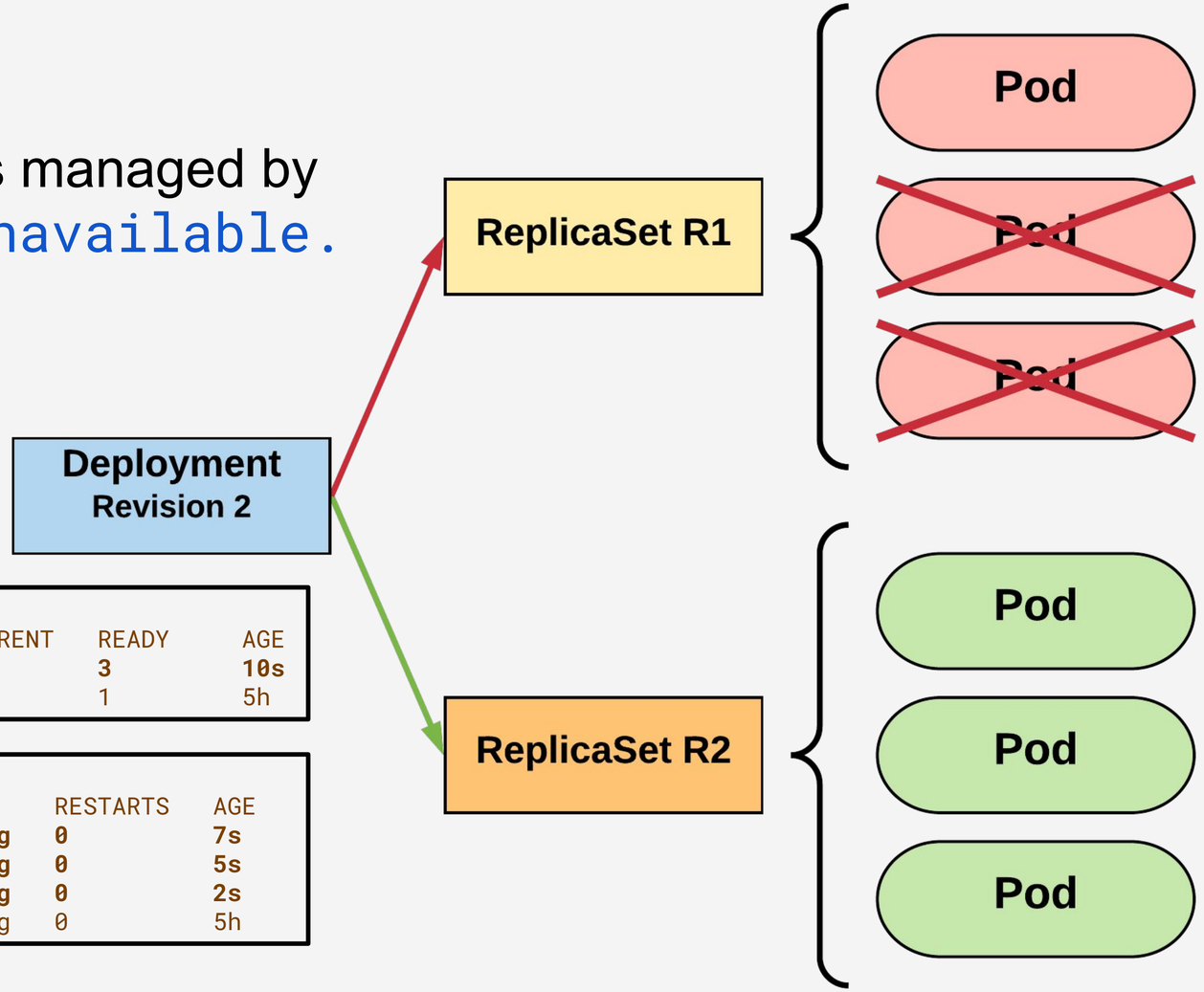
```
$ kubectl get pods
NAME                          READY     STATUS    RESTARTS   AGE
mydep-54f7ff7d6d-9gvll        1/1       Running   0          5s
mydep-54f7ff7d6d-cqvlq        1/1       Running   0          2s
mydep-6766777fff-9r2zn        1/1       Running   0          5h
mydep-6766777fff-hsfz9        1/1       Running   0          5h
```

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RollingUpdate Deployment

Phase out of old Pods managed by `maxSurge` and `maxUnavailable`.

R1 pod-template-hash:
`676677fff`
R2 pod-template-hash:
`54f7ff7d6d`




```
$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY     AGE
mydep-54f7ff7d6d              3         3         3         10s
mydep-6766777fff              0         1         1         5h
```

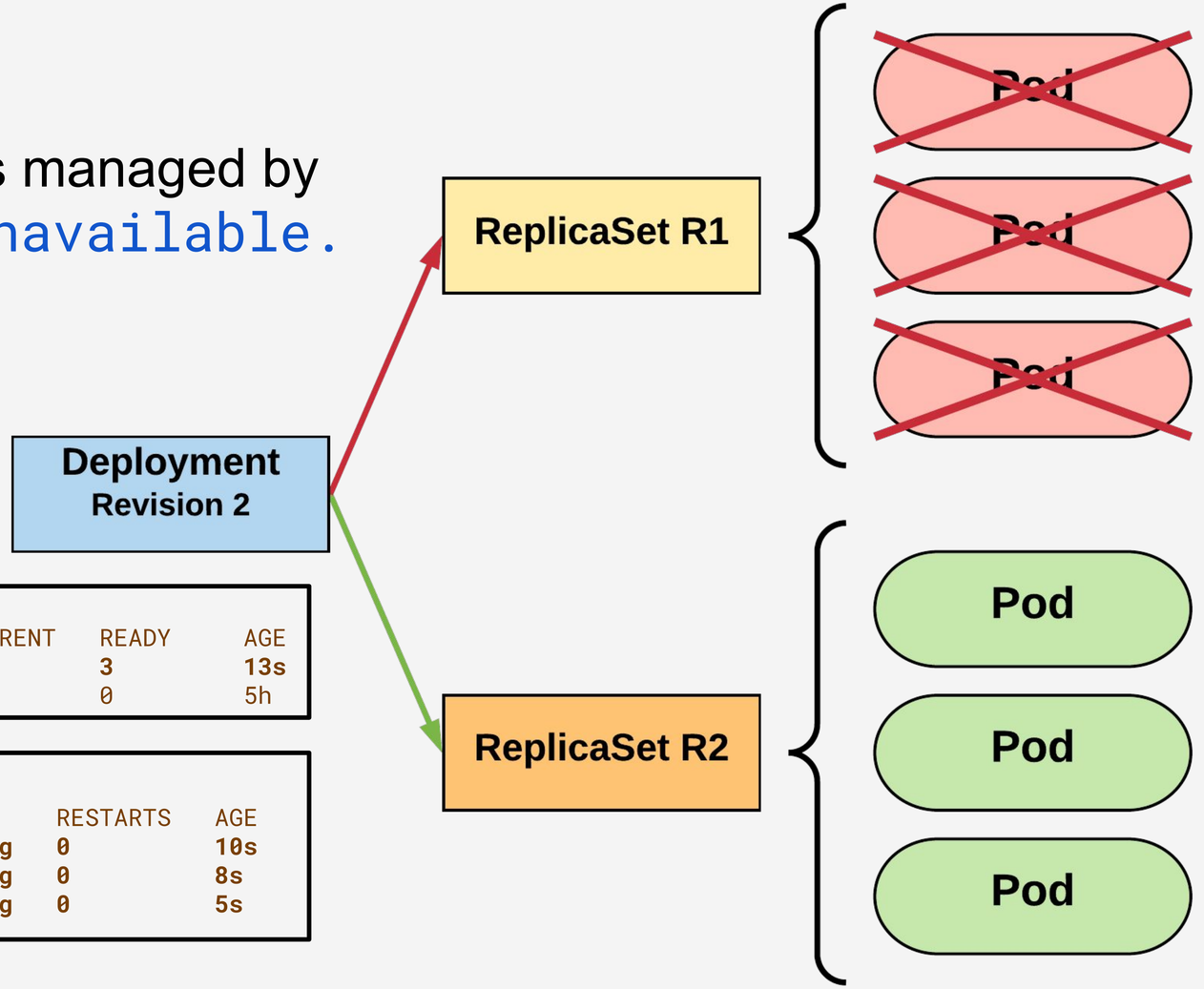
```
$ kubectl get pods
NAME                          READY     STATUS    RESTARTS   AGE
mydep-54f7ff7d6d-9gvll        1/1       Running   0          7s
mydep-54f7ff7d6d-cqvlq        1/1       Running   0          5s
mydep-54f7ff7d6d-gccr6        1/1       Running   0          2s
mydep-6766777fff-9r2zn        1/1       Running   0          5h
```

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RollingUpdate Deployment

Phase out of old Pods managed by `maxSurge` and `maxUnavailable`.

R1 pod-template-hash:
676677fff
R2 pod-template-hash:
54f7ff7d6d




```
$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY     AGE
mydep-54f7ff7d6d              3         3         3         13s
mydep-6766777fff              0         0         0         5h
```

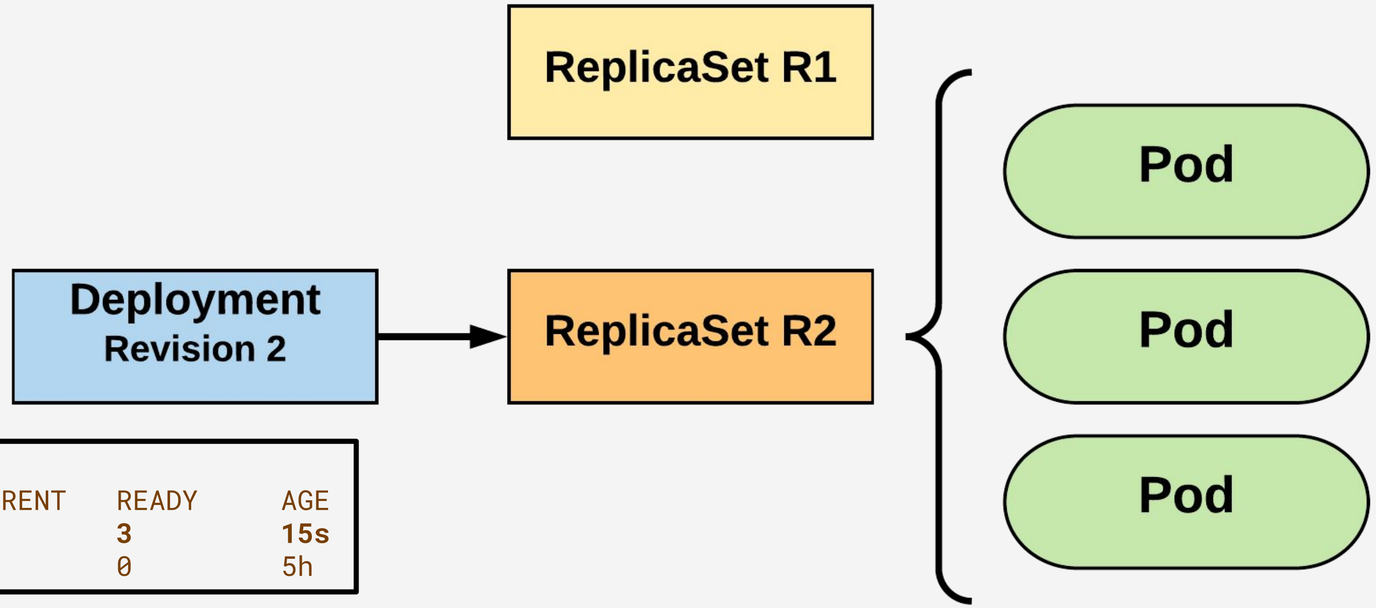
```
$ kubectl get pods
NAME                          READY     STATUS    RESTARTS   AGE
mydep-54f7ff7d6d-9gvll        1/1       Running   0          10s
mydep-54f7ff7d6d-cqvlq        1/1       Running   0          8s
mydep-54f7ff7d6d-gccr6        1/1       Running   0          5s
```

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RollingUpdate Deployment

Updated to new deployment revision completed.

R1 pod-template-hash:
`676677fff`
R2 pod-template-hash:
`54f7ff7d6d`




```
$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY     AGE
mydep-54f7ff7d6d              3         3         3         15s
mydep-6766777fff              0         0         0         5h
```

```
$ kubectl get pods
NAME                          READY     STATUS    RESTARTS   AGE
mydep-54f7ff7d6d-9gvll        1/1       Running   0          12s
mydep-54f7ff7d6d-cqvlq        1/1       Running   0          10s
mydep-54f7ff7d6d-gccr6        1/1       Running   0          7s
```

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StatefulSet

- Tailored to managing Pods that must persist or maintain state.
- Pod lifecycle will be ordered and follow consistent patterns.
- Assigned a unique ordinal name following the convention of '*<statefulset name>-<ordinal index>*'.



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StatefulSet

```yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: sts-example
spec:
  replicas: 2
  revisionHistoryLimit: 3
  selector:
    matchLabels:
      app: stateful
  serviceName: app
  updateStrategy:
   type: RollingUpdate
   rollingUpdate:
     partition: 0
  template:
    metadata:
      labels:
        app: stateful
```

<continued>

<continued>

```yaml
    spec:
      containers:
      - name: nginx
        image: nginx:stable-alpine
        ports:
        - containerPort: 80
        volumeMounts:
        - name: www
          mountPath: /usr/share/nginx/html
  volumeClaimTemplates:
  - metadata:
      name: www
    spec:
      accessModes: [ "ReadWriteOnce" ]
      storageClassName: standard
      resources:
        requests:
          storage: 1Gi
```

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StatefulSet

- revisionHistoryLimit: The number of previous iterations of the StatefulSet to retain.
- serviceName: The name of the associated headless service; or a service without a **ClusterIP**.

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: sts-example
spec:
  replicas: 2
  revisionHistoryLimit: 3
  selector:
    matchLabels:
      app: stateful
  serviceName: app
  updateStrategy:
   type: RollingUpdate
   rollingUpdate:
     partition: 0
  template:
    <pod template>
```

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containers are created from container images

Storage

- Volumes
- Persistent Volumes
- Persistent Volume Claims
- StorageClass

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Storage

Pods by themselves are useful, but many workloads require exchanging data between containers, or persisting some form of data.

For this we have **Volumes**, **PersistentVolumes**, **PersistentVolumeClaims**, and **StorageClasses**.

StorageClass

- Storage classes are an abstraction on top of an external storage resource (PV)
- Work hand-in-hand with the external storage system to enable **dynamic provisioning** of storage by eliminating the need for the cluster admin to pre-provision a PV

StorageClass

- provisioner: Defines the '*driver*' to be used for provisioning of the external storage.
- parameters: A hash of the various configuration parameters for the provisioner.
- reclaimPolicy: The behaviour for the backing storage when the PVC is deleted.
  - Retain - manual clean-up
  - Delete - storage asset deleted by provider

```
kind: StorageClass
apiVersion: storage.k8s.io/v1
metadata:
  name: standard
provisioner: kubernetes.io/gce-pd
parameters:
  type: pd-standard
  zones: us-central1-a, us-central1-b
reclaimPolicy: Delete
```

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Volumes

- Storage that is tied to the **Pod’s Lifecycle**.
- A pod can have one or more types of volumes attached to it.
- Can be consumed by any of the containers within the pod.
- Survive Pod restarts; however their durability beyond that is dependent on the Volume Type.

Volumes

- volumes: A list of volume objects to be attached to the Pod. Every object within the list must have it’s own unique name.
- volumeMounts: A container specific list referencing the Pod volumes by name, along with their desired mountPath.

```
apiVersion: v1
kind: Pod
metadata:
  name: volume-example
spec:
  containers:
  - name: nginx
    image: nginx:stable-alpine
    volumeMounts:
    - name: html
      mountPath: /usr/share/nginx/html
      ReadOnly: true
  - name: content
    image: alpine:latest
    command: ["/bin/sh", "-c"]
    args:
      - while true; do
          date >> /html/index.html;
          sleep 5;
        done
    volumeMounts:
    - name: html
      mountPath: /html
  volumes:
  - name: html
    emptyDir: {}
```

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Volumes

- volumes: A list of volume objects to be attached to the Pod. Every object within the list must have it’s own unique name.
- volumeMounts: A container specific list referencing the Pod volumes by name, along with their desired mountPath.

```
apiVersion: v1
kind: Pod
metadata:
  name: volume-example
spec:
  containers:
  - name: nginx
    image: nginx:stable-alpine
    volumeMounts:
    - name: html
      mountPath: /usr/share/nginx/html
      ReadOnly: true
  - name: content
    image: alpine:latest
    command: ["/bin/sh", "-c"]
    args:
      - while true; do
          date >> /html/index.html;
          sleep 5;
        done
    volumeMounts:
    - name: html
      mountPath: /html
  volumes:
  - name: html
    emptyDir: {}
```

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Volumes

- volumes: A list of volume objects to be attached to the Pod. Every object within the list must have it's own unique name.
- volumeMounts: A container specific list referencing the Pod volumes by name, along with their desired mountPath.

```
apiVersion: v1
kind: Pod
metadata:
  name: volume-example
spec:
  containers:
  - name: nginx
    image: nginx:stable-alpine
    volumeMounts:
    - name: html
      mountPath: /usr/share/nginx/html
      ReadOnly: true
  - name: content
    image: alpine:latest
    command: ["/bin/sh", "-c"]
    args:
      - while true; do
          date >> /html/index.html;
          sleep 5;
        done
    volumeMounts:
    - name: html
      mountPath: /html
  volumes:
  - name: html
    emptyDir: {}
```

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Persistent Volumes

- A **PersistentVolume** (PV) represents a storage resource.
- PVs are a **cluster wide resource** linked to a backing storage provider: NFS, GCEPersistentDisk, RBD etc.
- Generally provisioned by an administrator.
- Their lifecycle is handled independently from a pod
- **CANNOT** be attached to a Pod directly. Relies on a **PersistentVolumeClaim**

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PersistentVolumeClaims

- A **PersistentVolumeClaim** (PVC) is a **namespaced** request for storage.
- Satisfies a set of requirements instead of mapping to a storage resource directly.
- Ensures that an application's '*claim*' for storage is portable across numerous backends or providers.

PersistentVolume

- `capacity.storage:` The total amount of available storage.
- `volumeMode:` The type of volume, this can be either `Filesystem` or `Block`.
- `accessModes:` A list of the supported methods of accessing the volume. Options include:
  - `ReadWriteOnce`
  - `ReadOnlyMany`
  - `ReadWriteMany`

```yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: nfsserver
spec:
  capacity:
    storage: 50Gi
  volumeMode: Filesystem
  accessModes:
    - ReadWriteOnce
    - ReadWriteMany
  persistentVolumeReclaimPolicy: Delete
  storageClassName: slow
  mountOptions:
    - hard
    - nfsvers=4.1
  nfs:
    path: /exports
    server: 172.22.0.42
```

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PersistentVolume

- persistentVolumeReclaimPolicy: The behaviour for PVC's that have been deleted. Options include:
  - Retain - manual clean-up
  - Delete - storage asset deleted by provider.
- storageClassName: Optional name of the storage class that PVC's can reference. If provided, **ONLY** PVC's referencing the name consume use it.
- mountOptions: Optional mount options for the PV.

```yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: nfsserver
spec:
  capacity:
    storage: 50Gi
  volumeMode: Filesystem
  accessModes:
    - ReadWriteOnce
    - ReadWriteMany
  persistentVolumeReclaimPolicy: Delete
  storageClassName: slow
  mountOptions:
    - hard
    - nfsvers=4.1
  nfs:
    path: /exports
    server: 172.22.0.42
```

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PersistentVolumeClaim

- accessModes: The selected method of accessing the storage. This **MUST** be a subset of what is defined on the target PV or Storage Class.
  - ReadWriteOnce
  - ReadOnlyMany
  - ReadWriteMany
- resources.requests.storage: The desired amount of storage for the claim
- storageClassName: The name of the desired Storage Class

```
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: pvc-sc-example
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
  storageClassName: slow
```

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PVs and PVCs with Selectors

```yaml
kind: PersistentVolume
apiVersion: v1
metadata:
  name: pv-selector-example
  labels:
    type: hostpath
spec:
  capacity:
    storage: 2Gi
  accessModes:
    - ReadWriteMany
  hostPath:
    path: "/mnt/data"
```

```yaml
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: pvc-selector-example
spec:
  accessModes:
    - ReadWriteMany
  resources:
    requests:
      storage: 1Gi
  selector:
    matchLabels:
      type: hostpath
```

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PVs and PVCs with Selectors

```yaml
kind: PersistentVolume
apiVersion: v1
metadata:
  name: pv-selector-example
  labels:
    type: hostpath
spec:
  capacity:
    storage: 2Gi
  accessModes:
    - ReadWriteMany
  hostPath:
    path: "/mnt/data"
```

```yaml
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: pvc-selector-example
spec:
  accessModes:
    - ReadWriteMany
  resources:
    requests:
      storage: 1Gi
  selector:
    matchLabels:
      type: hostpath
```

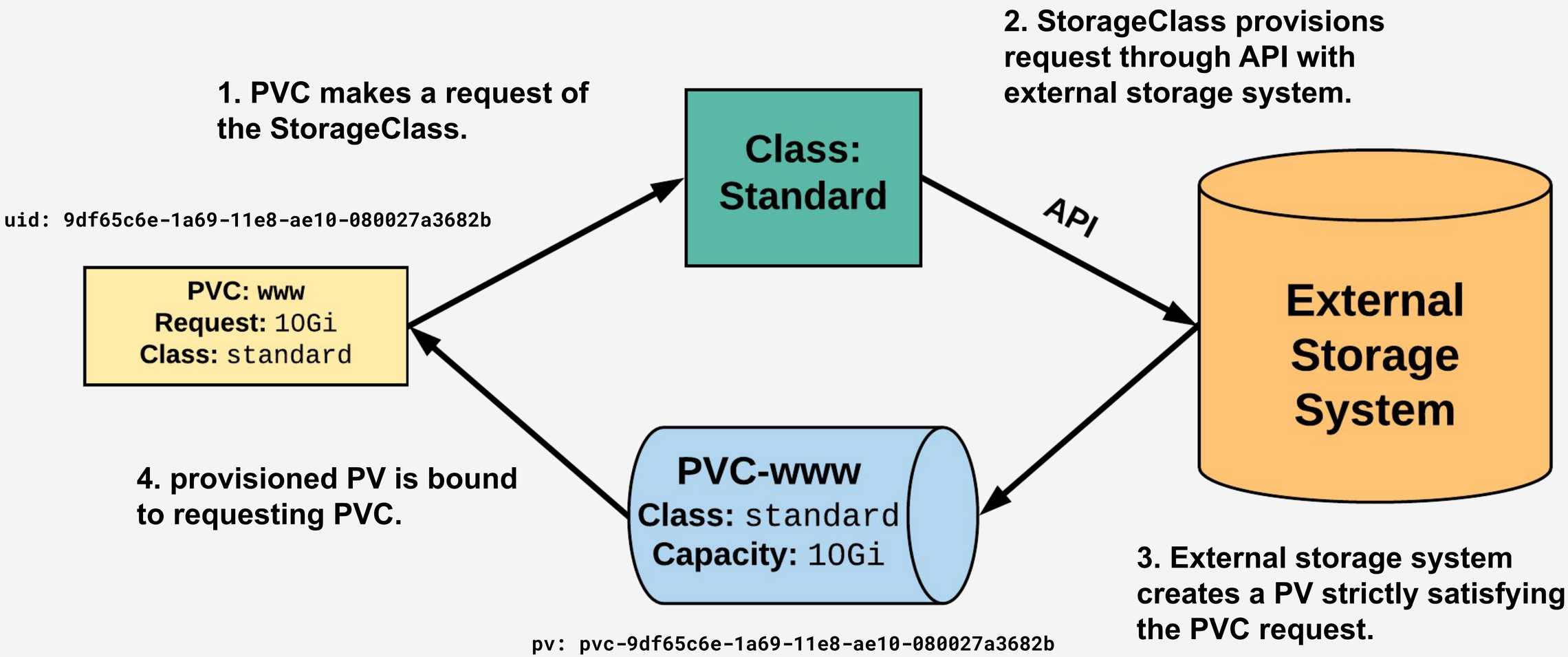
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PV Phases

Available	Bound	Released	Failed
PV is ready and available to be consumed.	The PV has been bound to a claim.	The binding PVC has been deleted, and the PV is pending reclamation.	An error has been encountered.

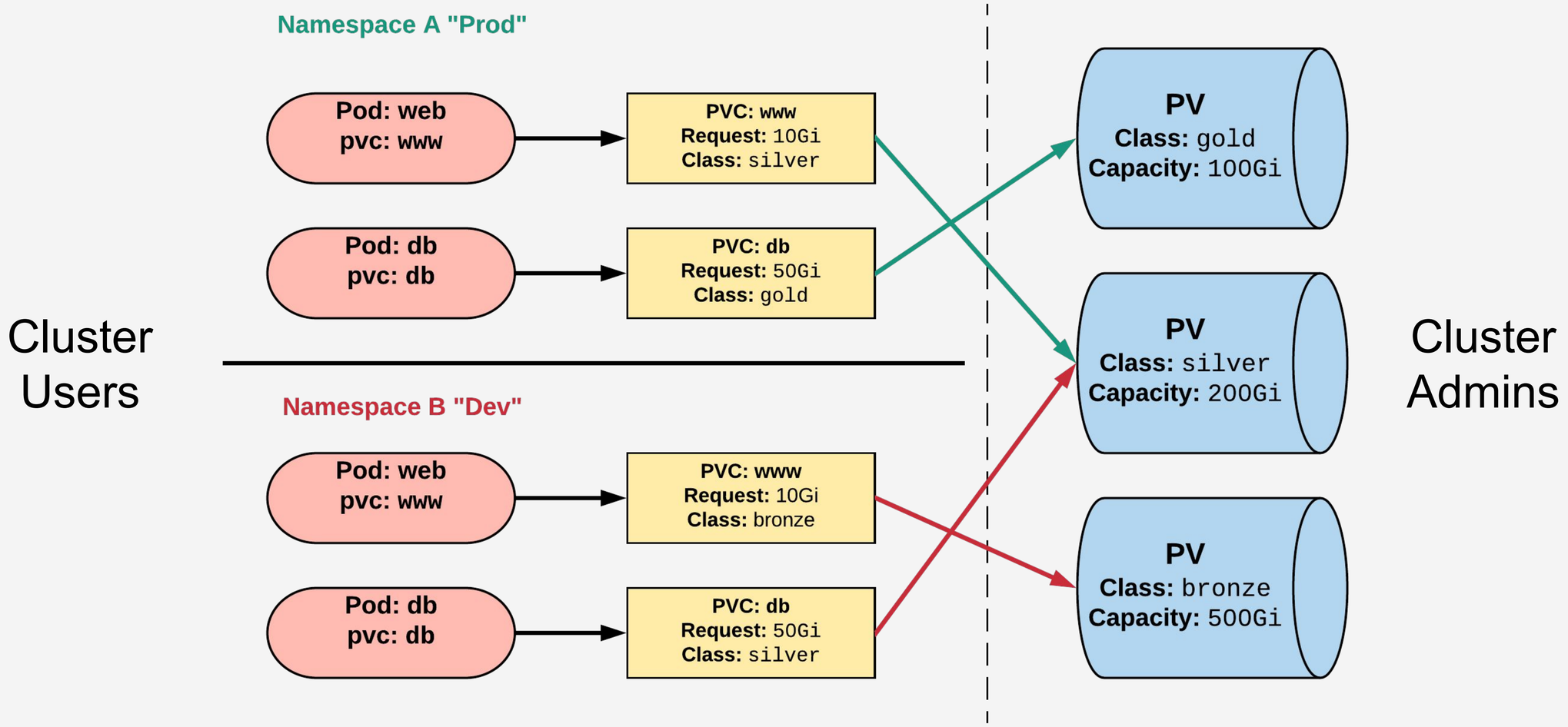
Red Hat IBM

StorageClass



Red Hat IBM

Persistent Volumes and Claims



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containers are created from container images

Working with Volumes

Concepts and resources

Red Hat IBM

containers are created from container images

Configurations

- ConfigMap
- Secret

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Configuration

Kubernetes has an integrated pattern for decoupling configuration from application or container.

This pattern makes use of two Kubernetes components: **ConfigMaps** and **Secrets.**

ConfigMap

- Externalized data stored within kubernetes.
- Can be referenced through several different means:
  - environment variable
  - a command line argument (via env var)
  - injected as a file into a volume mount
- Can be created from a manifest, literals, directories, or files directly.

ConfigMap

data: Contains key-value pairs of ConfigMap contents.

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: manifest-example
data:
  state: Michigan
  city: Ann Arbor
  content: |
    Look at this,
    its multiline!
```

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ConfigMap Example

All produce a **ConfigMap** with the same content!

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: manifest-example
data:
  city: Ann Arbor
  state: Michigan
```

```
$ kubectl create configmap literal-example \
> --from-literal="city=Ann Arbor" --from-literal=state=Michigan
configmap “literal-example” created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap dir-example --from-file=cm/
configmap "dir-example" created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap file-example --from-file=cm/city --from-file=cm/state
configmap "file-example" created
```

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ConfigMap Example

All produce a **ConfigMap** with the same content!

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: manifest-example
data:
  city: Ann Arbor
  state: Michigan
```

```
$ kubectl create configmap literal-example \
> --from-literal="city=Ann Arbor" --from-literal=state=Michigan
configmap "literal-example" created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap dir-example --from-file=cm/
configmap "dir-example" created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap file-example --from-file=cm/city --from-file=cm/state
configmap "file-example" created
```

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ConfigMap Example

All produce a **ConfigMap** with the same content!

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: manifest-example
data:
  city: Ann Arbor
  state: Michigan
```

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$ kubectl create configmap literal-example \
> --from-literal="city=Ann Arbor" --from-literal=state=Michigan
configmap “literal-example” created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap dir-example --from-file=cm/
configmap "dir-example" created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap file-example --from-file=cm/city --from-file=cm/state
configmap "file-example" created
```

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ConfigMap Example

All produce a **ConfigMap** with the same content!

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: manifest-example
data:
  city: Ann Arbor
  state: Michigan
```

```
$ kubectl create configmap literal-example \
> --from-literal="city=Ann Arbor" --from-literal=state=Michigan
configmap “literal-example” created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap dir-example --from-file=cm/
configmap "dir-example" created
```

```
$ cat info/city
Ann Arbor
$ cat info/state
Michigan
$ kubectl create configmap file-example --from-file=cm/city --from-file=cm/state
configmap "file-example" created
```

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Secret

- Functionally identical to a ConfigMap.
- Stored as **base64 encoded content.**
- Encrypted at rest within etcd (**if configured!**).
- Stored on each worker node in tmpfs directory.
- Ideal for username/passwords, certificates or other sensitive information that should not be stored in a container.

Secret

- type: There are three different types of secrets within Kubernetes:
  - docker-registry - credentials used to authenticate to a container registry
  - generic/Opaque - literal values from different sources
  - tls - a certificate based secret
- data: Contains key-value pairs of base64 encoded content.

```
apiVersion: v1
kind: Secret
metadata:
  name: manifest-secret
type: Opaque
data:
  username: ZXhhbXBsZQ==
  password: bXlwYXNzd29yZA==
```

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Secret Example

All produce a **Secret** with the same content!

```yaml
apiVersion: v1
kind: Secret
metadata:
  name: manifest-example
type: Opaque
data:
  username: ZXhhbXBsZQ==
  password: bXlwYXNzd29yZA==
```

```
$ kubectl create secret generic literal-secret \
> --from-literal=username=example \
> --from-literal=password=mypassword
secret "literal-secret" created
```

```
$ cat info/username
example
$ cat info/password
mypassword
$ kubectl create secret generic dir-secret --from-file=secret/
Secret "file-secret" created
```

```
$ cat secret/username
example
$ cat secret/password
mypassword
$ kubectl create secret generic file-secret --from-file=secret/username --from-file=secret/password
Secret "file-secret" created
```

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Secret Example

All produce a **Secret** with the same content!

```
apiVersion: v1
kind: Secret
metadata:
  name: manifest-example
type: Opaque
data:
  username: ZXhhbXBsZQ==
  password: bXlwYXNzd29yZA==
```

```
$ kubectl create secret generic literal-secret \
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> --from-literal=password=mypassword
secret "literal-secret" created
```

```
$ cat info/username
example
$ cat info/password
mypassword
$ kubectl create secret generic dir-secret --from-file=secret/
Secret "file-secret" created
```

```
$ cat secret/username
example
$ cat secret/password
mypassword
$ kubectl create secret generic file-secret --from-file=secret/username --from-file=secret/password
Secret "file-secret" created
```

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All produce a **Secret** with the same content!

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$ kubectl create secret generic literal-secret \
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secret "literal-secret" created
```

```
$ cat info/username
example
$ cat info/password
mypassword
$ kubectl create secret generic dir-secret --from-file=secret/
Secret "file-secret" created
```

```
$ cat secret/username
example
$ cat secret/password
mypassword
$ kubectl create secret generic file-secret --from-file=secret/username --from-file=secret/password
Secret "file-secret" created
```

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Secret Example

All produce a **Secret** with the same content!

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$ kubectl create secret generic literal-secret \
> --from-literal=username=example \
> --from-literal=password=mypassword
secret "literal-secret" created
```

```
$ cat info/username
example
$ cat info/password
mypassword
$ kubectl create secret generic dir-secret --from-file=secret/
Secret "file-secret" created
```

```
$ cat secret/username
example
$ cat secret/password
mypassword
$ kubectl create secret generic file-secret --from-file=secret/username --from-file=secret/password
Secret "file-secret" created
```

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Injecting as Environment Variable

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: cm-env-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["printenv CITY"]
        env:
        - name: CITY
          valueFrom:
            configMapKeyRef:
              name: manifest-example
              key: city
      restartPolicy: Never
```

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: secret-env-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["printenv USERNAME"]
        env:
        - name: USERNAME
          valueFrom:
            secretKeyRef:
              name: manifest-example
              key: username
      restartPolicy: Never
```

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Injecting as Environment Variable

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: cm-env-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["printenv CITY"]
        env:
        - name: CITY
          valueFrom:
            configMapKeyRef:
              name: manifest-example
              key: city
      restartPolicy: Never
```

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: secret-env-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["printenv USERNAME"]
        env:
        - name: USERNAME
          valueFrom:
            secretKeyRef:
              name: manifest-example
              key: username
      restartPolicy: Never
```

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Injecting in a Command

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: cm-cmd-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["echo Hello $(CITY)!"]
        env:
        - name: CITY
          valueFrom:
            configMapKeyRef:
              name: manifest-example
              key: city
      restartPolicy: Never
```

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: secret-cmd-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["echo Hello $(USERNAME)!"]
        env:
        - name: USERNAME
          valueFrom:
            secretKeyRef:
              name: manifest-example
              key: username
      restartPolicy: Never
```

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Injecting in a Command

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: cm-cmd-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["echo Hello ${CITY}!"]
        env:
        - name: CITY
          valueFrom:
            configMapKeyRef:
              name: manifest-example
              key: city
      restartPolicy: Never
```

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: secret-cmd-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: ["/bin/sh", "-c"]
        args: ["echo Hello ${USERNAME}!"]
        env:
        - name: USERNAME
          valueFrom:
            secretKeyRef:
              name: manifest-example
              key: username
      restartPolicy: Never
```

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Injecting as a Volume

```
apiVersion: batch/v1
kind: Job
metadata:
  name: cm-vol-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: [“/bin/sh”, “-c”]
        args: [“cat /myconfig/city”]
        volumeMounts:
        - name: config-volume
          mountPath: /myconfig
      restartPolicy: Never
      volumes:
      - name: config-volume
        configMap:
          name: manifest-example
```

```
apiVersion: batch/v1
kind: Job
metadata:
  name: secret-vol-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: [“/bin/sh”, “-c”]
        args: [“cat /mysecret/username”]
        volumeMounts:
        - name: secret-volume
          mountPath: /mysecret
      restartPolicy: Never
      volumes:
      - name: secret-volume
        secret:
          secretName: manifest-example
```

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Injecting as a Volume

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: cm-vol-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: [“/bin/sh”, “-c”]
        args: [“cat /myconfig/city”]
        volumeMounts:
        - name: config-volume
          mountPath: /myconfig
      restartPolicy: Never
      volumes:
      - name: config-volume
        configMap:
          name: manifest-example
```

```yaml
apiVersion: batch/v1
kind: Job
metadata:
  name: secret-vol-example
spec:
  template:
    spec:
      containers:
      - name: mypod
        image: alpine:latest
        command: [“/bin/sh”, “-c”]
        args: [“cat /mysecret/username”]
        volumeMounts:
        - name: secret-volume
          mountPath: /mysecret
      restartPolicy: Never
      volumes:
      - name: secret-volume
        secret:
          secretName: manifest-example
```

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Questions?

Adapted from slides by Joe Beda (Gluecon 2017).

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Changes were made.